SPECIAL PROJECT

# Standardizing Rigid Inclusion Soil Improvement Data with DIGGS

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**Abstract**

Rigid inclusions (RI), including controlled modulus columns, are a widely used ground-improvement technology for buildings and infrastructure on soft ground, yet the data that describe an RI project remain fragmented across incompatible, proprietary formats and across competing design standards. Rigid-inclusion installation relies on a variety of equipment and techniques, often marketed under different proprietary trade names. Depending on the equipment, the installation method and the soil, the process can partially or fully displace the surrounding ground and disturb adjacent soils, neighboring inclusions and nearby structures, even though the columns are typically installed beneath a load transfer platform (LTP) that supports the structure or embankment. Because these methods are proprietary and closely guarded for commercial advantage, the scope and quality of ground improvement work remain highly dependent on contractors for design and implementation, in addition to that at present no universally accepted design methodologies or construction specifications exist to evaluate the load-transfer mechanism or the installation-induced effects on surrounding soils, adjacent inclusions and existing structures. This report establishes the technical basis for standardizing rigid-inclusion data using DIGGS, the open, XML-based Data Interchange for Geotechnical and Geoenvironmental Specialists. It defines the rigid-inclusion system and its load-transfer mechanics, compares the principal design standards (ASIRI, BS 8006-1, EBGEO and FHWA GEC-13) and the installation practices of the specialty contractors, and reviews the recent state-of-the-practice survey. Building on this review, the data are organized into a four-phase framework (pre-installation, design, installation and post-installation) and a standardized parameter set is assembled that describes an RI project across its whole life cycle. Each parameter is then mapped to the DIGGS schema, distinguishing the data that existing DIGGS features already carry from the data that require a minimal, backward-compatible extension of new rigid-inclusion features, and a worked case study demonstrates that the framework can capture a complete project and feed a DIGGS instance file.

**Key Words**

Rigid inclusions; Controlled modulus columns; Ground improvement; Data standardization; DIGGS; XML schema; Load-transfer platform; Soil improvement.

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NOTATION

# List of Symbols

*This notation lists the symbols used throughout the report, grouped by the four project phases that organise the essential-parameter set: site investigation, design, installation and post-installation verification.*

Units follow SI conventions; a dash (–) denotes a dimensionless quantity, and alternative units in common practice are given in parentheses.

## A. Site Investigation

| Symbol | Definition | Unit |
|---|---|---|
| **Index & classification properties** | | |
| $w$ | Natural (in-situ) water content | % |
| $w_L$ | Liquid limit | % |
| $w_P$ | Plastic limit | % |
| $I_P$ | Plasticity index ($w_L - w_P$) | % |
| $\gamma$ | Bulk (total) unit weight of soil | kN/m³ |
| $\gamma_d$ | Dry unit weight of soil | kN/m³ |
| $G_s$ | Specific gravity of soil solids | – |
| $e_0$ | Initial (in-situ) void ratio | – |
| **Strength properties** | | |
| $c_u$ ($s_u$) | Undrained shear strength of fine-grained soil | kPa |
| $c'$ | Effective cohesion intercept | kPa |
| $\varphi'$ | Effective (drained) angle of shearing resistance | ° |
| $\varphi'_{cv}$ | Critical-state / constant-volume friction angle | ° |
| $S_t$ | Sensitivity (undisturbed/remoulded strength ratio) | – |
| $D_r$ | Relative density of granular / bearing strata | % |
| $K_0$ | Coefficient of earth pressure at rest | – |
| **Compressibility & consolidation (oedometer)** | | |
| $C_c$ | Compression index | – |
| $C_s$ ($C_r$) | Recompression / swelling index | – |
| $C_\alpha$ | Secondary-compression (creep) index | – |
| $\sigma'_p$ | Preconsolidation (yield) pressure | kPa |
| OCR | Overconsolidation ratio | – |
| $m_v$ | Coefficient of volume compressibility | $MPa^{-1}$ ($m^2/MN$) |
| $E_{oed}$ | Constrained (oedometric) modulus | MPa |
| $c_v$ | Coefficient of consolidation (vertical) | m²/yr |
| $c_h$ | Coefficient of horizontal consolidation | m²/yr |
| $k$ ($k_v$) | Vertical permeability (hydraulic conductivity) | m/s |
| $k_h$ | Horizontal permeability | m/s |
| **In-situ test data** | | |
| $q_c$ ($q_t$) | CPT / CPTu (corrected) cone tip resistance | MPa |
| $f_s$ | CPT sleeve friction | kPa |
| $R_f$ | CPT friction ratio ($f_s$ / $q_c$) | % |

| Symbol | Definition | Unit |
|---|---|---|
| $u_2$ | CPTu pore pressure measured behind the cone | kPa |
| N ($N_{60}$) | SPT blow count ($N_{60}$ = energy-corrected) | blows/300 mm |
| $E_M$ | Ménard pressuremeter modulus | MPa |
| $p_l$ ($p_l$*) | Pressuremeter limit pressure ($p_l$* = net) | kPa |
| $c_{u,vane}$ | Field-vane undrained shear strength | kPa |
| $V_s$ | Shear-wave velocity | m/s |
| $G_{max}$ | Small-strain (maximum) shear modulus | MPa |
| **Environmental / durability inputs** | | |
| $SO_4^{2-}$ | Sulfate content of soil and groundwater | mg/kg, mg/L |
| $Cl^-$ | Chloride content of soil and groundwater | mg/kg, mg/L |
| pH | pH of soil / groundwater | – |
| **Seismic & special inputs** | | |
| $a_g$ (PGA) | Design ground acceleration / peak ground acceleration | g |

## B. Design

| Symbol | Definition | Unit |
|---|---|---|
| **Design basis & safety format** | | |
| $t_L$ | Design working life | yr |
| $\gamma_F$, $\gamma_M$, $\gamma_R$ | Partial factors on actions, materials and resistances | – |
| f | Global factor of safety (allowable-stress format) | – |
| $k_{s,T}/k_s$ | Method-applicability gate (stiffness ratio) and geometry envelope | – |
| **Geometry & layout** | | |
| $A_{RI}$ | Support (treated) plan area of the improved ground | m² |
| s | Column spacing, centre-to-centre | m |
| $s_x$, $s_y$ | Column spacing in the x- and y-directions | m |
| D (d) | Inclusion (column) diameter | m |
| L | Inclusion length / toe depth | m |
| $d_{Ers}$ | Equivalent diameter of the inclusion (EBGEO) | m |
| $D_e$ | Equivalent / tributary (unit-cell) diameter (FHWA) | m |
| a | Head / cap equivalent width | m |
| α ($a_s$) | Area replacement ratio (inclusion area / tributary area) | % (–) |
| $H_{crit}$ | Critical height for full soil arching | m |
| n | Embankment side-slope ratio (horizontal:vertical) | – |
| β | Embankment batter (side-slope) angle | ° |
| **Column material & structural** | | |
| $f_{ck}$ ($f_{ck}$*) | Characteristic ($f_{ck}$* = reduced) compressive strength of the column material | MPa |
| $f_{cd}$ | Design compressive strength of the column material | MPa |
| $\bar{\sigma}_{lim}$ | Mean-stress limit (fixed cap, e.g. 7 MPa) | MPa |
| E | Elastic (Young's) modulus of the column material | GPa |
| $R_c$ | Column axial (structural) capacity | kN |
| N | Column axial load (structural demand) | kN |
| $N/R_c$ | Column axial utilization ratio | – |
| $\tau_{cp}$ | Column shear stress (shear check) | MPa |
| $\rho_s$ | Longitudinal steel reinforcement ratio | % |
| $A_s$ | Area of steel reinforcement | mm² |
| $L_{reinf}$ | Length of column reinforcement (Domain 1 / tension zone) | m |

| Symbol | Definition | Unit |
|---|---|---|
| $M$ | Bending-moment demand on the inclusion | kN·m |
| $H_{lat}$ | Lateral-force demand on the inclusion | kN |
| **Shaft, interface & neutral plane** | | |
| $k_{\tau}$ | Shaft load-transfer (t–z) modulus | kN/m³ |
| $k_q$ | Tip load-transfer (q–z) modulus | kN/m³ |
| $q_s$ | Negative skin-friction limit (unit shaft resistance) | kPa |
| $h_c$ ($z_n$) | Depth of the neutral plane | m |
| $Q_{max}$ | Maximum axial force (downdrag) at the neutral plane | kN |
| **Toe / base resistance** | | |
| $R_b$ | Toe / base (end-bearing) resistance | kN |
| $R_s$ | Shaft (positive) resistance | kN |
| $q_p$ | Limit unit tip (base) resistance | kPa |
| $Q_r$ | Design axial load required per column (demand) | kN |
| **Load-transfer platform (LTP)** | | |
| $H_M$ ($h$, $h^*$) | LTP thickness | m |
| $\varphi'_{LTP}$ | LTP fill friction angle | ° |
| $\gamma_{fill}$ | LTP / embankment fill unit weight | kN/m³ |
| $C_{c,arch}$ | Arching coefficient (BS 8006 modified Marston) | – |
| $p'_c/\sigma'_v$ | Cap-stress ratio (BS 8006) | – |
| $E_{crown}$ | Hewlett–Randolph crown arching efficiency | – |
| $E_{cap}$ | Hewlett–Randolph cap arching efficiency | – |
| $E_{min}$ | Governing (minimum) arching efficiency | – |
| $K_p$ | Passive earth-pressure coefficient | – |
| $K_{crit}$ | Critical stress ratio (EBGEO) | – |
| $\sigma_{zo,k}$ | Characteristic vertical stress on the soft soil between elements | kPa |
| $\sigma_{zs,k}$ | Characteristic vertical stress on the element (head) | kPa |
| $K$ | Adapted-Terzaghi earth-pressure coefficient | – |
| $N_q$ | Head-punching bearing-capacity factor (ASIRI) | – |
| $q_p^+$ | Head stress on the inclusion | kPa |
| **Geosynthetic reinforcement (conditional)** | | |
| $n_{geo}$ | Number of geosynthetic reinforcement layers | – |
| $J$ ($J_k$) | Tensile stiffness of the reinforcement | kN/m |
| $T_D$ ($R_{B,d}$) | Long-term design tensile strength of the reinforcement | kN/m |
| A1–A5 | Reduction factors (creep, installation damage, weathering, joints/seams) | – |
| $f_m$ | Partial material factor on the reinforcement | – |
| $T_{rp}$ | Reinforcement design tension supporting the embankment | kN/m |
| $E_{M,geo}$ | Membrane (tensile) force in the reinforcement (EBGEO) | kN/m |
| $T$ | Reinforcement tensile force (generic) | kN/m |
| $L_e$ | Reinforcement anchorage (embedment) length | m |
| $L_b$ ($L_A$) | Reinforcement bond length | m |
| $\varepsilon$ | Allowable / design tensile strain of the reinforcement | % |
| $\Delta\varepsilon_{kr}$ | Creep-strain limit over the design life | % |
| **Soft soil, ground model & loading** | | |
| $k_s$ | Modulus of subgrade reaction (Winkler) | kN/m³ |
| $H$ | Embankment height | m |
| $w_s$ ($p_k$) | Surcharge / live load | kPa |
| $q$ | Applied structure / slab load | kPa |

| Symbol | Definition | Unit |
|---|---|---|
| **Load transfer, efficacy & stability (ULS)** | | |
| $E$ | Load (stress) efficacy — fraction of load carried by the inclusions | – |
| $E_{tass}$ | Settlement efficacy / settlement-reduction ratio | – |
| $K_a$ | Active earth-pressure coefficient of the fill | – |
| FS | Global / slope-stability factor of safety | – |
| $T_{ds}$ | Lateral-spreading (thrust) demand | kN/m |
| $P_{Lat}$ | Lateral thrust on the inclusions (FHWA) | kN/m |
| $FS_{ls}$ | Lateral-spreading factor of safety | – |
| $L_p$ | Pile-group lateral extent at the embankment toe | m |
| **Serviceability & acceptance** | | |
| $s_{tot}$ | Total settlement | mm |
| $\Delta s$ | Differential settlement | mm |
| $y$ | Reinforcement mid-span (sag) deflection | mm |
| **Auxiliary symbols in the design equations** | | |
| $A_c$ | Column (inclusion) cross-sectional area | m² |
| $A$ | Unit-cell (tributary) area | m² |
| $A_{cell}$ | Unit-cell area (FHWA) | m² |
| $A_{grid}$ | Grid tributary area (ASIRI) | m² |
| $A_p$ | Inclusion cross-sectional (tip) area | m² |
| $A_E$ | Influence area per element (EBGEO) | m² |
| $A_S$ | Cross-sectional area of the element / column (EBGEO) | m² |
| $A_{soil}$ | Soft-soil plan area within the unit cell | m² |
| $n$ | Stress concentration ratio ($\sigma_c$ / $\sigma_s$) | – |
| $\sigma_c$ ($\sigma_{col}$) | Vertical stress on the column / inclusion | kPa |
| $\sigma_s$ ($\sigma_{soil}$) | Vertical stress on the soft soil | kPa |
| $\sigma_{avg}$ | Average applied (design) vertical stress | kPa |
| $\sigma'_v$ | Vertical effective stress | kPa |
| $\sigma_{max}$ | Maximum compressive stress in the column | MPa |
| $q_0$ | Applied vertical stress at the base of the LTP | kPa |
| $q_s^+$ | Vertical stress on the soft soil at the inclusion head | kPa |
| SRR | Stress-reduction ratio (soil stress / average stress) | – |
| $D_s$ | Shaft diameter used in the t–z law | m |
| $D_p$ | Tip (base) diameter used in the q–z law | m |
| $m_\tau$, $m_q$ | Frank–Zhao shaft and tip rheological coefficients | – |
| $k_c$ | Cone (penetrometer) bearing factor | – |
| $k_p$ | Pressuremeter bearing factor | – |
| $\delta$ | Soil–inclusion interface friction angle | ° |
| $z$ | Depth below the platform / ground surface | m |
| $f_n$ | Unit negative skin friction | kPa |
| $F_N$ | Downdrag (negative-friction) force above the neutral plane | kN |
| $F_P$ | Positive shaft resistance below the neutral plane | kN |
| $y_0$ | Settlement of the untreated soil (no inclusions) | mm |
| $y_M$ | Settlement of the improved ground | mm |
| $\beta_{HR}$ | Hewlett–Randolph arching-efficiency factor | – |
| $W_T$ | Distributed tensile load carried by the reinforcement (BS 8006) | kN/m |
| $s_d$ | Clear spacing between elements (s − d) | m |
| $h_g$ | Arch height (EBGEO) | m |

| Symbol | Definition | Unit |
|---|---|---|
| $\varepsilon_{max}$ | Maximum reinforcement strain | % |
| $E_s$ | Constrained modulus of the soft soil (subgrade) | kPa |
| $t_w$ | LTP thickness on the Winkler basis (EBGEO) | m |
| $N_c$, $N_\gamma$ | Cohesion and self-weight bearing-capacity factors | – |
| $s_c$, $s_q$, $s_\gamma$ | Bearing-capacity shape factors | – |
| $\theta_p$ | Slip-plane inclination (45° − $\varphi'_{cv}$/2) | ° |
| W | Embankment (fill) self-weight per unit cell | kN |
| Q | Applied surcharge / structural load per unit cell | kN |
| $Q_p$ | Load carried by the inclusion (column) head | kN |
| $f_{fs}$ | Partial load factor on soil self-weight (BS 8006) | – |
| $f_q$ | Partial load factor on surcharge (BS 8006) | – |
| $\gamma_G$, $\gamma_Q$ | Partial factors on permanent and variable actions | – |
| $\gamma_C$ | Partial factor on concrete | – |
| $\eta_M$ | Model factor (EBGEO) | – |
| $\chi$, $\lambda_1$, $\lambda_2$ | EBGEO arching auxiliary coefficients | – |
| $L_s$ | Length of soil providing lateral resistance (FHWA) | m |
| $c_{i,emb}$ | Embankment interface adhesion coefficient (FHWA) | – |
| $V_{Ed}$ | Design shear force on the column | kN |
| $k_1$, $k_2$, $k_3$ | ASIRI strength-reduction coefficients (concrete / soil-mix) | – |
| $\alpha_{cc}$ | Concrete long-term strength coefficient | – |
| $C_{max}$ | Maximum allowable characteristic strength (ASIRI) | MPa |

## C. Installation

| Symbol | Definition | Unit |
|---|---|---|
| **As-built geometry** | | |
| D | As-built inclusion diameter | mm |
| L | As-built inclusion length | m |
| $z_{max}$ | Maximum depth reached by the tool | m |
| $t_{wp}$ | Working-platform thickness | m |
| **Process record (MWD / drilling coverage)** | | |
| $v_{pen}$ | Penetration rate versus depth | m/h (cm/min) |
| $v_{ext}$ | Extraction rate versus depth | m/h |
| $n_{rot}$ | Rotation speed | rev/min |
| C ($M_t$) | Rotation torque versus depth | kN·m (t·m) |
| $F_{crowd}$ | Crowd / down-thrust force | kN (bar) |
| $P_B$ | Grout / concrete injection pressure versus depth | bar (kPa) |
| V | Grout / concrete volume versus depth | m³ |
| **Concrete / grout as-placed** | | |
| $V_{tot}$ | Total concrete / grout volume placed in the column | m³ |
| $V_{ws}$ | Wet-squeeze / grout-return volume | m³ |

## D. Post-installation

| Symbol | Definition | Unit |
|---|---|---|
| **Integrity & material verification** | | |

| Symbol | Definition | Unit |
|---|---|---|
| $f_c$ | Concrete cylinder / core compressive strength (7 d / 28 d) | MPa |
| **Load testing (performance verification)** | | |
| Q–s | Static load–settlement response of a single column | kN, mm |
| $R_{c,cr}$ | Critical creep load | kN |
| **Load-transfer platform verification** | | |
| $D_{Pr}$ | LTP degree of compaction (Proctor) | % |
| $\varphi'_{LTP}$ | Verified LTP fill friction angle | ° |
| **Instrumentation & monitoring** | | |
| $s_{meas}$ | Measured settlement | mm |
| $\Delta s_{meas}$ | Measured differential settlement | mm |
| u | Pore-water pressure (piezometer) | kPa |
| $\varepsilon_{meas}$ | Measured reinforcement / geosynthetic strain | % |
| $V_{s,comp}$ | Composite shear-wave velocity of the improved ground (post-installation) | m/s |

## CHAPTER 1
# Introduction

*Rigid inclusions are one of the most widely used ground-improvement solutions for building and infrastructure on soft ground, yet the data that describe an inclusion project, the ground model, the design intent, the as-built execution and the verification testing, remain scattered across incompatible, proprietary formats. This first chapter sets out the problem this fragmentation creates and explains why an open data standard, such as DIGGS, is the right instrument to solve it for rigid inclusions. It introduces the sponsoring special project, states the problem on its two fronts, the design record and the construction record, describes what DIGGS is and why it matters specifically for rigid-inclusion data, and fixes the objective and scope of the report.*

## 1.1 Background and the Special Project

This report is a technical product of the ASCE Geo-Institute (GI) special project "Standardizing Rigid Inclusion Soil Improvement Data with DIGGS," sponsored by the Technical Coordination Council (TCC) through the Soil Improvement Committee (SIC) for the 2025–2026 program year. The project is led by Dr. Mohamed Ezzat with subject-matter experts Mr. Daniel Ponti (U.S. Geological Survey, DIGGS development) and Mr. Allen Cadden (Schnabel Engineering), supported by graduate-student participants.

Rigid inclusions (RI), including controlled modulus columns, have become one of the most common ground-improvement technologies for construction on soft ground, because they reduce and homogenize settlement, increase bearing capacity and enhance stability quickly and economically. Their growing use has produced a growing volume of data, generated from the ground investigation, the design, the installation of every column and the verification testing. That data is today fragmented: it is recorded on paper, in spreadsheets and in proprietary formats that do not talk to one another, and no common, machine-readable structure exists in which to record, exchange and compare it. The special project exists to close that gap by developing a standardized rigid-inclusion data set and mapping it onto the DIGGS open standard, so that rigid-inclusion projects can be recorded once, compared consistently, and connected from design intent through construction to performance.

## 1.2 Problem Statement

The design and execution of rigid-inclusion systems are strongly influenced by the standards and specifications adopted by each designer and each soil-improvement contractor. This produces fragmentation on two fronts, the design record and the construction record, and it is the root cause of the data-management problem this project addresses. This is not only an academic concern: rigid-inclusion practitioners have also reported that "there currently exists no industry-wide standard for the selection, design, and verification of RIs" (Smith et al. 2023a, 2023b), and that because a rigid inclusion is deliberately separated from the structure by the load-transfer platform, its design falls outside the minimum requirements of Chapter 18 (Soils and Foundations) of the 2024 International Building Code (ICC 2024), so the technology has grown without the common data framework a standard would provide.

#### 1.2.1 Fragmentation of the design record

Several design methods have been developed to analyze rigid-inclusion-supported ground, each based on different simplifying assumptions for load transfer, arching and reinforcement behavior. In the most recent independent evaluation, Han et al. (2025) compared four popular methods, BS 8006-1, EBGEO, CUR226 and the Federal Highway Administration (FHWA) method, against a database of 24 full-scale experiments and four model tests for three key design parameters: load efficacy, differential settlement and reinforcement strain. The comparison revealed variations and inconsistencies of the calculated results among the design methods; the FHWA and CUR226 methods predicted the measured behavior comparatively more accurately, while BS 8006-1 overestimated all three parameters (Han et al. 2025). Comparable divergence appears in stability analysis, where the Column-Wall, Equivalent-Strength, Stress-Reduction and Pile-Support methods return materially different factors of safety for the same embankment (Han et al. 2025). The practical consequence is that two competent engineers, applying two accepted standards to the same site, can report different design quantities under different names, with no common structure in which to record, audit or compare them.

#### 1.2.2 Fragmentation of the construction record

On the construction side, rigid-inclusion information is today captured in proprietary contractor logs and standard-specific formats. Each specialty contractor records essentially the same physical quantities (depth, penetration and withdrawal rate, torque, crowd, grout or concrete pressure and volume, and refusal or termination criteria) but under different field names, units and file layouts generated by different data-acquisition systems. As a result, rigid-inclusion information cannot readily be aggregated across projects, and owners, consultants and researchers are prevented from building reliable case-history databases, comparing performance, or feeding analytics and digital-twin models.

Underlying both problems is the absence of a consensus data model. Even where mature formats exist for site-investigation and monitoring data, there is no rigid-inclusion object into which design intent and as-built execution can be written. Each project therefore becomes a data island: its data are not comparable, and its lessons are not carried forward. A single, standardized and openly-defined data dictionary, one that names each parameter unambiguously, records its units and provenance, and is portable across the competing standards and contractors, is the missing piece, and the motivation for this work.

### 1.3 Why DIGGS Matters for Rigid Inclusions

DIGGS, the Data Interchange for Geotechnical and Geoenvironmental Specialists, is an open, non-proprietary, XML-based standard for transferring geotechnical and geoenvironmental data between organizations and software. Because it is a published, vendor-neutral schema rather than a product format, data written to DIGGS remain readable and exchangeable independently of any single software vendor, and they can be validated automatically against the schema. Its releases have progressively broadened coverage of ground-improvement technologies: the DIGGS 2.6 schema added support for deep soil mixing and grouting. Despite that progress, the schema still lacks a dedicated data model for rigid inclusions, which is precisely the gap this special project addresses, by reusing the features DIGGS already provides and adding a small, backward-compatible rigid-inclusion module, developed in the DIGGS 3.0 Extension and set out in Chapter 6.

DIGGS is the right instrument for rigid-inclusion data for three reasons. First, a rigid-inclusion project has exactly the life cycle DIGGS already models well elsewhere: a ground investigation, a set of materials and their tests, an installed physical element with a construction activity and a process record, and a program of verification tests and monitoring. Much of an RI dataset, the borings and in-situ and laboratory tests, the fresh-material tests, the monitoring records and the documents, can therefore reuse features the schema already defines, and the extension needed for the rigid-inclusion-specific objects is correspondingly small and backward-compatible. Second, standardizing RI data on the same open schema as the rest of geotechnical practice makes rigid-inclusion records interoperable with site investigation, deep foundations and monitoring, rather than siloed in a specialty format. Third, because a rigid-inclusion project is a soil-improvement work whose performance depends on the composite behaviour of column and soil, its design record, its as-built record and its verification record only have meaning together, and an open standard is what lets them be linked, by a common identifier and a common vocabulary, across the whole project.

The benefits of standardizing rigid-inclusion data on DIGGS follow directly from the problems above. A standardized, machine-readable record fixes each design parameter once, with its unit, definition and source, so that a design is reproducible and auditable and can be cross-checked against a second method, the very exercise that exposed the divergence between standards. It lets every installed column be tied by a unique identifier to its design and its tests, so that as-built values can be compared against design targets column by column and anomalies flagged during construction rather than discovered afterwards. It lets verification data and acceptance limits share one structure, so that conformance can be evaluated automatically and the acceptance trail preserved for audit. And beyond the single project, it enables the case-history databases that let design methods be calibrated against measured performance, the analytics and machine learning that depend on aggregated data, and the integration with building-information-modeling (BIM), geographic-information-system (GIS) and digital-twin workflows that increasingly underpin asset management, while reducing the manual re-entry that wastes effort and introduces error.

## 1.4 Objective and Scope of This Report

The objective of this report is to develop the technical, that is, geotechnical, basis for standardizing rigid-inclusion data with DIGGS: to establish what rigid-inclusion data must be described, why it matters, where the discrepancies between the standards and the contractors lie that a standard must reconcile, and how the resulting parameter set maps onto the DIGGS schema. The emphasis is on the engineering meaning of the data, the rigid-inclusion system and the parameters that describe it, rather than on any design calculation: the report does not re-implement or arbitrate between the design methods, but names, defines and organizes the parameters they produce. Building on that parameter set, the report goes on to present the rigid-inclusion extension the schema requires and to demonstrate the whole framework on a worked case study, which is encoded as a complete, validating DIGGS instance file delivered with the report. The detailed authoring of the schema itself, the XML-schema (XSD) elements of the rigid-inclusion module in the DIGGS 3.0 Extension, is the continuing work of the DIGGS development effort, which Chapter 6 describes and to which this report is the geotechnical input.

The scope is bounded to four internationally recognized design standards, ASIRI (2012, France), BS 8006-1 (2010, United Kingdom), EBGEO (2011, Germany) and FHWA GEC No. 13 (2017, United States), chosen to span the main design philosophies; to two representative specialty contractors, Menard (controlled modulus columns) and Keller / Hayward Baker (rigid inclusions); to the drilled displacement column as the primary technique, being the most widely used in current practice; and to the four project phases of pre-installation investigation, design, installation and post-installation verification. Within those bounds the report aims to be complete, carrying a real rigid-inclusion project from the ground model through the design and the as-built columns to their verification, and the standardized parameter set it develops is kept general enough to describe the other inclusion types as well.

The remainder of the report is organized as follows. Chapter 2 reviews the rigid-inclusion technology, its four design standards and the differences between them, its installation methods and contractors, and the United States state-of-the-practice survey, establishing both the physical system to be described and the fragmentation a standard must reconcile. Chapter 3 sets out the four-phase standardization methodology that distils, from those standards and contractors, a single method-agnostic and contractor-agnostic parameter set. Chapter 4 presents that set of essential data elements phase by phase and explains how each maps onto DIGGS, reusing an existing feature or requiring a new rigid-inclusion one. Chapter 5 applies the whole framework to a real controlled-modulus-column project, a substation on soft, aggressive ground improved with about 1,075 columns. Chapter 6 sets out the rigid-inclusion schema developed in the DIGGS 3.0 Extension and the complete DIGGS instance file that encodes the case study. Together they carry the argument from the problem stated here to a demonstrated, standardized and openly-defined data set for rigid inclusions.

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**CHAPTER 2**

# Methodology

*This chapter sets out the method by which the essential parameters of a rigid-inclusion project were identified, organized and prepared for standardization, so that they can be mapped onto the DIGGS schema. Where the literature review in Chapter 3 establishes what rigid inclusions are and documents the fragmentation across design standards and contractors that a standard must reconcile, this chapter describes the systematic process built to overcome that fragmentation: a four-phase framework, fed by a comprehensive literature review and structured stakeholder engagement, that distils from the competing standards and contractor practices a single, method-agnostic and contractor-agnostic parameter set, and then maps that set onto DIGGS. Throughout, the focus is the drilled displacement column, the dominant rigid-inclusion technique in current practice, while the method is kept general enough to describe the other inclusion types.*

## 2.1 A Phased Standardization Framework

A rigid-inclusion system is only safe when the design intent, the ground data, the as-built execution and the verification all refer to the same, unambiguous parameters. The standardization method is therefore organized around the natural life cycle of a rigid-inclusion project and divided into four phases: pre-installation (the geotechnical investigation), design, installation, and post-installation (acceptance and verification). Each phase produces a distinct dataset with a distinct provenance, the ground model, the design record, the as-built production record and the verification record, and each is treated as a separate but linked body of parameters. Figure 2-1 shows the framework end to end: the two inputs that feed it, the needs assessment that consolidates them, the four phases into which the parameters are organized, the standardized data dictionary they produce, and the mapping of that dictionary onto DIGGS.

Two principles govern the framework throughout. The first is that it is method-agnostic in design and contractor-agnostic in installation: the standardized set is deliberately the union of the parameters required by the competing standards and contractors, not any single method or firm, so that any legitimate design method or installation technique can be reported completely, together with the design method, the parameter provenance and the factor set that make a result reproducible and comparable. The second is that verification is anchored to the design: the acceptance limits published in the standards and codes define what must be measured after construction and how it is judged, so that the post-installation record closes the loop back to the design intent. Applying the framework yields a single standardized rigid-inclusion data dictionary, giving for each parameter its unit, its definition, its relationship to the rigid-inclusion system, its data source and its governing standard, which is then mapped onto DIGGS, reusing existing schema features where a home already exists and identifying the gaps that require new rigid-inclusion features.

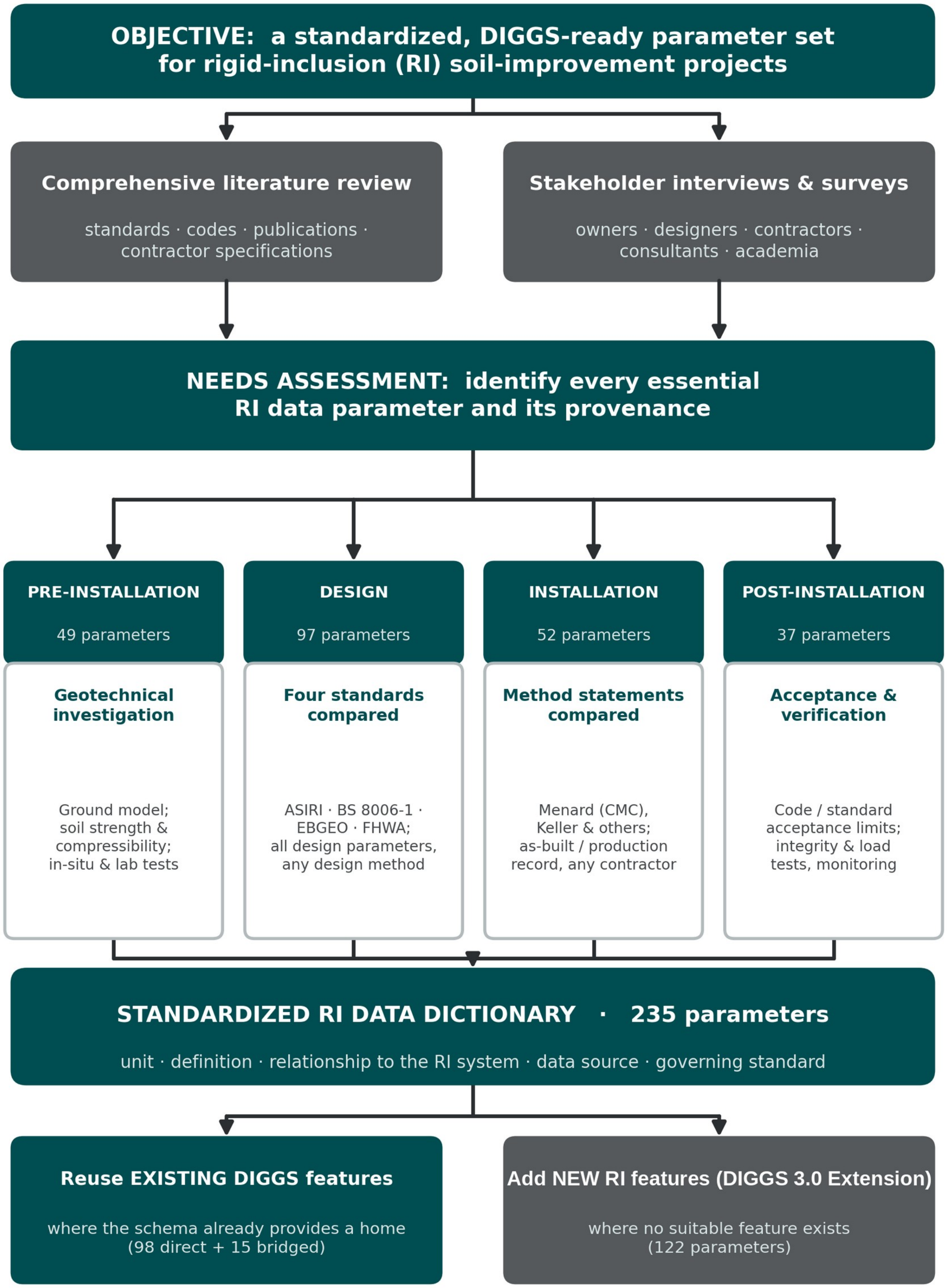


**Figure 2-1.** The four-phase methodology. A comprehensive literature review and structured stakeholder engagement feed a needs assessment that identifies every essential rigid-inclusion parameter and its provenance; the parameters are developed across the four project phases into a standardized data dictionary of 235 parameters, which is then mapped onto DIGGS, reusing existing features where the schema already provides a home and adding new rigid-inclusion features where it does not. *(original)*

## 2.2 Scope and Focus: the Drilled Displacement Column and the Elements of the System

Two scoping decisions shape the parameter set. The first concerns the installation technique. As the state-of-the-practice survey reviewed in Chapter 3 shows, the drilled displacement column is the most widely used rigid-inclusion system in United States practice, ahead of every other technique, and the two representative contractors examined, Menard and Keller / Hayward Baker, both install by drilled displacement. The drilled displacement column is therefore adopted as the focus of the standardized set, and the installation-phase parameters are drawn primarily from the depth-indexed production record that a displacement rig generates. The framework nonetheless remains general: the installation technique and the tool are carried as explicit coded attributes, so the other inclusion types the survey identifies can be described within the same structure without redesigning it.

The second decision concerns what, within a rigid-inclusion project, the parameters must describe. The standardized set is built around the physical elements of the rigid-inclusion system identified in Chapter 3: the inclusion column body, the shaft and its neutral plane, the toe or base, the head or cap and its transfer zone, the load-transfer platform, the geosynthetic reinforcement, the soft soil between inclusions, and the inclusion spacing that fixes the global geometry. Each element is a locus of parameters, geometric, material and performance, and organizing the data dictionary around these physical elements, rather than around any one standard's calculation, is what lets the same record serve every design method and every contractor and, later, map cleanly onto an object-based schema.

## 2.3 The Two Inputs to the Needs Assessment

The needs assessment, which identifies every essential rigid-inclusion parameter and its provenance, draws on two inputs that complement each other: the codified knowledge in the literature and the working knowledge held by practitioners.

### 2.3.1 Comprehensive literature review

The first input is a comprehensive review of the design standards, codes of practice, published research and contractor specifications that govern rigid-inclusion systems, assembled to serve two goals: to gather the full population of parameters used anywhere in current practice, and to expose the terminological and methodological differences a standard must reconcile.

The terminology itself is a barrier. The same system is known by many names across countries and eras, as vertical or columnar inclusions, as composite ground and rigid-column composite ground, as an unconnected, disconnected or non-connected piled raft, and simply as columns or, when it carries fill, as a column-supported embankment; commercially it appears as the controlled modulus column and, classified by installation technique, as the drilled displacement column, the auger cast-in-place or continuous-flight-auger column, the vibro concrete column and the driven or precast inclusion. This fragmentation of names is precisely why a controlled vocabulary, not merely a set of fields, is required. On the substance, four design standards are carried through the study as the primary basis for the design-phase parameters, ASIRI, BS 8006-1, EBGEO and FHWA GEC No. 13, supplemented by the independent state-of-the-practice synthesis of Han et al. (2025), which also evaluates the Dutch CUR226 method, by the two-part United States industry commentary of the DFI Ground Improvement Committee's Rigid Inclusion Task Force on the selection, design, construction and verification of rigid inclusions (Smith et al. 2023a, 2023b), and by the contractor literature of Menard and Keller / Hayward Baker for the installation phase. As Chapter 3 documents, the standards agree on the mechanisms but differ in the parameters they require, the symbols they use and the limits they set, and that difference is the raw material the design-phase comparison works from.

### 2.3.2 Stakeholder engagement: interviews and surveys

Literature captures codified practice but not the working knowledge held by the people who commission, design and build rigid inclusions. The second input is therefore structured engagement with stakeholders, owners, designers, contractors, consultants and academics, through interviews and surveys, to learn which parameters practitioners actually record and need, to surface proprietary practices that never reach the published literature, and to validate the emerging data dictionary against real project files. There is direct precedent: Han et al. (2025) surveyed owners, engineers and researchers across the United States and summarized the special provisions of four state departments of transportation to identify gaps in current practice. That survey is used here in a specific and bounded way, to enumerate the range of options a standardized vocabulary must accommodate, the inclusion types in use, whether they are reinforced, the design methods adopted and the quality-control methods applied, as set out in Chapter 3; the present project complements it by interviewing owners, designers and contractors and reviewing representative project files, including a complete controlled-modulus-column case study, as part of its needs assessment. Several such interviews were conducted for this study; as reported in Chapter 3, the working knowledge they captured proved largely consistent with the published survey while adding practical refinements the literature omits, among them the treatment of the support area, the load-transfer platform and the working platform. Two inputs from specialty-contractor practice were especially useful in structuring the needs assessment: a complete controlled-modulus-column project, drawn on for the case study, and a structured breakdown of the activities that constitute a rigid-inclusion project, contributed by a practitioner at Menard, which helped organize the parameter set into the four life-cycle phases used throughout this report.

## 2.4 Distilling the Essential Parameters, Phase by Phase

With the inputs assembled, the essential parameters are distilled phase by phase. Each phase applies the same logic, take the union of what the sources require, name each parameter unambiguously, and record its provenance, but each draws on a different body of source material, summarized in Chapter 4.

### 2.4.1 Pre-installation: the geotechnical investigation

The pre-installation phase captures the ground model against which the system is designed and its performance later judged. Its parameters, the stratigraphy and groundwater regime, the soft-soil strength and compressibility, and the in-situ and laboratory tests that supply them, are largely those of a conventional geotechnical investigation, and they are the least contested of the four phases, because site-characterization data are already well described in practice and, as Section 2.6 shows, already well supported by the DIGGS schema. The one rigid-inclusion-specific requirement is that each soil parameter carries its provenance, because the design standards draw on different tests and correlations for the same nominal property.

### 2.4.2 Design: the four standards compared

The design phase is where the standards diverge most and where standardization delivers the most value. The four standards were compared parameter by parameter with two aims: to expose the discrepancies between them, developed in Chapter 3, and to ensure the standardized set covers every parameter any of the four requires, whichever a designer adopts. Because the standards disagree, the set cannot follow any single method's list; instead it takes the union of their requirements, organized into thematic groups spanning the design basis and method, the geometry and layout, the inclusion's material and structural design, the shaft and neutral plane, the toe and base, the load-transfer platform, the geosynthetic reinforcement, the soft soil and loading, the load-transfer verifications, and serviceability and acceptance. A cross-standard requirement matrix records, for each parameter, which standards require it, making the common core and the method-specific parameters visible at a glance, and every parameter is stored with the design method, the parameter provenance and the factor set so that a design is reproducible and can be compared with one performed to a different standard. CUR226 was reviewed but not carried as a fifth basis, because the four retained standards were found to cover the essential parameters.

### 2.4.3 Installation: the contractor records compared

The installation phase captures what was actually built, column by column. The method statements, data-acquisition outputs and per-column records of the two representative drilled-displacement contractors were compared to identify the essential installation parameters independent of contractor. As Chapter 3 shows, Menard and Keller / Hayward Baker record essentially the same physical quantities, depth and penetration and withdrawal rate, torque and crowd, grout or concrete pressure and volume, stroke count and the termination or anchorage criterion, but under different proprietary field names, units and layouts, and each derives its own signature quality metric from the placed-versus-theoretical volume. The standardized installation record therefore captures the union of these depth-indexed channels and maps them to a single neutral vocabulary; it adds the installation technique and tool as coded attributes and the installation effects and their monitoring, so that an as-built inclusion can be reported identically whoever installs it and by whatever displacement tool.

### 2.4.4 Post-installation: acceptance ranges and verification

The post-installation phase asks whether the constructed system meets the design intent, and its parameters come from the acceptance ranges the design standards and codes publish. Verification is organized into the categories reviewed in Chapter 3: the as-built survey of position and verticality against tolerance; integrity testing of the columns; material-strength testing of cylinders or cores; load testing, static maintained-load, plate and zone tests, taken in United States practice to prescribed multiples of the design load; load-transfer-platform verification of thickness, compaction and fill strength; instrumentation and monitoring of settlement, differential settlement, pore-water pressure, load on heads and soil, reinforcement strain and lateral movement; and the acceptance and non-conformance records that document the outcome. Because these limits are fixed at the design stage, the post-installation parameters close the loop back to the design record, and the standardized set stores each acceptance criterion alongside the value it judges.

## 2.5 The Standardized Rigid-Inclusion Data Dictionary

Applying the framework across the four phases yields the standardized rigid-inclusion data dictionary, the central product of the method and the input to the schema work. It comprises 235 parameters, 49 for pre-installation, 97 for design, 52 for installation and 37 for post-installation, and for each it records the name, the symbol, the unit, the definition, the relationship to the rigid-inclusion system, the data source and the governing standard. The dictionary is deliberately method-agnostic and contractor-agnostic: it is the union of what the four standards and the two contractors require, so that any rigid-inclusion project, designed to any of the standards and built by any displacement contractor, can be described in full within one structure. It is this dictionary, not any single standard, that is carried forward to the schema.

## 2.6 Mapping the Parameters onto DIGGS

The final step of the method maps the standardized dictionary onto the DIGGS schema, in a deliberate order of preference intended to keep the extension minimal and backward-compatible. Each parameter is tested first against the features DIGGS already defines: much of a rigid-inclusion dataset, the borings and in-situ and laboratory tests, the fresh-material tests, the monitoring records and the project documents, reuses features the schema already provides, and where an existing feature carries the parameter directly it is used unchanged. Where the schema has a closely related feature that can host the parameter with a minor, backward-compatible addition, the parameter is bridged onto it rather than given a wholly new home. Only where no suitable feature exists is a new rigid-inclusion feature defined. Applied to the 235 parameters, this order places 98 on existing

DIGGS features directly, bridges a further 15 onto related features, and identifies 122 that require new rigid-inclusion features, concentrated, as Chapter 4 shows, in the design and installation phases that are specific to the technology.

A guiding decision underlies the mapping: DIGGS is extended to describe the rigid-inclusion system, not to re-implement the design calculations. The examination of completed DIGGS instance files for other applications shows that the schema models the physical element, the activity that creates or tests it, the resulting record, and the associated tests and materials, and that it carries design intent as a lean specification rather than reproducing each method's internal computation. The rigid-inclusion extension follows the same pattern: it gives the physical elements of the system, the inclusion and its head, the load-transfer platform, the reinforcement and the treated ground, together with their construction activity, their production record and their verification tests, a home in the schema, and attaches the design and acceptance parameters to those objects as attributes. The design-method internals, the arching formulae, the membrane models and the partial-factor arithmetic that differ from standard to standard, are not encoded in the schema; what is standardized and carried is the parameter each method produces, with its provenance and governing standard, so that the record stays method-agnostic and the schema stays compact.

## 2.7 Validation Through a Case Study

The method is validated against a complete rigid-inclusion case study, a controlled-modulus-column project for which the geotechnical report, the design, the as-built installation records and the verification testing were all available. Populating the standardized dictionary from the project confirmed that the four-phase set can capture a real project end to end, from the ground model through the design to every installed column and its acceptance testing, and that the resulting record can feed a DIGGS instance file. The case study is presented in full in a later chapter; here it serves as the check that the parameter set is complete and workable, not merely comprehensive on paper.

## 2.8 Summary

The methodology is a four-phase standardization framework, fed by a comprehensive literature review and structured stakeholder engagement and consolidated in a needs assessment, that distils from four design standards and two drilled-displacement contractors a single, method-agnostic and contractor-agnostic set of 235 rigid-inclusion parameters, each with its unit, definition, relationship to the system, provenance and governing standard. Mapping that set onto DIGGS, in the order existing feature, bridged feature, then new feature, reuses the schema wherever it already provides a home and confines the extension to the rigid-inclusion-specific objects, describing the system rather than the calculations. The chapters that follow apply this framework phase by phase, developing the parameters of the pre-installation, design, installation and post-installation phases and their mapping onto DIGGS, and demonstrating the result on the case study.

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## CHAPTER 3

# Literature Review

*Chapter 2 set out the method of this study: a four-phase framework, fed by a comprehensive review of the standards and literature and by a state-of-the-practice survey and structured practitioner interviews, that distils a single, method-agnostic and contractor-agnostic set of rigid-inclusion parameters. This chapter is that review. It first states the scope and the sources of information, then follows the natural life cycle of a rigid-inclusion project through five parts, the geotechnical investigation of the soft soil, the rigid-inclusion system itself, its design, its construction, and its post-construction verification, so that the reader meets the parameters of each phase in the order a project generates them. Throughout, the review has two aims: to establish the physical system that must be described, and to expose the fragmentation, across four design standards and across specialty contractors, that a standardized data set must reconcile.*

### 3.1 Scope, Sources of Information and the State of the Practice

As explained in the methodology of Chapter 2, the essential parameters of a rigid-inclusion project were identified from two complementary bodies of knowledge: the codified knowledge in the design standards, codes of practice and published research, and the working knowledge held by the practitioners who commission, design and build these systems. Four design standards are carried through the study as the primary basis for the design phase, the French ASIRI recommendations (2012), the British Standard BS 8006-1 (2010), the German EBGEO recommendations (2011) and the United States FHWA Geotechnical Engineering Circular No. 13 (2017), supplemented by the recent United States state-of-the-practice synthesis of Han et al. (2025), the two-part industry commentary of the DFI Ground Improvement Committee's Rigid Inclusion Task Force (Smith et al. 2023a, 2023b), and the contractor literature of Menard and Keller / Hayward Baker for the installation phase. This section introduces the state-of-the-practice survey and the practitioner interviews first, because their findings recur throughout the review and fix the focus of the study.

To establish the current state of the practice, the project team combined the literature review with a series of interviews and informal consultations with rigid-inclusion contractors, designers and owners, and drew on the most recent and comprehensive published survey of United States practice, the Phase-I transportation study led by Jie Han and colleagues (Han et al. 2025). That survey comprised twenty-five multiple-choice questions answered by thirty-six respondents (from sixty-seven invitations) drawn from government agencies (about twenty-eight percent), contractors (twenty-five percent), consultants (nineteen percent) and academia (nineteen percent), the great majority reporting expertise in rigid-inclusion design, installation, quality control or performance evaluation. Most of the responses obtained in the interviews were consistent with the published survey, which lends confidence to its findings and allows this review to draw on it directly; the interviews were likewise consistent with the DFI two-part commentary (Smith et al. 2023a, 2023b), several of whose authors are drawn from the same specialty contractors consulted here. Figure 3-1 shows the survey's headline result on why rigid inclusions are used: overwhelmingly to reduce settlement (about ninety-four percent of respondents), followed by increasing bearing capacity, enhancing slope stability and mitigating liquefaction.

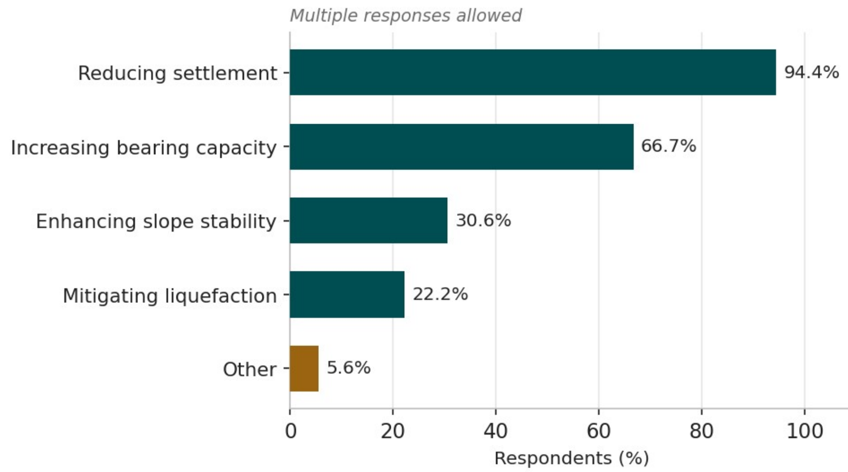



**Figure 3-1.** Ground-improvement objectives for which rigid inclusions are used in United States transportation practice; reducing settlement dominates. Respondents could select more than one objective. *redrawn from Han et al. (2025), Figure 3.5.*

Beyond corroborating the published survey, the interviews conducted for this study drew out three practical refinements that the standardized record must accommodate, each raised independently by experienced designers and installers. The support area, the plan extent actually treated with inclusions, was described as a quantity to be defined in its own right, because it is commonly made wider than the footprint of the structure or embankment it supports. The load-transfer platform was characterised as a conditional and variable element rather than a fixed granular cushion: it is provided where a shallow bearing capacity must be developed between the heads and may, in rare cases, be dispensed with, while a demand for higher shallow bearing capacity is met by reinforcing the fill with geogrid, by mixing or cementing the platform soil, or, in the extreme, by a structural concrete slab. And the working platform, the compacted mat built to carry the installation plant, was noted to be a near-universal precursor to construction that ordinarily survives as the first layer of the load-transfer platform, so that its material, thickness and level are themselves part of the as-built record. These points are reflected in the system description of Section 3.3 and in the parameters carried forward into the standardized set.

Each of the survey's questions enumerates the options available for one attribute of a rigid-inclusion project. Table 3-1 collects those options, the choices offered to respondents, for the attributes most relevant to design, construction and quality control. The point of the table is not how respondents answered but how many distinct, named alternatives exist for every attribute: it is this breadth, confirmed in the interviews, that a standardized vocabulary must capture without ambiguity. The specific survey results that bear on each phase are presented in the relevant section below.

**Table 3-1.** Attributes of a rigid-inclusion project and the options enumerated in the state-of-the-practice survey (options only; response frequencies are given by phase in the sections that follow). After Han et al. (2025), Chapter 3.

| Project attribute (survey question) | Options presented to respondents |
|---|---|
| **Inclusion type / technique** | Drilled displacement columns (DDC); auger cast-in-place (ACIP); vibro-displacement columns; vibro-compaction concrete columns; grouted gravel/stone columns; timber piles; steel piles; other (e.g. precast concrete, soil-mix) |
| **Application** | Embankments; retaining / sound-barrier walls; box culverts / drainage pipes; bridge foundations; building foundations |
| **Ground-improvement objective** | Reduce settlement; improve bearing capacity; enhance slope stability; mitigate liquefaction |
| **Soil type improved** | Clay; organic soil; silt; peat; loose sand or gravel; other (e.g. municipal solid waste, unsatisfactory fill) |
| **Design approach** | Analytical method; numerical analysis; personal experience; other (e.g. load testing) |
| **Analytical design method** | FHWA beam; FHWA strain-compatibility; BS 8006-1; EBGEO; CUR226; Nordic; other |
| **Inclusion diameter** | Less than 14 in; 14–18 in; 19–24 in; greater than 24 in |

| Project attribute (survey question) | Options presented to respondents |
|---|---|
| **Inclusion spacing** | Less than 6 ft; 6–9 ft; 10–12 ft; greater than 12 ft |
| **Area replacement ratio** | Less than 2.5%; 2.5–5.0%; 6–10%; 11–20%; greater than 20% |
| **Cap on inclusion head** | No cap; a range of percentage coverage up to and beyond 30% |
| **Steel reinforcement** | Never; rarely; sometimes; often; always |
| **Condition triggering reinforcement** | Horizontal loads (slope stability, seismic); eccentric / lateral loading; others |
| **LTP fill material** | Gravel / graded aggregate base; sand; other (cement-treated soil, lean concrete) |
| **Quality / performance method** | Static load test on individual inclusion; pile integrity testing; thermal integrity profiling; composite / group load test; test on surrounding soil; other |
| **Instrumentation & monitoring** | Always; often; sometimes; rarely; never |
| **Downdrag treatment** | Neutral-plane method; downdrag in limit-strength design; by experience; not considered; other |
| **Contract method** | Design-build (DB); design-bid-build (DBB); other |
| **Governing specification** | Owner or agency provisions; contractor specifications; none; other |

## 3.2 Geotechnical Investigation of the Soft Soil

Rigid inclusions exist to solve one recurring problem: building on soft, compressible ground with controlled settlement. Every design therefore begins from a geotechnical investigation that characterises the ground the inclusions must reinforce and against which their performance is later judged. The soils that call for the technique are the weakest and most compressible in the ground-improvement repertoire. In the state-of-the-practice survey, rigid inclusions were reported to be used most often in clay (about ninety-seven percent of respondents), and frequently in organic soils, silts, peats and loose sands and gravels, as shown in Figure 3-2 (Han et al. 2025); the documented case studies reviewed later in this chapter include peats and organic clays with water contents above three hundred percent and undrained shear strengths of only a few kilopascals. These are grounds on which a conventional shallow foundation would settle excessively and on which staged preloading would be slow.

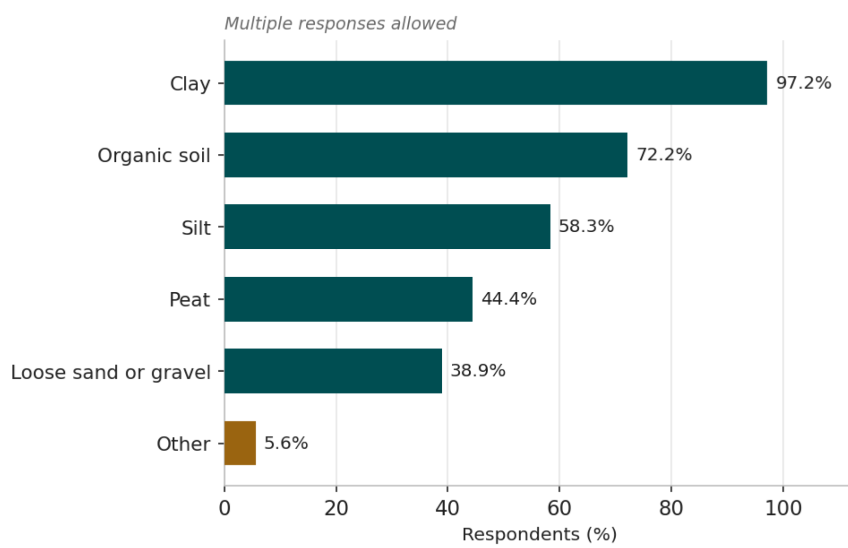



**Figure 3-2.** Soil types improved with rigid inclusions in United States transportation practice; soft clays dominate, with organic soils, silts, peats and loose granular soils also common. Respondents could select more than one soil type. *redrawn from Han et al. (2025), Figure 3.6.*

The investigation must therefore quantify two families of soil behaviour, strength and compressibility, together with the groundwater regime and, where relevant, the seismic and durability conditions. Strength is characterised by the undrained shear strength ($c_u$) of fine-grained soils and by the effective-stress parameters ($c'$, $\varphi'$) of the soils and fills, obtained from the standard penetration test (N), the cone penetration test ($q_c$, $f_s$, $u_2$), the field vane, the

Ménard pressuremeter ($E_M$, $p_l$) and laboratory triaxial and shear-box tests. Compressibility, which governs the settlement the inclusions are installed to control, is characterised by the compression and recompression indices ($C_c$, $C_s$), the preconsolidation pressure ($\sigma'_p$) and the oedometric (constrained) modulus ($E_{oed}$), together with the coefficient of consolidation ($c_v$) that sets the time-rate of settlement, all obtained from the oedometer. The stiffness of the bearing stratum into which the inclusions are founded, and the small-strain stiffness ($V_s$, $G_{max}$) where seismic performance matters, complete the ground model.

A single rigid-inclusion-specific requirement runs through this otherwise conventional investigation: each soil parameter must carry its provenance, because the four design standards draw on different tests and correlations for the same nominal property. ASIRI is built around the Ménard pressuremeter and derives the soil-inclusion load-transfer laws and the tip resistance from the pressuremeter modulus and limit pressure; BS 8006-1, EBGEO and FHWA rely more on the standard penetration test, the cone penetration test and laboratory oedometer and triaxial data. The same nominal value, the undrained strength or the constrained modulus, therefore carries a different pedigree and a different valid correlation depending on the standard that will use it, which is why the standardized ground model records not only the value but the test and correlation that produced it. The full set of pre-installation parameters, with their symbols, units and provenance, is developed in Chapter 4.

## 3.3 Rigid Inclusions: Components, Mechanism, Types and Applications

A rigid inclusion is a slender, usually unreinforced, cementitious column that is far stiffer than the soil around it, installed in a regular grid through soft or compressible ground and taken down to a more competent bearing stratum. The defining feature that distinguishes a rigid inclusion from a pile is that the inclusion is not structurally connected to the structure it supports; instead a granular load-transfer platform (LTP), or cushion, is placed over the inclusion heads, and it is within this platform, through soil arching and, where provided, geosynthetic reinforcement, that load is shared between the stiff columns and the intervening soft soil. The improved volume therefore behaves as a stiffer composite ground rather than as a group of piles, and it is designed by treating the columns and soil together (Han 2021; ASIRI 2012).

Figure 3-3 identifies the components of the rigid-inclusion system that recur throughout this report and that the standardized data set must describe. They are: (1) the rigid inclusion column body carried through the soft ground to the stiff substratum; (2) the shaft and the neutral plane that separates negative skin friction above from positive skin friction below; (3) the toe or base bearing in the competent stratum; (4) the head or cap and its head-transfer zone; (5) the load-transfer platform, in which arching develops over a critical height; (6) the geosynthetic reinforcement, where provided; (7) the soft soil between inclusions; and (8) the inclusion spacing that fixes the global system geometry. Each of these is a locus of parameters, geometric, material, and performance, that a complete record must carry.

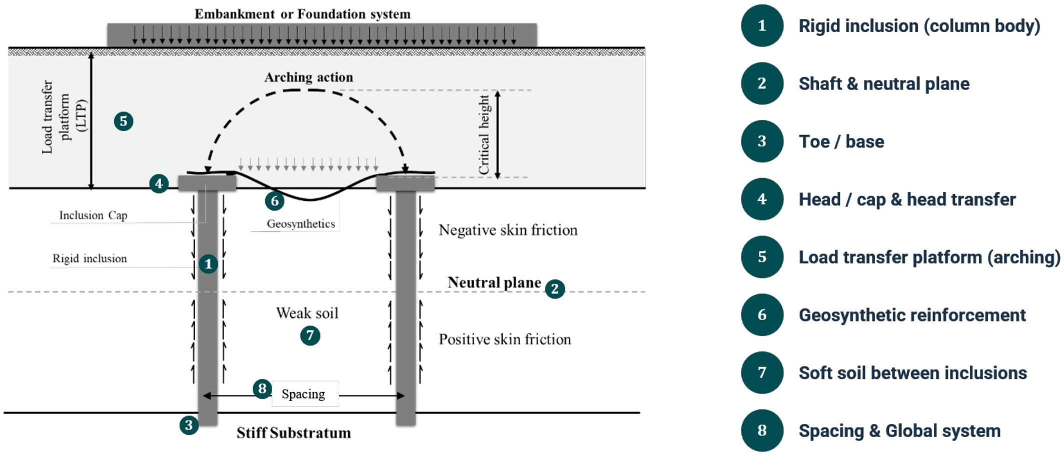



**Figure 3-3.** The rigid-inclusion system and its essential components: (1) the rigid inclusion column body carried through soft ground to a stiff substratum; (2) the shaft and the neutral plane that separates negative skin friction above from positive skin friction below; (3) the toe or base bearing in the competent stratum; (4) the head or cap and its head-transfer zone; (5) the load-transfer platform (LTP), in which arching develops over a critical height; (6) the geosynthetic reinforcement; (7) the soft soil between inclusions; and (8) the inclusion spacing that fixes the global system geometry. *(original)*

Two further features of the system, easily overlooked because they are not always drawn on a type section, must nonetheless be defined and recorded. The first is the support area, the plan extent over which the inclusions are actually installed; it is not required to coincide with the footprint of the structure or embankment above, and in practice it is frequently made wider, so that the improved block extends beyond the loaded area to carry edge loads, to spread load onto the soft soil and to safeguard stability. The second is the working platform, a compacted granular or cemented-soil layer placed over the ground before installation begins so that the heavy drilling and pumping plant can operate safely; because it is left in place, the working platform commonly becomes the lowest layer of the load-transfer platform, and its level fixes the datum from which every column depth is later measured. The load-transfer platform itself is less uniform than a single granular cushion suggests: it is provided wherever a shallow bearing capacity has to be developed between the inclusion heads, and where a higher capacity is needed it is strengthened by degrees, a granular fill reinforced with geogrid, a soil-mixed or cemented-soil raft, or, in the most demanding cases, a structural concrete slab. The platform is therefore better understood as a zone of defined plan extent, thickness and material, ranging from an unreinforced granular fill to a reinforced or cemented raft.

Rigid inclusions are used because they solve, economically and quickly, the recurring problem of building on soft ground. They reduce total settlement and, just as importantly, homogenize it so that differential settlement stays within tolerance; they increase the bearing capacity of the composite ground; they improve global and slope stability; and, in some conditions, they help mitigate liquefaction (the objectives of Figure 3-1). Compared with the alternatives, large-scale excavation and replacement, staged preloading with wick drains, or a full piled foundation with a structural cap, inclusions are frequently faster to install, generate little or no spoil when installed by displacement, and use less material because they exploit the strength of the soil rather than bypassing it (Han 2021; Han et al. 2025; ASIRI 2012).

### 3.3.1 Types of rigid inclusions

Rigid inclusions are classified in two complementary ways: by the material that forms the column, and, more importantly for both construction and data, by the technique used to install it.

**By material.** Most inclusions are cast-in-place concrete, grout or mortar columns. The family also includes soil-mix elements, in which a binder is mixed with the in-situ soil, and grouted stone columns, in which a granular column is bound with grout. The material fixes the column's stiffness, compressive strength and durability, and it determines which fresh-state quality tests apply (for example slump for concrete, or flow and density for a fluid grout).

**By installation technique.** The installation technique is the primary classifier because it governs spoil, disturbance and the quality-control data the rig produces. Displacement techniques laterally displace the soil rather than removing it, so they generate no spoil and tend to densify the surrounding ground; they include drilled displacement columns (DDC), of which the controlled-modulus-column (CMC) technique is the best-known example, screw- and auger-displacement tools, vibro concrete columns (VCC), and driven or precast inclusions. Replacement techniques remove soil as the column is formed; they include continuous-flight-auger (CFA) and auger-cast-in-place (ACIP) columns and conventional bored columns. The distinction matters for standardization because the two families produce different as-built records, a point developed in Section 3.5.

**Reinforced or unreinforced.** Rigid inclusions are usually unreinforced, because the composite-ground concept assumes the column carries essentially concentric axial load. Steel reinforcement is added only where the inclusion must carry bending or tension, typically the edge columns of an embankment, columns beneath slopes or retaining structures, and any inclusion subjected to significant horizontal or seismic loading, which can cause rigid inclusions to fail by bending or rotation (Han et al. 2025). Whether an inclusion is reinforced, over what length and under what condition are therefore genuine project variables that a standardized dataset must record unambiguously; the frequency of reinforcement in practice is examined with the design phase in Section 3.4.

### 3.3.2 Applications

Rigid inclusions are used wherever soft or compressible ground must carry new load with controlled settlement. For the purposes of standardization it is useful to organize this variety by the source of the load into three fundamental support scenarios, because that source governs the load model, whether a footing configuration or a full load-transfer platform applies, the governing design method and the settlement criterion (Figure 3-4): inclusions supporting isolated or strip footings, where a load-transfer platform is mandatory and a footing is never placed directly on an inclusion (ASIRI 2012); inclusions supporting a mat or raft foundation, a distributed structural load over a full grid; and inclusions supporting an embankment, the column-supported-embankment case, frequently with basal geosynthetic reinforcement. Across these scenarios the principal applications are embankments for road and rail, slab- and raft-type structures, tanks, silos and heavy storage facilities, footings for buildings and wind turbines, and the retaining walls, box culverts and approach structures of transportation projects (Han 2021; Han and Gabr 2002; Han et al. 2025).

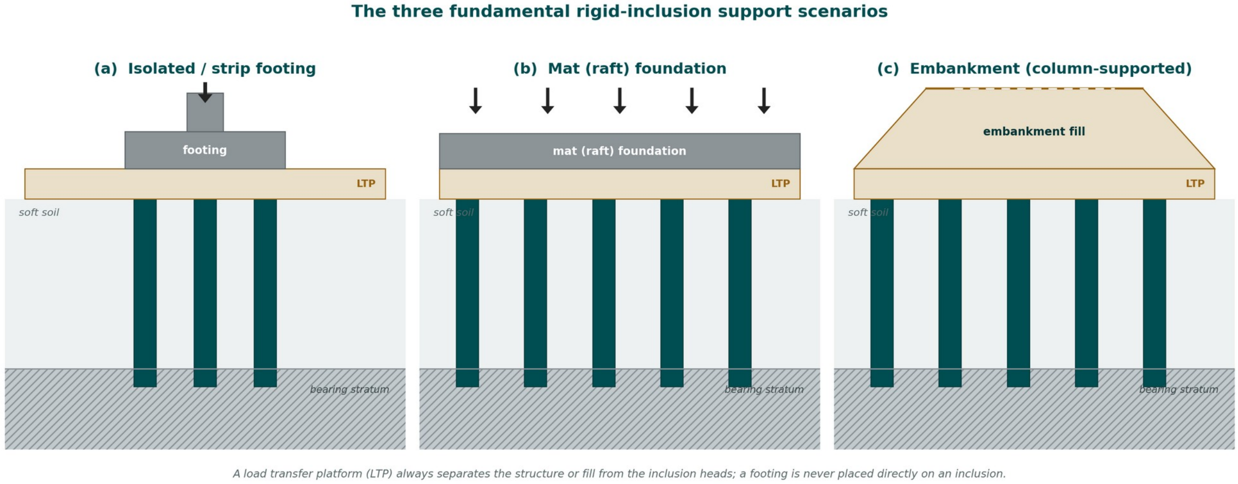



**Figure 3-4.** The three fundamental rigid-inclusion support scenarios that organize the standardized dataset: (a) isolated / strip footing, (b) mat (raft) foundation, and (c) column-supported embankment. *(original)*

### 3.3.3 Case Studies

To ground the review in real projects, the team reviewed a set of documented rigid-inclusion case studies from the engineering literature, chosen to span these application types and a range of ground conditions and design standards. Table 3-2 collects a representative sample: above-ground storage and liquefied-natural-gas tanks, highway embankments over extremely soft peat and organic clay, and a heavily-loaded industrial slab in a seismic region, designed variously to ASIRI, to Eurocode and to United States practice. Two observations bear directly on this report. First, the same compact family of parameters recurs in every case, the inclusion diameter and spacing, the length and area replacement ratio, the platform thickness and material, the column strength or modulus, and the total, differential and residual settlement, which is the set the following chapters standardize. Second, the cases report that family inconsistently, several omitting the diameter, the grid or the area replacement ratio and some giving only predicted rather than measured settlement, which is precisely the fragmentation a standardized, machine-readable record is meant to remove.

**Table 3-2.** Representative documented rigid-inclusion and controlled-modulus-column case studies reviewed for this report, spanning the main application types.

| Application (project, location) | Structure and ground | Rigid-inclusion solution and reported performance | Source |
|---|---|---|---|
| **Above-ground storage tanks — Quebec City and Texas City, USA** | Petroleum tanks up to 150 ft in diameter on soft clay and organic fill | 12.5 in CMC on a ~5 ft grid with a perimeter ring; 50–95% of the load to the columns; measured hydro-test settlement 1–1.4 in | Masse and Pearlman (2009) |
| **LNG tanks — Al-Zour, Kuwait** | Eight prestressed tanks, 96 m diameter × 45 m, 225,000 m³ each, on reclaimed marine sand; strong seismic (2,475-year return) | 9,024 unreinforced CFA columns, Ø0.8 m, 2.8/2.4 m dual mesh, 2 m gravel platform; ASIRI; differential ≤ 1/300 | Bernuy et al. (2018) |
| **Highway embankments and box bridges — Highway 2000, Jamaica** | 2.5–12 m embankments over peat and organic clay more than 20 m deep (water content > 300%) | CMC on 1.4–2.0 m grids, σ ≤ 5 MPa, no tension; residual settlement ≤ 40–80 mm over 35 years; liquefaction FoS 0.42 → 1.03 | Plomteux and Lacazedieu (Menard) |
| **Heavy industrial slab-on-grade — glass factory, Calaraşi, Romania** | 21,500 m² slab, live load 80 kN/m², seismic (peak ground acceleration 0.26 g); loess over alluvial clay | Ø360 mm CMC on a 2.25 m grid, about 17 m long; grout modulus 5,000 MPa; differential < 1/500; unreinforced, seismic-verified | Plomteux and Liausu (Menard) |
| **Highway embankment, hybrid scheme — B5 near Husum, Germany** | 7 m embankment on soft fat clay and peat (undrained strength 6–20 kPa) beside a live highway | CMC (500 kN per column) in the softest zone with wick drains elsewhere; steel-grid reinforced-earth platform; settlement in the CMC zone < 2 cm | Kirstein and Wittorf (2013) |
| **Airport apron — Terminal 3, Soekarno-Hatta International Airport, Jakarta, Indonesia (2011–2013)** | New aircraft apron, Phase 1 65,174 m² and Phase 2 346,075 m²; loose silty sand and silty clay with marginal layers at 0–3 m and 6–9 m, water table 2–3 m; 7 t/m² Boeing 747 loading | Two-grid Ø0.32 m CMC (long primary and short secondary columns) with a 0.6 m compacted-sand load-transfer platform; met CBR ≥ 6% and 60 mm allowable settlement while the airport stayed operational | Menard (2022) |
| **Automated container terminal — Webb Dock East, Port of Melbourne, Australia (2014–2015)** | Container-stacking yard and automated stacking-crane rails on reclaimed ground; container 178 kN per corner casting, crane rail 172 kN/m | 3,663 CMC, Ø0.36 m, average 16 m deep (58,608 lm) over 22,000 m²; settlement limited to 100 mm vertical and 1/500 differential; scope completed in 67 days | Menard (2022) |
| **Refinery tank farm — Nghi Son Refinery and Petrochemical Complex, Vietnam (2014)** | Storage tanks 24–69 m diameter, tank loads 139–350 kPa | CMC over 49,500 m² (about 225,000 lm executed) to limit tank settlement and differential settlement | Menard (2022) |
| **Petrochemical storage tank — PT Chandra Asri Petrochemicals, Indonesia (2014)** | Naphtha storage tank, 80 m diameter, tank load 224 kPa | CMC over a 5,542 m² treated area to control tank settlement | Menard (2022) |
| **Wind-turbine foundations — Phuong Mai 3 Wind Farm, Vietnam (2019)** | Six gravity turbine foundations, 21–26 m diameter; SLS bearing capacity greater than 200 kPa | CMC (about 1,600 lm) beneath the foundations to provide bearing capacity and control settlement | Menard (2022) |

| Application (project, location) | Structure and ground | Rigid-inclusion solution and reported performance | Source |
|---|---|---|---|
| **Hospital spread footings — Fort Belvoir Community Hospital, Virginia, USA (2008)** | Spread footings carrying 335–383 kPa uniform load | 4,900 CMC to a maximum depth of 9 m, installed in 29 weeks | Menard (2022) |

Beyond the North American record, the controlled modulus column technique has been documented on large international infrastructure projects that broaden the range of applications and performance criteria represented in the dataset. At Terminal 3 of Soekarno-Hatta International Airport in Jakarta, a new aircraft apron of more than 0.4 million square metres was supported on a two-grid arrangement of 0.32 m controlled modulus columns, a primary grid of long columns and a secondary grid of short columns, beneath a 0.6 m compacted-sand load-transfer platform (Figure 3-5). The scheme was required to meet an allowable settlement of 60 mm and a California-bearing-ratio target of 6 per cent under a 7 t/m$^2$ Boeing 747 wheel loading, and the columns were installed from the working platform while the airport remained in operation (Figure 3-6; Menard 2022).

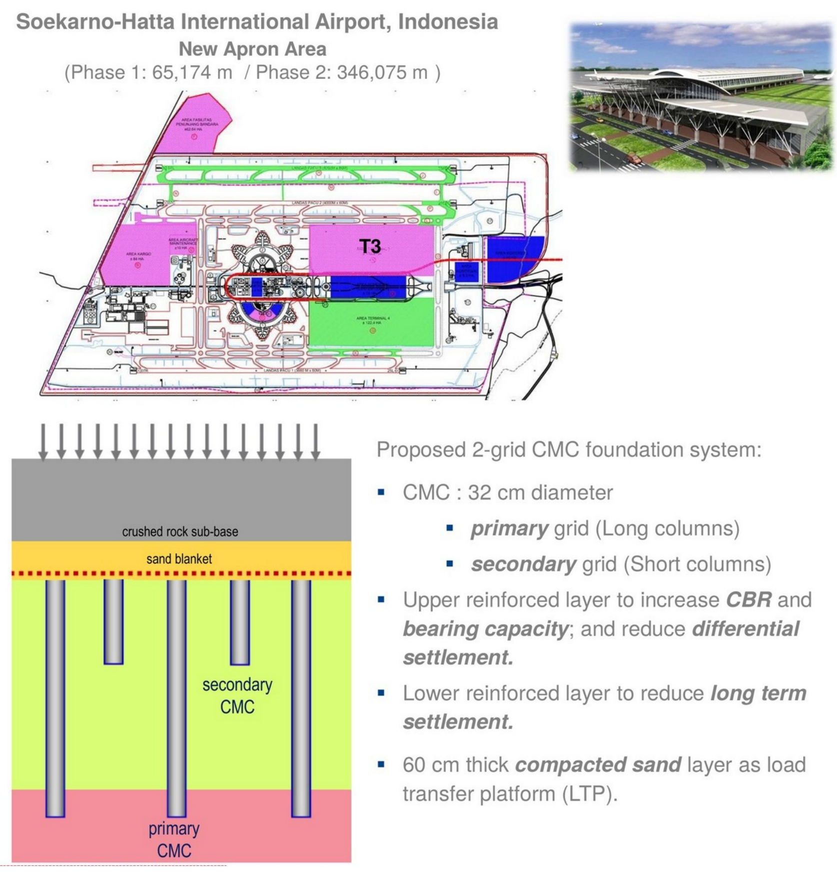



**Figure 3-5.** Terminal 3, Soekarno-Hatta International Airport, Jakarta: project layout and new apron area (top) and the two-grid controlled-modulus-column scheme with primary and secondary columns beneath a compacted-sand load-transfer platform (bottom). *(Menard 2022)*


**Figure 3-6.** Controlled-modulus-column installation for the Terminal 3 apron, carried out within the operating airport during Phase 1 (2011) and Phase 2 (2013). *(Menard 2022)*

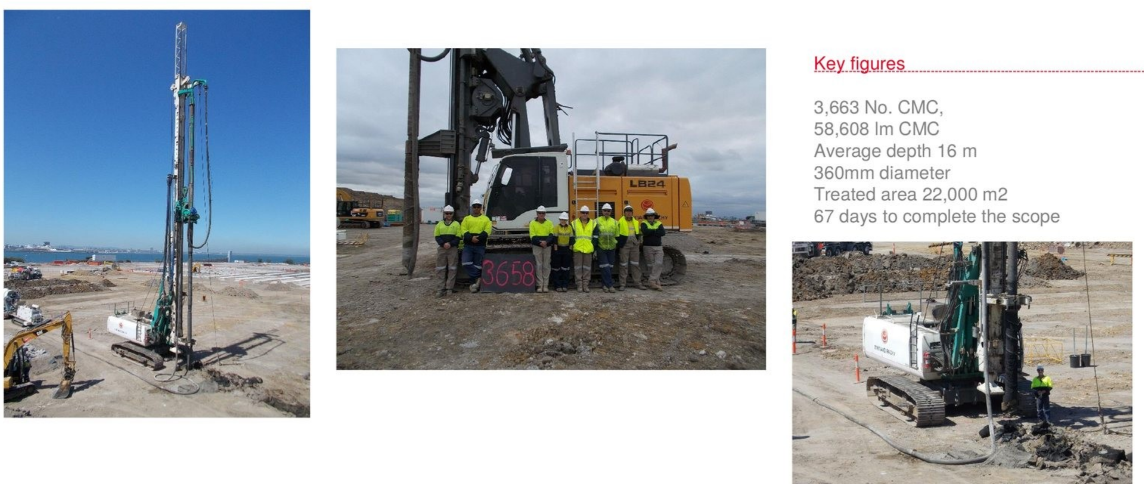
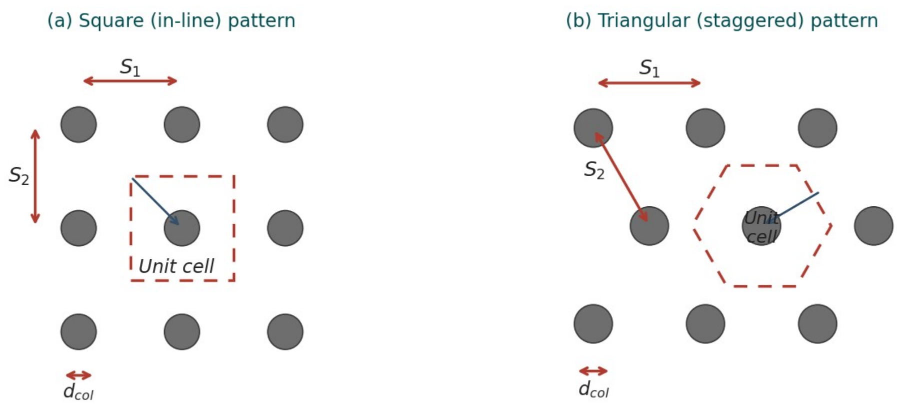



**Figure 3-7.** Webb Dock East, Port of Melbourne: controlled-modulus-column rigs and the reported key figures for the container-terminal ground improvement. *(Menard 2022)*

### 3.3.4 Load-transfer mechanics

The behaviour of a rigid-inclusion system is governed by how the applied load divides between the stiff inclusions and the soft soil, and by how that division changes with depth. For a regular grid, analysis is usually reduced to a unit cell: one inclusion and the tributary area of soil it supports, as shown in Figure 3-8. The single most important geometric parameter is the area replacement ratio ($\alpha$, also written $a_s$), the ratio of the inclusion (or cap) cross-section ($A_c$) to the tributary area ($A$). For a circular column of diameter ($d_{col}$) on a grid of centre-to-centre spacings ($s_x$) and ($s_y$), the tributary area is $A = s_x\ s_y$ (reducing to $s^2$ on a square grid), and the area replacement ratio follows from Equation 3-1. This geometric definition is common to all four standards; what differs, as the following sections show, is the arching model each uses to split the load.




$$S' = \frac{\sqrt{S_1^2 + S_2^2}}{2} - \frac{d_{col}}{2}$$

$S_1$, $S_2$: centre-to-centre spacing between adjacent inclusions. $d_{col}$: inclusion diameter. $S'$: half clear diagonal span.

**Figure 3-8.** The unit cell and the area replacement ratio for the two common layouts: (a) a square (in-line) pattern and (b) a triangular (staggered) pattern, with the centre-to-centre spacings ($s_x$, $s_y$), the inclusion diameter ($d_{col}$) and the half clear diagonal span (S'). The area replacement ratio is the ratio of the inclusion cross-section to the tributary area of one unit cell. *(original; layout after ASIRI 2012 and FHWA GEC-13 2017).*

$$\alpha = \frac{A_c}{A} = \frac{\pi}{4}\left(\frac{a}{s}\right)^2 \tag{3-1}$$

*ASIRI (2012), §3, p. 22; FHWA GEC-13 (2017), p. 6-30.*

Because the column is much stiffer than the soil, it attracts a disproportionate share of stress. The ratio of the vertical stress on the column ($\sigma_c$) to that on the soil ($\sigma_s$) is the stress-concentration ratio (n) of Equation 3-2. The fraction of the total applied load that reaches the inclusions rather than the soft soil is the load efficacy (E), the single headline measure of how well the system is working; it follows directly from the area replacement ratio and the stress-concentration ratio (Equation 3-3). A higher area replacement ratio and a stiffer column raise the efficacy and cut the settlement, but they also raise the stress the column and its head must carry.

$$n = \frac{\sigma_c}{\sigma_s} \tag{3-2}$$

*General composite-ground definition; see Han et al. (2025) and FHWA GEC-13 (2017), Ch. 6.*

$$E = \frac{\alpha\,\sigma_c}{\alpha\,\sigma_c + (1-\alpha)\,\sigma_s} = \frac{\alpha\,n}{1+(n-1)\,\alpha} \tag{3-3}$$

*General composite-ground definition; see FHWA GEC-13 (2017), Ch. 6, and Han et al. (2025).*

The four design standards reviewed in Section 3.4 all rest on this same load-sharing idea, that the applied stress ($\sigma_{avg}$) divides between the inclusions and the soft soil (Equation 3-4), but they compute the split by different mechanisms, and it is this difference that makes them diverge. Above the heads, load is concentrated onto the inclusions by soil arching: as the soft soil settles more than the columns, shear stresses in the granular platform transfer part of the overburden and surcharge onto the stiffer heads, and the arch becomes fully effective once the platform is thicker than a critical height related to the clear spacing between heads. Where a basal geosynthetic is present, it carries the residual load as a tensioned membrane spanning between heads. Below the platform, the same relative movement governs the shaft: over the upper part of the inclusion the soil moves down relative to the column and drags it down through negative skin friction (downdrag), while below a neutral plane the column moves down relative to the soil and is supported by positive skin friction. The axial load therefore grows from the applied head load to a maximum at the neutral plane and is then shed by positive friction to the base, as shown in Figure 3-9.

$$\sigma_{avg} = (\gamma H + q) = a_s\,\sigma_{col} + (1 - a_s)\,\sigma_{soil} \tag{3-4}$$

*FHWA GEC-13 (2017), Eq. 6-11, p. 6-30.*

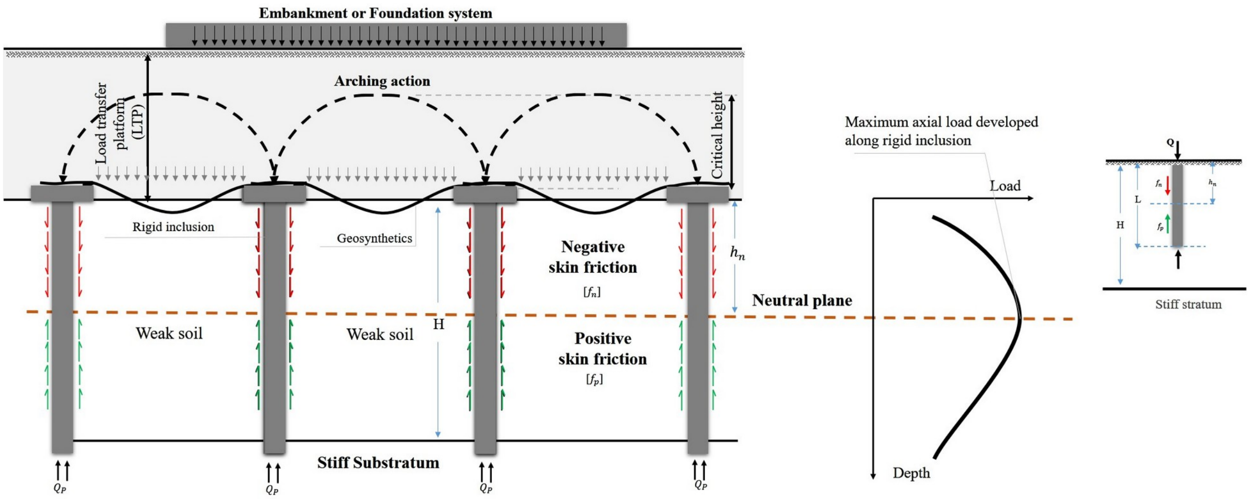



**Figure 3-9.** Load transfer in a rigid-inclusion system. Soil arching over the critical height concentrates the embankment or foundation load onto the inclusion heads through the load-transfer platform and its geosynthetic; below, negative (downdrag) skin friction ($f_n$) develops above the neutral plane and positive skin friction ($f_p$) below it, so that the axial load in the inclusion is greatest at the neutral plane before being shed to the base ($Q_P$). The inset shows the corresponding axial-load-versus-depth distribution. *provided by the project team.*

The combined effect of stress concentration, arching and shaft transfer is to route most of the load to the bearing stratum through the inclusions, cutting both the total and the differential settlement at the surface, the settlement efficacy that is the system's reason for being. Crucially, the load share and the settlement are sensitive to a small set of interacting parameters: the area replacement ratio (α), the platform thickness ($H_M$), the inclusion stiffness (its modulus E and characteristic strength $f_{ck}$), the grid spacing (s), and the tensile stiffness of any geosynthetic (J). A modest change in one of these can shift the load efficacy by several percent and change the computed settlement materially (ASIRI 2012; FHWA GEC-13 2017). It is this sensitivity that makes the accurate, unambiguous capture of each parameter, with its correct units and provenance, a determinant of whether the design and its verification are sound, and it is a principal reason the standards, which model arching and downdrag differently, diverge.

Survey practice on the geometry that these mechanics depend on confirms both a common core and a wide spread. The centre-to-centre spacing is most often between six and nine feet, and the area replacement ratio most often between 2.5 and five percent, with a substantial minority of respondents unable to state the ratio they use; column caps are frequently not used at all, and where they are, the percentage coverage varies widely (Figures 3-10, 3-11 and 3-12). This variability, together with the different ranges quoted by the standards themselves, is compared next.

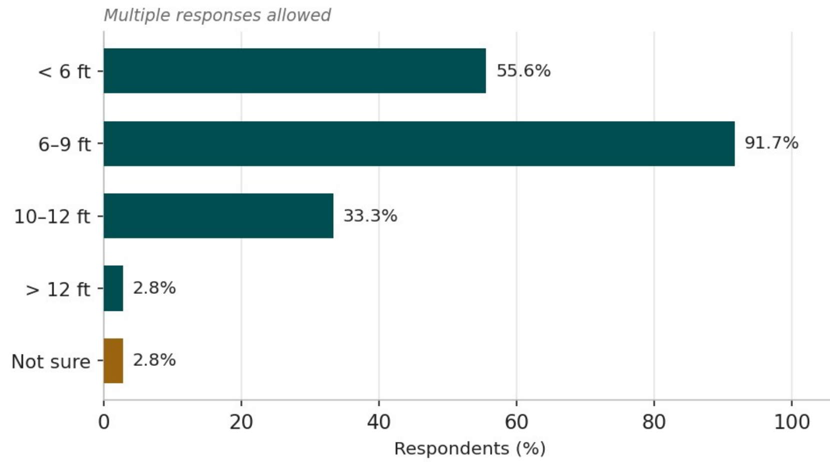



**Figure 3-10.** Commonly adopted centre-to-centre spacing of rigid inclusions in United States practice. Respondents could select more than one range. *redrawn from Han et al. (2025), Figure 3.10.*

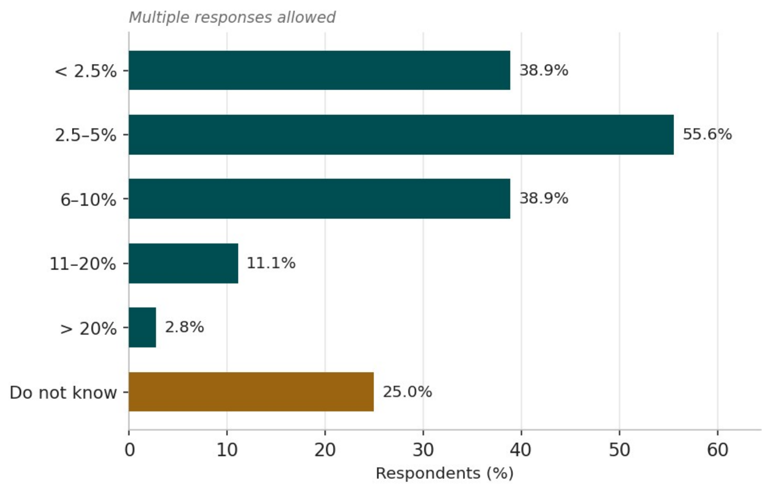



**Figure 3-11.** Commonly adopted area replacement ratios in United States practice; a quarter of respondents did not know the ratio used. Respondents could select more than one range. *redrawn from Han et al. (2025), Figure 3.11.*

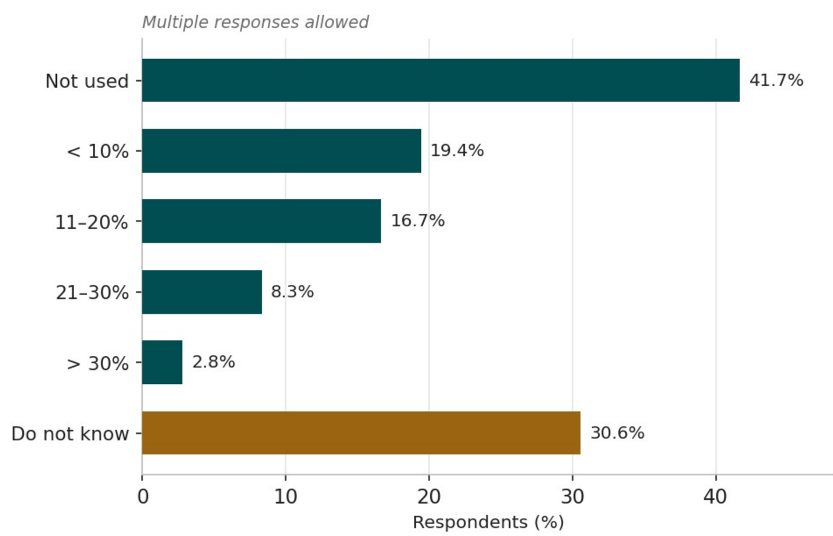



**Figure 3-12.** Commonly adopted percentage coverage of caps on top of rigid inclusions; caps are frequently not used. Respondents could select more than one range. *redrawn from Han et al. (2025), Figure 3.12.*

The area replacement ratio is defined the same way by all four standards but framed and bounded differently, and Table 3-3 sets the ranges side by side. FHWA gives the clearest guidance, an area replacement ratio between 3.5 and ten percent, occasionally as low as 2.5 percent, and adds that where caps are used the ratio must be computed on the cap area (FHWA GEC-13 2017, p. 6-11). ASIRI notes that the coverage ratio commonly ranges between two and ten percent and is an essential factor of reinforcement efficiency (ASIRI 2012, p. 22). BS 8006-1 and EBGEO do not quote a numerical range but bound the geometry indirectly, BS 8006-1 through the arching condition on the clear spacing and cap width and EBGEO through a geometry envelope on the diameter-to-spacing ratio. The United States survey range, 2.5 to five percent most commonly, sits at the lower end of the FHWA band.

**Table 3-3.** Area replacement ratio across the four design standards and United States practice: definition, typical range and how each source bounds it.

| Source | Definition | Typical range | How it is bounded (reference) |
|---|---|---|---|
| **ASIRI (2012)** | $\alpha = A_p / A_{grid}$ (cap area / grid area) | 2–10% | Stated as the common range and “an essential factor of reinforcement efficiency” (p. 22) |
| **BS 8006-1 (2010)** | Cap area / spacing (a/s) | Not quoted | Bounded indirectly by the arching condition on H, s and cap width a (cl. 8.3.3.6, p. 184) |
| **EBGEO (2011)** | Head area / influence area | Not quoted | Bounded by the geometry envelope $d/s \geq 0.15$ and the arching-height check (§9, pp. 156–157) |
| **FHWA GEC-13 (2017)** | $a_s = A_c / A_{unitcell}$ (on cap area where caps are used) | 3.5–10% (as low as 2.5%) | Stated range from documented case histories (p. 6-11); definition p. 6-30 |
| **US survey (Han et al. 2025)** | As reported by practitioners | 2.5–5% (most common) | Empirical; a quarter of respondents did not state a ratio (Fig. 3-11) |

## 3.4 Design of Rigid Inclusions

The design of rigid-inclusion systems is governed internationally by a small number of recommendations and codes. Four are adopted here as the basis for the study, the French ASIRI recommendations (2012), the British Standard BS 8006-1 (2010), the German EBGEO recommendations (2011) and the United States FHWA Geotechnical Engineering Circular No. 13 (2017), because between them they span the main design philosophies in current use and invoke the principal load-transfer models, and because they are the standards most frequently cited in practice, including in the recent United States state-of-the-practice review (Han et al. 2025) and the recent contractor literature (Smith et al. 2023a; Swift and Pearlman 2024; Wehr et al. 2012). The four agree on the underlying physics, arching over stiffer inclusions, a load-transfer platform, optional basal reinforcement, and load shed to a bearing stratum, but they differ in the analytical method they prescribe, the parameters they require, the symbols and units they use, the factors they apply, and the acceptance limits they set. This variety is mirrored in practice: in the survey, designers most often used an analytical method or numerical analysis (Figure 3-13), and, among the analytical methods, the FHWA beam and strain-compatibility methods and BS 8006-1 were the most used, though a substantial fraction relied on “other” methods (Figure 3-14).

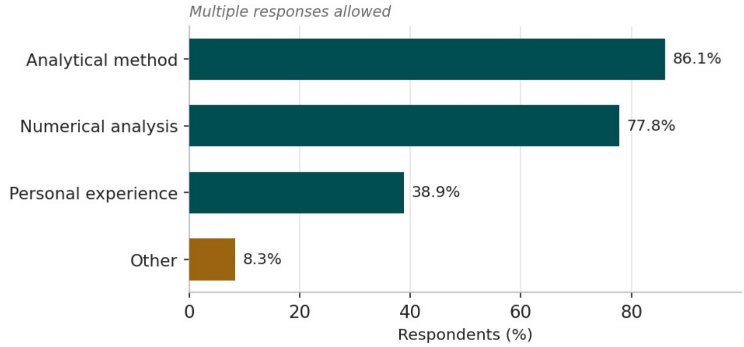



**Figure 3-13.** Design approaches used for rigid inclusions in United States practice. Respondents could select more than one approach.
*redrawn from Han et al. (2025), Figure 3.7.*

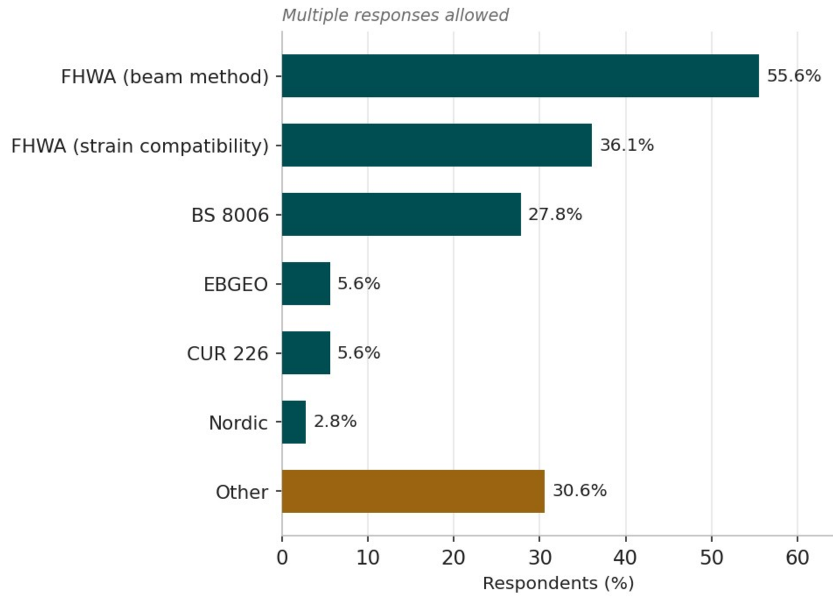



**Figure 3-14.** Analytical design methods used for rigid inclusions in United States practice. Respondents could select more than one method. *redrawn from Han et al. (2025), Figure 3.8.*

### 3.4.1 The four design philosophies compared

Although the four standards agree on the physics, each rests on a different analytical philosophy, and it is worth setting out each in turn, because the philosophy determines the parameters the standard requires, the symbols it uses and the acceptance limits it applies. Alongside these four formal standards, United States practice is now summarized in a two-part industry commentary by the DFI Ground Improvement Committee's Rigid Inclusion Task Force (Smith et al. 2023a, 2023b), which does not prescribe a new method but records the United States consensus: rigid inclusions are designed by allowable-stress design against the material stresses of the International Building Code, with the structural and geotechnical resistance both checked at the neutral plane; global stability is assessed by a stress reduction compatible with the load transferred to the inclusions, never by a composite shear strength, because unreinforced inclusions fail in bending rather than shear; and seismic design must consider kinematic flexure and the effect of the improved ground on the seismic Site Class.

**ASIRI (France, 2012).** ASIRI is built around the pressuremeter and a unit-cell analysis. Soil-inclusion interaction is modelled with load-transfer (t–z) laws derived from the Ménard pressuremeter modulus (the Frank and Zhao formulation), and the stress delivered to the inclusion head is governed by a Prandtl punching mechanism through the platform. A load-transfer platform is mandatory, a footing is never placed directly on an inclusion, and the design is organized into domains according to the load-transfer configuration and the stress state at the head. Its headline serviceability output is the settlement efficacy.

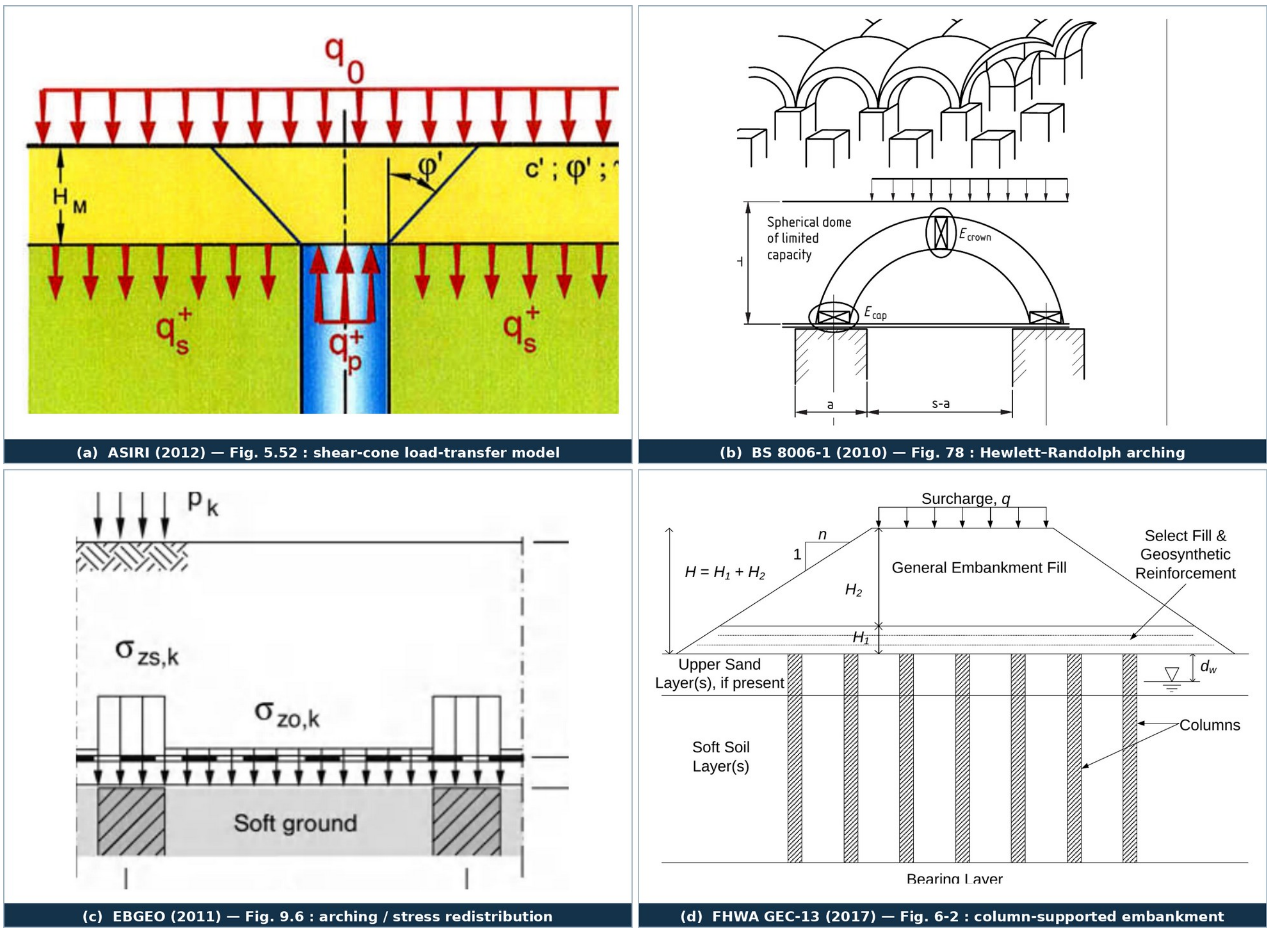
**BS 8006-1 (United Kingdom, 2010).** BS 8006-1 treats the system in a limit-state framework with partial factors. Arching is described by a Marston planar relationship or, in the form now most used, by the three-dimensional Hewlett and Randolph dome model, and any residual load the arch does not carry is taken by a basal geosynthetic acting as a tensioned, parabolic membrane. The standard requires a minimum fraction of the load to be carried by the reinforcement and limits the mid-span deflection of the membrane at the serviceability limit state. The support of the membrane by the soft soil beneath is usually neglected, which is conservative.

**EBGEO (Germany, 2011).** EBGEO adopts the Zaeske multi-shell arching model, in which a series of nested arches transfers load to the inclusions, and, distinctively, it supports the basal geosynthetic on the soft soil represented as a bed of Winkler springs (a subgrade-reaction modulus). Actions are organized into load cases.

Because it credits the soil beneath the membrane with carrying part of the load, EBGEO generally predicts lower reinforcement forces than a method that ignores that support.

**FHWA GEC No. 13 (United States, 2017).** The FHWA circular uses a load-displacement-compatibility (LDC) method, in which the load carried by the inclusions, the soil and the reinforcement is found by matching their deformations, with arching represented by an adapted-Terzaghi formulation using a fixed lateral earth-pressure coefficient (K = 0.75). It offers a beam method and a strain-compatibility variant and, reflecting its transportation focus, adds explicit checks for lateral spreading and global stability, working in allowable-stress terms with factors of safety.

Figure 3-15 contrasts the load-transfer idealizations these four methods use, and Table 3-4 summarizes their differences at a glance.




(a) ASIRI (2012) — Fig. 5.52 : shear-cone load-transfer model
(b) BS 8006-1 (2010) — Fig. 78 : Hewlett-Randolph arching
(c) EBGEO (2011) — Fig. 9.6 : arching / stress redistribution
(d) FHWA GEC-13 (2017) — Fig. 6-2 : column-supported embankment

**Figure 3-15.** Load-transfer and arching idealizations reproduced from each standard's own document: (a) ASIRI's shear-cone punching model; (b) BS 8006-1's Hewlett–Randolph arching domes; (c) EBGEO's arching / contact-stress redistribution; and (d) FHWA's column-supported embankment with geosynthetic reinforcement. *Sources: ASIRI (2012); BS 8006-1 (2010); EBGEO (2011); FHWA GEC-13 (2017).*

**Table 3-4.** Comparative overview of the four rigid-inclusion design standards.

| Aspect | ASIRI (France) | BS 8006-1 (UK) | EBGEO (Germany) | FHWA GEC-13 (USA) |
|---|---|---|---|---|
| **Basis / approach** | Pressuremeter-based unit-cell with load-transfer (t–z) laws | Limit-state arching plus tensioned membrane | Multi-shell arching plus membrane on elastic support | Load-displacement-compatibility (LDC) method |
| **Arching model** | Prandtl punching at the head | Marston / Hewlett–Randolph | Zaeske multi-shell arches | Adapted Terzaghi (K = 0.75) |
| **Geosynthetic role** | Optional | Essential (parabolic membrane) | Essential (membrane on Winkler springs) | Essential (beam / strain-compatibility) |

| Aspect | ASIRI (France) | BS 8006-1 (UK) | EBGEO (Germany) | FHWA GEC-13 (USA) |
|---|---|---|---|---|
| **Soil support of membrane** | Considered (soil reaction) | Usually neglected (conservative) | Explicitly included (subgrade reaction) | Included in LDC |
| **Safety format** | EC7 design approaches / partial factors | Limit-state partial factors | Load cases (LC1–LC3) | Allowable stress with factors of safety |
| **Settlement basis** | Settlement efficacy (Domain 2) | Mid-span deflection limit | Charts / membrane deflection | LDC settlement compatibility |
| **Distinctive feature** | Ménard-modulus t–z transfer; LTP mandatory | Minimum reinforcement load fraction | Elastic soil support beneath membrane | Explicit lateral-spreading & stability checks |

### 3.4.2 ASIRI (France, 2012): unit-cell load-transfer method

ASIRI is fundamentally a unit-cell, load-transfer (t–z) method built on deep-foundation transfer functions, not an embankment-arching method. The reinforced ground is reduced to a cylindrical unit cell, one inclusion and its tributary soil, and the design first evaluates the structure without inclusions and then quantifies the contribution of the inclusions, following the sequence of Figure 3-16. At the base of the load-transfer platform the applied stress ($q_0$) is shared between the inclusion head and the soil according to the cell equilibrium of Equation 3-5, in which α is the area replacement ratio, $q_p$+ the stress on the head and $q_s$+ the stress on the surrounding soil.

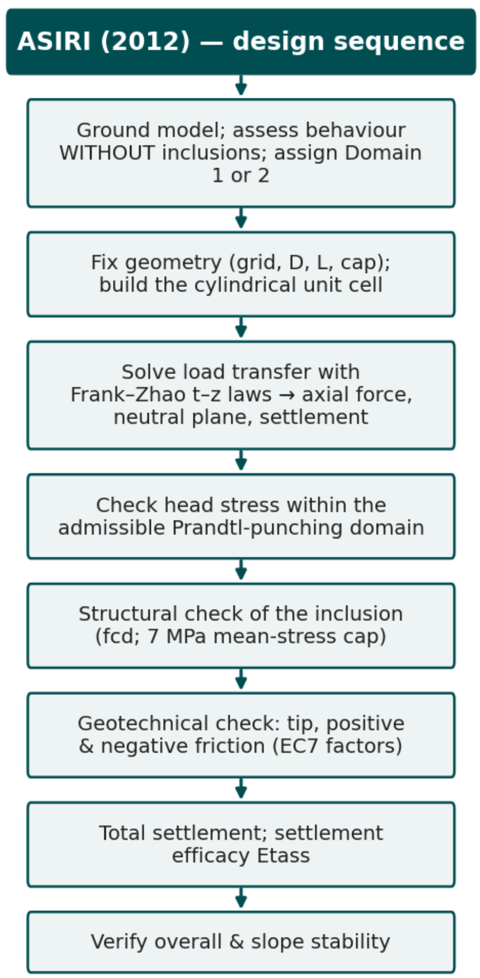



**Figure 3-16.** The ASIRI (2012) design sequence, from the ground model and the assignment of the design domain through the unit-cell load-transfer solution to the structural, geotechnical, settlement and stability checks. *(after ASIRI 2012).*

$$q_0 = \alpha\, q_p^+ + (1-\alpha)\, q_s^+ \tag{3-5}$$

*ASIRI (2012), Eq. 5.36, p. 218.*

Along the shaft and at the base, the mobilization of friction and end bearing follows the bilinear Frank and Zhao (1982) load-transfer laws, whose initial slopes are proportional to the Ménard pressuremeter modulus ($E_M$) and inversely proportional to the inclusion diameter (Equation 3-6, with $m_\tau$ and $m_q$ soil-and-installation coefficients tabulated in ASIRI). The unit base resistance is taken from the pressuremeter limit pressure or the cone resistance through a bearing factor (Equation 3-7).

$$k_\tau = m_\tau \frac{E_M}{D_s}\ ; \qquad k_q = m_q \frac{E_M}{D_p} \tag{3-6}$$

*ASIRI (2012), Eqs. 3.1–3.2, pp. 93–94 (Frank and Zhao 1982).*

$$R_b = k_c\, q_c\, A_p \ =\ k_p\, p_l^*\, A_p \tag{3-7}$$

*ASIRI (2012), §4.1.1, p. 214.*

The stress the granular platform can concentrate onto the inclusion head is not unlimited: it is capped by a Prandtl punching mechanism in the platform, so that $q_p$+ cannot exceed $N_q$ times the soil stress (Equation 3-8), where the bearing factor $N_q$ is evaluated from the critical-state friction angle of the fill. Below the platform, because the soft soil settles more than the inclusion, negative skin friction develops above a neutral plane (Equation 3-9, with K tanδ a single empirical downdrag coefficient); the axial force grows from the head load to a maximum at the neutral plane, where inclusion and soil settle equally, and is then shed by positive friction to the base (Equation 3-10). Serviceability is expressed through the settlement efficacy, the fractional reduction of surface settlement relative to the untreated soil (Equation 3-11).

$$q_p^+ = N_q\, q_s^+\ , \qquad N_q = \tan^2\left(\frac{\pi}{4} + \frac{\phi'}{2}\right) e^{\pi \tan\phi'} \tag{3-8}$$

*ASIRI (2012), Eq. 5.37, p. 218 (N_q, p. 223).*

$$f_n(z) = K\, \sigma_v'(z) \tan\delta \tag{3-9}$$

*ASIRI (2012), Eq. 2.4, p. 60 (K tanδ, p. 63).*

$$Q_p(0) + F_N = F_P + Q_p(L) \tag{3-10}$$

*ASIRI (2012), §2.2.2, p. 56.*

$$E_{tass} = 1 - \frac{y_M}{y_0} \tag{3-11}$$

*ASIRI (2012), Eq. 2.23, p. 79.*

ASIRI organizes the design into two domains, decided by checking stability while ignoring the inclusions: in Domain 1 the inclusions are needed for bearing or stability and are verified as deep foundations under Eurocode 7, and must be reinforced over any non-compressed length; in Domain 2 they only reduce settlement and may be left unreinforced. At the ultimate limit state the maximum compressive stress in the inclusion is limited to the design value $f_{cd}$, and the mean stress over the compressed section is capped at a fixed 7 MPa (ASIRI 2012, p. 179). Table 3-5 lists the parameters this method requires.

**Table 3-5.** Principal design parameters required by ASIRI (2012).

| Parameter | Symbol | Role in the design | Source / test |
|---|---|---|---|
| **Ménard pressuremeter modulus** | $E_M$ | Scales the Frank and Zhao t–z slopes (Eq 3-6); settlement | Ménard pressuremeter |
| **Ménard limit pressure (net)** | $p_l$* | Unit tip resistance and the soil head-stress limit | Pressuremeter |
| **Cone resistance** | $q_c$ | Alternative unit tip resistance $R_b = k_c\, q_c\, A_p$ | CPT |
| **Bearing factors** | $k_p$, $k_c$ | Convert $p_l$* or $q_c$ to tip resistance | EC7 / NF P94-262 |
| **Negative-friction coefficient** | K tanδ | Governs downdrag and the load reaching the head (Eq 3-9) | Empirical / measured |
| **Oedometric modulus** | $E_{oed}$ | Soft-soil compressibility in the unit cell; settlement | Oedometer |
| **Platform friction angle, cohesion** | φ', c' | Prandtl factor $N_q$ that caps the head stress (Eq 3-8) | Shear box / triaxial |
| **Area replacement ratio** | α | Fixes the load split at the platform base (Eq 3-5) | Geometry |
| **Concrete characteristic strength** | $f_{ck}$ | Structural check of the inclusion ($f_{cd}$; 7 MPa cap) | Cylinders, 7 and 28 d |

### 3.4.3 BS 8006-1 (United Kingdom, 2010): limit-state arching and membrane

BS 8006-1 treats the column-supported system in a limit-state framework and, in the form now most used, describes the load transfer by the three-dimensional Hewlett and Randolph (1988) arching model, in which a series of hemispherical domes spans between the inclusion heads (Figure 3-17 gives the design sequence). The passive earth-pressure coefficient of the fill (Equation 3-12, evaluated with the peak friction angle) sets the strength of the arch, and the arching efficacy, the fraction of the embankment load carried directly by the heads, is evaluated both at the crown of the arch and at the head or cap; the head-or-cap value is given by Equations 3-13 and 3-14, and the lesser of the two governs the design.

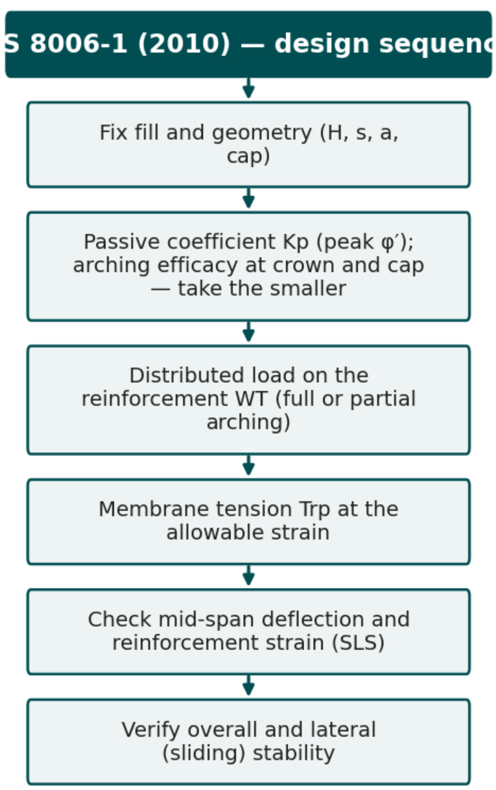



**Figure 3-17.** The BS 8006-1 (2010) design sequence, from the fill and geometry through the arching efficacy and the membrane tension to the serviceability and stability checks. *(after BS 8006-1 2010).*

$$K_p = \frac{1+\sin\phi'}{1-\sin\phi'} \tag{3-12}$$

*BS 8006-1 (2010), cl. 8.3.3.7.2, p. 187 (peak friction angle φ'_p).*

$$E_{c,c} = \frac{\beta}{1+\beta} \tag{3-13}$$

*BS 8006-1 (2010), cl. 8.3.3.7.2, p. 187.*

$$\beta = \frac{2K_p}{(K_p+1)(1+\frac{a}{s})}\left[(1-\tfrac{a}{s})^{-K_p} - (1+K_p\tfrac{a}{s})\right] \tag{3-14}$$

*BS 8006-1 (2010), cl. 8.3.3.7.2, p. 187.*

The load the arch does not carry is transferred to a basal geosynthetic acting as a tensioned membrane. The distributed load carried by the reinforcement between adjacent heads follows from the minimum arching efficacy (Equation 3-15), the tension in the reinforcement is found at the design strain (Equation 3-16), and the deformed shape is taken as parabolic, so that the maximum differential settlement between head and mid-span follows from the strain (Equation 3-17). The standard neglects any support of the membrane by the soft soil beneath, which is conservative; it requires the reinforcement to carry at least fifteen percent of the vertical load, and it caps the reinforcement strain (six percent in the short term, with a smaller creep strain over the design life) at the serviceability limit state.

$$W_T = \frac{s(\gamma H+q)(1-E_{min})\,s^2}{s^2-a^2} \tag{3-15}$$

*BS 8006-1 (2010), cl. 8.3.3.7.1, p. 186.*

$$T = \frac{W_T(s-a)}{2a}\sqrt{1+\frac{1}{6\varepsilon}} \tag{3-16}$$

*BS 8006-1 (2010), cl. 8.3.3.9, p. 188.*

$$y = (s-a)\sqrt{\frac{3\varepsilon}{8}} \tag{3-17}$$

*BS 8006-1 (2010), cl. 8.3.3, p. 188 (deflection); strain limits cl. 8.3.3.14, p. 195.*

The design sequence is: fix the fill and geometry; compute the passive coefficient and the arching efficacy at crown and cap and take the smaller; obtain the distributed load on the reinforcement; determine the tension at the allowable strain; check that the mid-span deflection and the reinforcement strain lie within their serviceability limits; and verify overall and lateral (sliding) stability, all in the partial-factor limit-state format. Table 3-6 lists the parameters required.

**Table 3-6.** Principal design parameters required by BS 8006-1 (2010).

| Parameter | Symbol | Role in the design | Source / test |
|---|---|---|---|
| **Embankment height** | H | Overburden driving the arch (Eq 3-15) | Design geometry |
| **Fill unit weight** | γ | Vertical load on the system | Compaction / lab |
| **Fill peak friction angle** | $\varphi'_p$ | Passive coefficient $K_p$ and arching (Eq 3-12) | Shear box / triaxial |
| **Cap width and spacing** | a, s | Arching geometry and area replacement ratio | Design |
| **Surcharge** | q | Applied traffic or structural load | Design |
| **Reinforcement tensile stiffness** | J | Membrane tension at the design strain (Eq 3-16) | Index test (ASTM D6637) |
| **Design strain** | ε | Membrane tension and deflection (Eq 3-16, 3-17) | Code limit (≤ 6%) |
| **Area replacement ratio** | α | Load split between heads and soil | Geometry |

### 3.4.4 EBGEO (Germany, 2011): multi-arch model with elastic soil support

EBGEO describes the load transfer with the Zaeske multi-arch model, a family of nested arches derived from lower-bound plasticity, pilot-scale tests and numerical analysis (Figure 3-18 gives the design sequence). The model returns the vertical stress on the soft soil between the inclusions and, from it, the stress carried by the inclusion head (Equation 3-18, in which A is the tributary area and $A_c$ the head area); the arching height that fixes the load spread is tied to the diagonal spacing between inclusions (Equation 3-19).

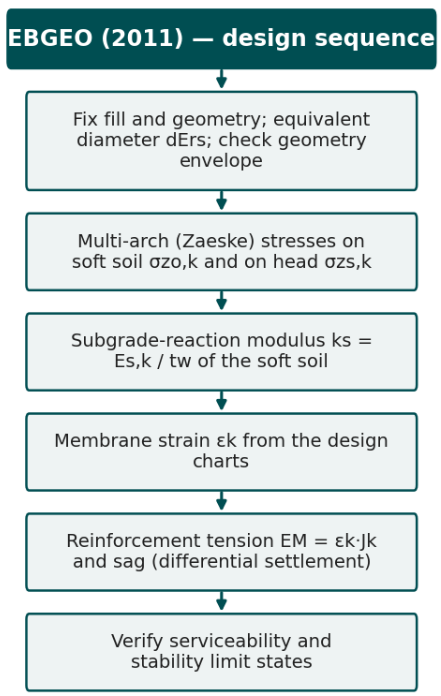



**Figure 3-18.** The EBGEO (2011) design sequence, from the geometry envelope and the multi-arch stresses through the Winkler soil support and the membrane strain to the serviceability and stability checks. *(after EBGEO 2011).*

$$\sigma_c^a = [(\gamma H + q) - \sigma_s^a]\frac{A}{A_c} + \sigma_s^a \tag{3-18}$$

*EBGEO (2011), Eq. 9.11, p. 164 (soil stress σ_zo,k, Eqs. 9.5–9.6, p. 162).*

$$h_g = \frac{s_d}{2} \quad (H \geq s_d/2) \tag{3-19}$$

*EBGEO (2011), §9, p. 162.*

The residual load is carried by the geosynthetic as a membrane whose maximum tension is the product of its tensile stiffness and the maximum strain read from the EBGEO design charts (Equation 3-20). The distinctive feature of EBGEO is that the membrane is not assumed to span a void but is supported on the soft soil, represented as a bed of Winkler springs whose subgrade-reaction modulus is the constrained modulus of the soil divided by its thickness (Equation 3-21). Because it credits the soil beneath the reinforcement with carrying part of the load, EBGEO generally predicts lower reinforcement forces than a method that ignores that support, and it limits the creep strain of the reinforcement to two percent over the service life.

$$T_{max} = J\,\varepsilon_{max} \tag{3-20}$$

*EBGEO (2011), Eqs. 9.24–9.25, p. 167 (creep-strain limit Δε_kr ≤ 2%, Eq. 9.42, p. 176).*

$$k_s = \frac{E_s}{t_w} \tag{3-21}$$

*EBGEO (2011), Eq. 9.26, p. 169.*

Actions are organized into load cases, and the design sequence is: fix the fill and geometry; compute the arching stresses on soil and head from the multi-arch model; evaluate the subgrade-reaction modulus of the soft soil; read the membrane strain from the design charts for that modulus; obtain the reinforcement tension and the sag (differential settlement); and verify the serviceability and stability limit states. Table 3-7 lists the parameters required.

**Table 3-7.** Principal design parameters required by EBGEO (2011).

| Parameter | Symbol | Role in the design | Source / test |
|---|---|---|---|
| **Embankment height, unit weight** | H, γ | Overburden for the multi-arch model | Design / lab |
| **Fill effective friction angle** | φ' | Multi-shell arching stresses | Shear box |
| **Cap width and spacings** | a, $s_x$, $s_y$ | Arching geometry and diagonal spacing $s_d$ | Design |
| **Reinforcement tensile stiffness** | J | Membrane tension $E_M = \varepsilon_k J_k$ (Eq 3-20) | Index test |
| **Soft-soil constrained modulus** | $E_s$ | Elastic (Winkler) support of the membrane | Oedometer |
| **Soft-soil thickness** | $t_w$ | Subgrade-reaction modulus (Eq 3-21) | Boring / CPT |
| **Subgrade-reaction modulus** | $k_s$ | Sets the membrane strain from the design charts | Derived (Eq 3-21) |
| **Maximum reinforcement strain** | $\varepsilon_{max}$ | Membrane tension and sag (≤ 2% creep) | EBGEO design charts |

### 3.4.5 FHWA GEC No. 13 (United States, 2017): load-displacement compatibility

The FHWA circular uses the load-displacement-compatibility (LDC) method, in which the load carried by the inclusions, the soil and the reinforcement is found by requiring their deformations to be compatible (Figure 3-19 gives the design sequence). Vertical equilibrium at the base of the embankment (Equation 3-22) states that the average applied stress is shared between the inclusions and the soft soil in proportion to the area replacement ratio, considering both soil arching and, where present, the tensioned geosynthetic. Arching is represented by an adapted-Terzaghi formulation that uses a fixed lateral earth-pressure coefficient (Equation 3-24), and the reduction of the vertical stress reaching the soil or the reinforcement is expressed as a stress-reduction ratio (Equation 3-23), from which the load efficacy and the differential settlement follow.

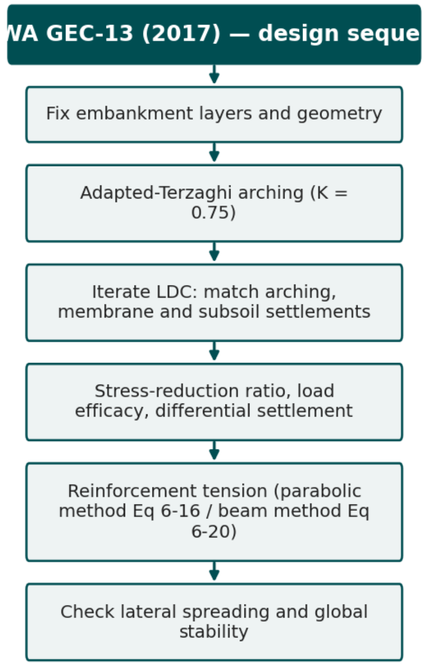



**Figure 3-19.** The FHWA GEC-13 (2017) design sequence, from the embankment geometry and the adapted-Terzaghi arching through the load-displacement-compatibility solution to the reinforcement tension and the lateral-spreading and stability checks. *(after FHWA GEC-13 2017).*

$$\sigma_{avg} = \gamma_1 H_1 + \gamma_2 H_2 + q = a_s\,\sigma_{col,\,geo} + (1 - a_s)\,\sigma_{soil,\,geo} \tag{3-22}$$

*FHWA GEC-13 (2017), Eq. 6-11, p. 6-30.*

$$\mathrm{SRR}_{net} = \frac{\sigma_{soil,\,geo}}{\sigma_{avg}} \tag{3-23}$$

*FHWA GEC-13 (2017), Ch. 6 (Adapted Terzaghi Method).*

$$K = 0.75 \tag{3-24}$$

*FHWA GEC-13 (2017), p. 6-31 (Russell and Pierpoint 1997).*

Displacement compatibility is enforced by requiring the differential settlement at the base of the embankment, computed from the elastic compression of the inclusions above the neutral plane and the settlement of the subsoil, to be the same in the arching, the membrane and the subsoil mechanisms; the nonlinear system is solved in a public spreadsheet. The tension in the reinforcement is then obtained by one of two methods the circular provides. The generalized parabolic method assumes a parabolic deformed shape of the reinforcement spanning between adjacent heads and solves a cubic for the tension (Equation 3-25, with the strain $\varepsilon = T/J$), and is recommended for rectangular arrays with a side ratio between one-half and two (FHWA GEC-13 2017, p. 6-35). The simpler beam method treats a minimum of three reinforcement layers as a platform beam and gives each layer's tension as its carried load times a membrane factor $\Omega$, tabulated against strain, and half the design span (Equation 3-26); it is used for preliminary design when the soft-soil characterisation is incomplete (FHWA GEC-13 2017, p. 6-39).

$$6T^3 - 6T\left(\frac{\sigma_{net} A_{soil}}{p}\right)^2 - J\left(\frac{\sigma_{net} A_{soil}}{p}\right)^2 = 0\,, \qquad \varepsilon = \frac{T}{J} \tag{3-25}$$

*FHWA GEC-13 (2017), Eq. 6-16, p. 6-35 (Filz and Smith 2006; Sloan 2011).*

$$T_{rpn} = W_{Tn}\,\Omega\,\frac{D}{2} \tag{3-26}$$

*FHWA GEC-13 (2017), Eq. 6-20 and Table 6-2, p. 6-39 (Young et al. 2003).*

Reflecting its transportation focus, the method adds explicit checks for lateral spreading, comparing the active thrust of the embankment with the available reinforcement tension and interface resistance and requiring a factor of safety of at least 1.5, and for global stability, and it works in allowable-stress terms with factors of safety. The design sequence is: fix the embankment layers and geometry; compute the adapted-Terzaghi arching stresses; iterate the LDC solution to match the arching, membrane and subsoil settlements; obtain the stress-reduction ratio, the efficacy and the differential settlement; determine the reinforcement tension by the parabolic or beam method; and check lateral spreading and global stability. Table 3-8 lists the parameters required.

**Table 3-8.** Principal design parameters required by FHWA GEC-13 (2017).

| Parameter | Symbol | Role in the design | Source / test |
|---|---|---|---|
| **Embankment layers (bridging, fill)** | $\gamma$, $H$ | Overburden and arching (Eq 3-22) | Design / lab |
| **Lateral earth-pressure coefficient** | $K$ | Adapted-Terzaghi arching, fixed at 0.75 (Eq 3-24) | Method constant |
| **Cap width, spacing, ARR** | $a$, $s$, $a_s$ | Geometry and load split (Eq 3-22) | Design |
| **Column modulus and length** | $E$, $L$ | Column compression above the neutral plane | Design / mix |
| **Reinforcement tensile stiffness** | $J$ | Tensioned-geosynthetic tension (Eq 3-25, 3-26) | Index test |
| **Membrane factor** | $\Omega$ | Beam-method tension vs strain (Eq 3-26) | Table 6-2 |
| **Subgrade modulus** | $k_s$ | Subsoil settlement in the compatibility solution | Oedometer / CPT |
| **Coefficient of consolidation** | $c_v$ | Time-rate of subsoil settlement | Oedometer |

### 3.4.6 Downdrag, reinforcement and the seismic case

Two design considerations cut across the four methods and were singled out in the survey. The first is downdrag. Because the soft soil settles more than the inclusion, negative skin friction drags the upper shaft down and raises the axial load to a maximum at the neutral plane, the depth at which inclusion and soil settle equally (Figure 3-20). Practitioners treat this effect unevenly: in the survey the neutral-plane method was the most common approach, but a substantial fraction accounted for downdrag only within a limit-strength design, by experience, or not at all (Figure 3-21). The neutral-plane construction that locates the maximum axial force is therefore among the parameters a standardized design record must carry explicitly, so that the structural and geotechnical checks refer to the same demand.

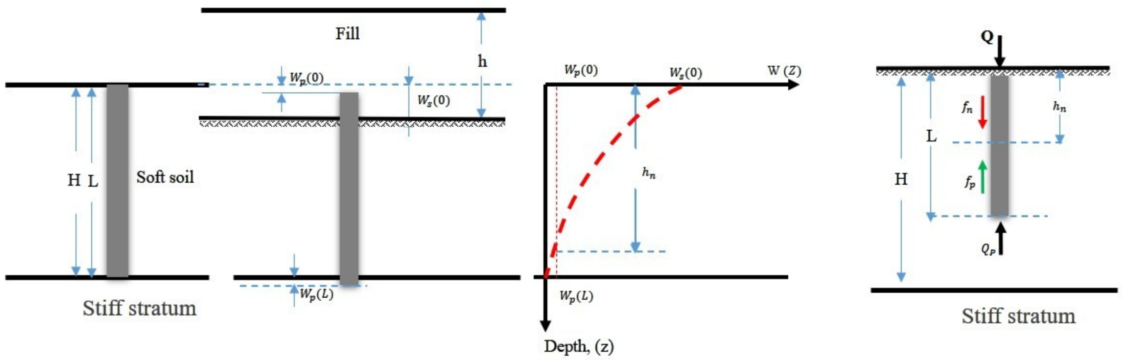



**Figure 3-20.** The settlement compatibility that produces downdrag: the fill and soft soil settle more than the stiff inclusion ($W_s(0) > W_p(0)$), so negative skin friction ($f_n$) develops over the depth ($h_n$) above the neutral plane and positive skin friction ($f_p$) below it, and the axial load is greatest at the neutral plane. *provided by the project team.*

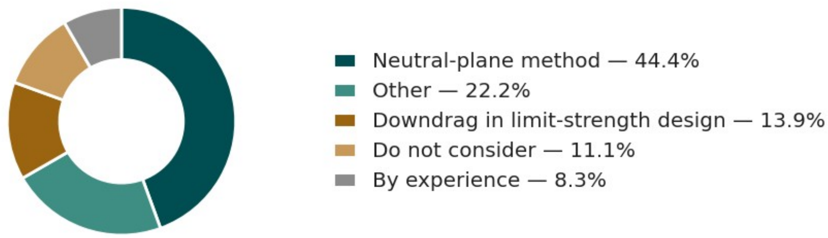



**Figure 3-21.** Methods used to account for downdrag (negative skin friction) along rigid inclusions in United States practice. *redrawn from Han et al. (2025), Figure 3.22.*

The second is reinforcement. Rigid inclusions are usually unreinforced, but steel is added where the inclusion must carry bending or tension, at embankment edges and beneath slopes and, above all, under seismic loading, which can impose kinematic flexure and cause unreinforced columns to fail by bending or rotation. In the survey, steel was used only sometimes or rarely by most respondents and never routinely (Figure 3-22), which makes whether, where and over what length an inclusion is reinforced a genuine and consequential project variable. The DFI commentary adds that the seismic case requires the effect of the improved ground on the seismic Site Class to be considered, so that a post-improvement shear-wave velocity may itself become a design and verification parameter (Smith et al. 2023a).

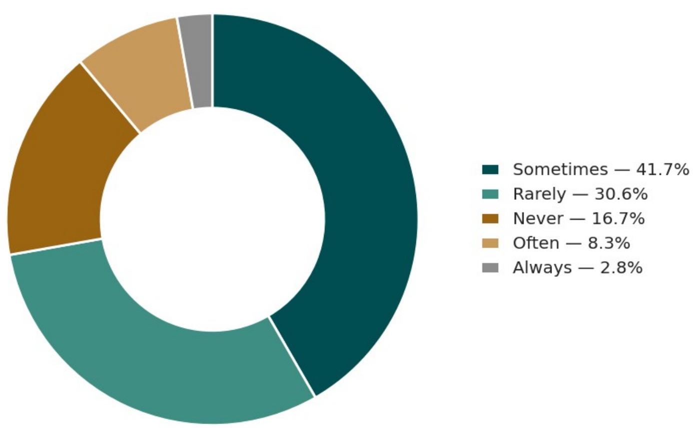



**Figure 3-22.** Frequency of use of steel reinforcement in rigid inclusions in United States practice; most inclusions are unreinforced. *redrawn from Han et al. (2025), Figure 3.14.*

### 3.4.7 Key discrepancies and acceptance limits

These differences of philosophy translate into concrete, quantifiable discrepancies. Because the four arching models partition the load differently, they return different head stresses, load efficacies and residual soil stresses for identical geometry, the single largest source of divergence. The basal geosynthetic is optional in ASIRI but essential in the other three, and the assumed membrane shape, the allowable strain and whether the soil beneath is credited with support all differ, changing the computed reinforcement tension. The safety formats are not directly comparable: ASIRI works within the Eurocode 7 design-approach framework, BS 8006-1 applies limit-state partial factors, EBGEO uses load cases, and FHWA uses allowable stresses with factors of safety. And the parameter provenance differs, ASIRI deriving soil parameters from the pressuremeter while the others rely more on SPT, CPT and laboratory tests, so the same nominal parameter carries a different pedigree and a different valid correlation depending on the standard.

This spread was confirmed quantitatively by the Phase-I study, which evaluated the methods against a database of twenty-four full-scale experiments and four model tests: BS 8006-1 overestimated load efficacy, differential settlement and reinforcement strain, whereas the FHWA and CUR226 methods predicted the measured behaviour comparatively more accurately, with EBGEO intermediate. A comparable spread appeared in stability analysis, where the Column-Wall, Equivalent-Strength, Stress-Reduction and Pile-Support methods returned factors of safety differing by more than ten percent for the same embankment (Han et al. 2025). The practical consequence is that no single method's parameter list suffices, and results are comparable only when the method, the provenance and the factors are recorded alongside them.

The differences extend to the acceptance limits each standard sets, and because these limits are fixed at the design stage they define what must be measured after construction, closing the loop to the verification of Section 3.6. Table 3-9 collects the serviceability and material limits of the four standards. They differ in both value and form: ASIRI expresses total and differential settlement through the French building tolerances (DTU 13.3) and caps the mean stress in the inclusion at 7 MPa; BS 8006-1 and EBGEO limit the reinforcement strain and its creep; and FHWA quotes post-construction settlement targets and a geosynthetic strain threshold. A standardized record must therefore store each acceptance criterion alongside the standard that sets it and the value it judges.

**Table 3-9.** Acceptance and serviceability limits across the four design standards (with the governing clause or page).

| Standard | Settlement / serviceability | Reinforcement / material | Reference |
|---|---|---|---|
| **ASIRI (2012)** | Total ≤ 20 + $L_1$/2000 mm; differential ≤ 10 + $L_2$/2000 mm (DTU 13.3) | ULS mean stress ≤ 7 MPa; max stress ≤ $f_{cd}$ | pp. 179, 182 |
| **BS 8006-1 (2010)** | Mid-span deflection of the reinforcement limited (parabolic sag) | Strain ≤ 6% short-term; creep ≤ 2%; reinforcement carries ≥ 15% of load | cl. 8.3.3.9–8.3.3.14, pp. 188–195 |
| **EBGEO (2011)** | Serviceability sag from the design charts | Creep strain $\Delta\varepsilon_{kr}$ ≤ 2% over the service life | Eq. 9.42, p. 176 |
| **FHWA GEC-13 (2017)** | Post-construction settlement < 2–4 in; differential < 1 in | Geosynthetic strain ≤ 5–6% | pp. 6-12, 6-21 |
| **US DOT special provisions** | Total ≤ 1.5 in; differential ≤ 1 in per 100 ft; FoS ≥ 1.5 | Per project; load test to 200% design load | Han et al. (2025); see Table 3-11 |

Tracing the four design sequences also exposes the full population of parameters a rigid-inclusion design generates: the ground model with its provenance; the geometry reduced to a unit cell; the arching split into head stress, residual soil stress and load efficacy; the platform thickness, critical height, fill strength and, where present, the reinforcement stiffness, design strength, tension, strain and anchorage; the structural design of the inclusion through its material, modulus, capacity and any steel; the shaft transfer through the negative friction, neutral-plane depth and maximum axial force; and the verifications of stability, lateral spreading and serviceability against the acceptance limits above. The design record is therefore not a single number but this whole chain, each link of which a standardized data set must name, with its unit, symbol, provenance and governing standard, the task of Chapter 4.

## 3.5 Construction of Rigid Inclusions

The installation technique is the primary way rigid inclusions are classified, because it governs the spoil, the ground disturbance and the quality-control data the rig produces. Following Basu et al. (2010) and Han et al. (2025), and the technical documentation of the two specialty contractors examined here (Menard 2022; Blackburn and Franz 2018), the techniques fall into two families, the displacement method and the partial or non-displacement method, and the difference between them is whether the soil is pushed aside or removed. As the practitioner interviews of Section 3.1 confirmed, both methods share a site-preparation step that precedes the first column and that the record must capture: because the rigs are heavy tracked machines, a working platform, a compacted granular or cemented-soil mat of engineered thickness and stiffness, is built over the soft ground to carry the plant, to keep it stable and level during drilling and to protect the surface soils. The platform level becomes the datum against which every installed depth, cut-off and toe elevation is referenced, and, because it is not removed, it usually remains in service as the lowest layer of the load-transfer platform. Its plan extent, thickness and material are therefore installation data in their own right.

### 3.5.1 Displacement and non-displacement methods

In the displacement method the tool is drilled, driven or jacked into the ground and the soil is displaced laterally rather than removed, so that little or no spoil is generated. The lateral displacement changes the stress state of the surrounding ground and, in granular soils, densifies it and raises its strength and modulus; in saturated soft clays the same displacement causes ground heave, lateral movement and a spike in pore-water pressure. The rigid inclusions installed this way include drilled-displacement columns (DDC), vibro-concrete columns (VCC), driven columns and grouted stone columns; drilled-displacement columns are now the most widely used in practice (Han et al. 2025). A drilled-displacement column is built in one continuous operation, a specially shaped tool on the end of a hollow stem rotated and pushed into the ground until a combined criterion of low advance rate and high torque confirms the founding stratum, after which concrete or grout is pumped down the stem under controlled pressure while the tool is withdrawn, forming a continuous column from the toe up. Within this family the shape of the tool itself varies widely, a further source of the fragmentation a standard must absorb; Figure 3-23 shows eight tools in common use, each with a soil-displacement body, a partial-flight auger segment and a sacrificial tip.

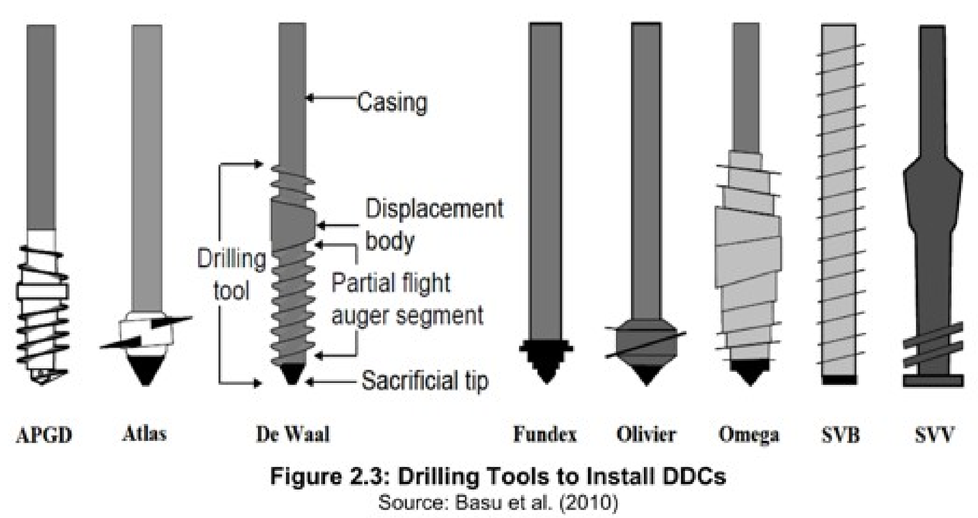



**Figure 3-23.** Common drilling tools used to install drilled-displacement rigid inclusions (APGD, Atlas, De Waal, Fundex, Olivier, Omega, SVB, SVV), showing the soil-displacement body, the partial-flight auger segment and the sacrificial tip. *reproduced from Han et al. (2025), Figure 2.3, after Basu et al. (2010).*

Figure 3-24 labels the functional parts of one such tool. Below the joint, a counter-rotating flight recompacts the soil, a stabilizing section steadies the tool in the bore, the main flight drills and displaces the soil laterally, and flat or bullet teeth loosen the ground at the tip, so that the shaped body and sacrificial tip leave a clean, spoil-free column (GIE Foundation).

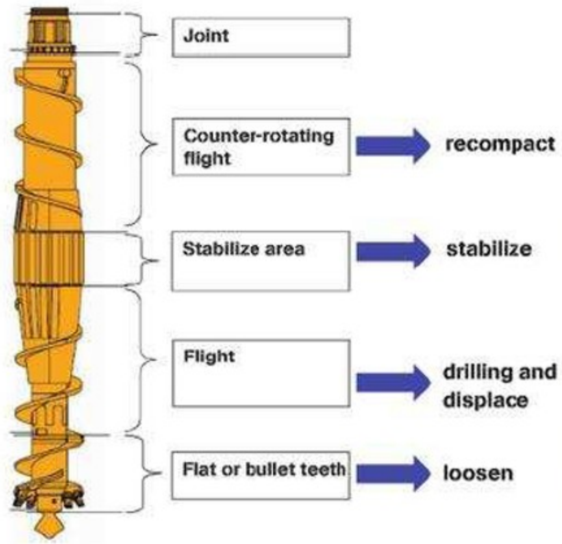



**Figure 3-24.** Anatomy of a drilled-displacement tool: the joint, the counter-rotating flight (recompact), the stabilizing section (stabilize), the main flight (drill and displace) and the flat or bullet teeth (loosen). *after GIE Foundation product literature (giefoundation.com).*

The column itself is formed in the continuous sequence of Figure 3-25. The hollow-stem displacement auger is rotated and pushed to the design depth, displacing the soil laterally with little or no spoil, and concrete or grout is then pumped through the hollow stem under controlled pressure as the auger is withdrawn, building the column from the toe upward.

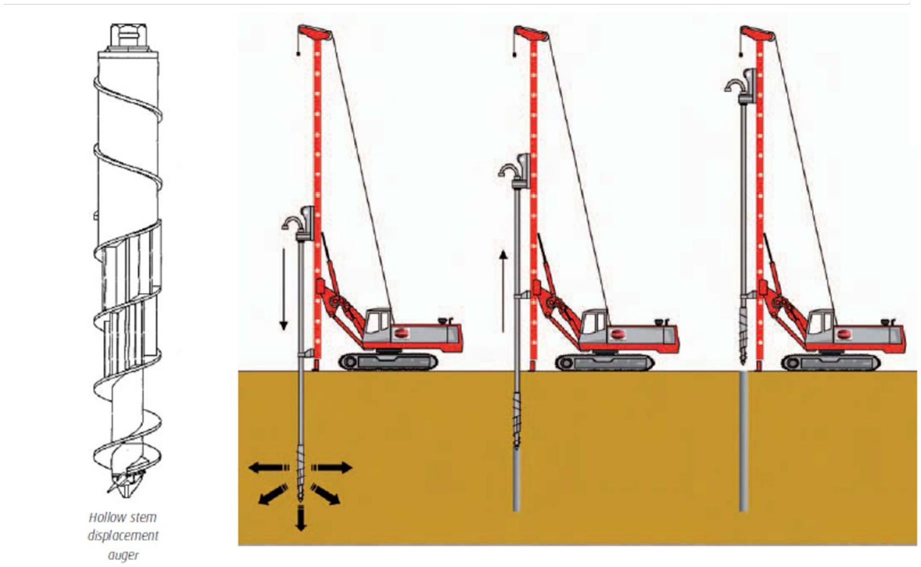



**Figure 3-25.** The drilled-displacement installation sequence: the hollow-stem displacement auger (left) is advanced to depth, displacing the soil, and the column is formed by grouting through the stem during controlled withdrawal. *after Menard controlled-modulus-column documentation.*

Figure 3-26 shows real displacement tooling: a set of displacement augers of the kind used in the displacement-auger method, and a conical vibratory displacement tool used in the vibratory method, each shaped to displace rather than remove the soil.

**Figure 3-26.** Real drilled-displacement tooling: (left) displacement augers used in the displacement-auger method; (right) a conical vibratory displacement tool used in the vibratory method. *displacement augers after SoilMec (soilmec.com); vibratory displacement tool, field photograph.*

In the partial or non-displacement method the inclusion is formed by excavating a cylinder of soil and filling the void with concrete; the side bond is lower and a large quantity of spoil is generated, but the method can be preferable where displacement would be obstructed or would disturb adjacent structures. The rigid inclusions installed this way are principally auger cast-in-place (ACIP) and continuous-flight-auger (CFA) columns; in the survey, auger cast-in-place columns were the second most-used system after drilled displacement (Han et al. 2025). The displacement method is nonetheless the dominant technique in United States transportation practice, selected by about sixty-nine percent of respondents, which is why it is adopted as the primary focus of this report.

### 3.5.2 The specialty contractors: Menard and Keller / Hayward Baker

The clearest way to expose what must be standardized is to place the practice of two established specialty contractors side by side. Menard installs controlled modulus columns (CMC) and Keller / Hayward Baker installs rigid inclusions (RI); both are drilled-displacement techniques, both pump a cementitious material through a hollow stem during controlled extraction, and both instrument the rig to record the installation in real time (Figures 3-27 and 3-28). Yet each records the process in its own proprietary system and reports it in its own document format. Menard installs the controlled modulus column with a purpose-built rig and a single displacement auger in three steps, penetration, withdrawal and completion, pumping ready-mixed concrete of controlled modulus, of the order of 5 to 30 MPa, down the hollow stem as the auger is withdrawn so that the column forms from the toe up; the technique is spoil-free and almost vibration-free, which Menard promotes as an urban method, and anchorage in the bearing stratum is declared by a combined criterion, an advance rate below 250 m/h together with a torque above 5 t·m sustained over half a metre (Menard 2022).

**Figure 3-27.** A Menard drilled-displacement (CMC) rig installing controlled modulus columns on an industrial site, showing the mast, the rotary head and the hollow displacement stem through which concrete is pumped on withdrawal. *reproduced from Menard controlled-modulus-column documentation (Menard 2022).*


**Figure 3-28.** A Keller / Hayward Baker drilled-displacement rig installing rigid inclusions, with the displacement tool and grout delivery; an in-cab data-acquisition system records the installation in real time. *reproduced from the Hayward Baker Rigid Inclusions brochure (Blackburn and Franz 2018).*

Keller / Hayward Baker installs the rigid inclusion, which it also markets under several trade names, as a grout or cemented-aggregate element, using either a single-pass displacement auger or a traction compacting tool that displaces the soil on both insertion and extraction, and it offers a vibratory displacement method as well. The work is carried out with third-party fixed or telescopic mast rigs (ABI, RTG, Bauer or SoilMec) at typical diameters of twelve to eighteen inches (about 0.30 to 0.46 m), and the element is grouted on withdrawal by a piston pump. Quality is managed through an in-cab data-acquisition system that can be fitted to any rig and that records depth, strokes, penetration rate and inclination, supported by daily slump tests and twenty-eight-day cylinders and a running check of the grout volume against the theoretical volume plus waste; verification is by single-element static load tests to ASTM D1143 and by pile-integrity testing, and, distinctively, the specialty contractor designs the load-transfer platform but does not construct it (Blackburn and Franz 2018). Table 3-10 sets the two practices side by side, showing that they build the same kind of column and capture the same physical quantities but under different names, units and layouts.

**Table 3-10.** Comparison of the Menard (CMC) and Keller / Hayward Baker (RI) installation practice, as set out in the two contractors' own technical documentation (Menard 2022; Blackburn and Franz 2018).

| Aspect | Menard, Controlled Modulus Columns (CMC) | Keller / Hayward Baker, Rigid Inclusions (RI) |
|---|---|---|
| **Trade name(s)** | Controlled Modulus Column (CMC) | Rigid Inclusion (RI); also marketed as GI, GeoConcrete Column (GCC), Controlled Stiffness Column (CSC) and Augered Pressure Grout Column (APGC) |
| **Installation family** | Drilled displacement | Drilled displacement |
| **Installation method / tooling** | Single displacement-auger cycle: penetration, withdrawal and completion | Displacement auger (single-pass) or Traction Compacting Tool (≈50% insertion / 50% extraction); a vibratory method is also offered |
| **Rig** | Purpose-built CMC rig with displacement auger | Third-party fixed / telescopic mast rigs (ABI, RTG, Bauer, SoilMec) |
| **Column material** | Ready-mixed concrete of controlled modulus, 5 to 30 MPa, delivered by a concrete pump | Grout or cemented aggregate delivered by a piston pump; slump typically 4 to 8 in |
| **Soil handling & effects** | Lateral displacement; spoil-free; densifies granular soils; heaves soft ground; virtually no vibration | Lateral displacement; little spoil; densifies granular soils; usually heaves the site |
| **Typical diameter** | Typically 0.30–0.45 m (12–18 in); 0.28 m in the project case study | Typically 0.30–0.46 m (12–18 in) |
| **Real-time monitoring** | In-cabin control panel with an on-board recorder | HBI data-acquisition system that can be fitted to any rig; electronic and paper QC reports |
| **Recorded channels** | Torque, speed, depth, down-pull force, grout pressure and pumped volume, against depth | Depth, strokes, penetration rate and inclination (with crowd and grout pressure), against depth |
| **Termination / anchorage criterion** | Advance rate < 250 m/h AND torque > 5 t·m sustained over 0.50 m into the bearing stratum | Refusal set to reflect the site geology and confirmed against the design depth; contractor-selected |
| **Signature QC metric** | Concrete / grout over-consumption ratio (actual vs theoretical volume) | Grout volume vs theoretical (plus waste); stroke count and grout factor |
| **Verification testing** | Plate load test (PLT) and shaft-integrity test | Single-element static load test (ASTM D1143) and pile-integrity test (PIT) |
| **Per-column deliverable** | “LOG CENTER” installation log, one per column | Installation / data-acquisition QC report, one per element |

### 3.5.3 The displacement production record

Because the displacement column is formed in one continuous, instrumented operation, the rig data-acquisition system records the installation against depth, so that every column leaves a complete production record rather than a single acceptance stamp. The quantities that record captures are essentially the same whichever contractor performs the work: the depth and the penetration and withdrawal rate, the rotation torque and the crowd or down-thrust, the grout or concrete pressure and the placed volume, the number of pump strokes, and the termination or anchorage criterion that ended the drilling. From these, each contractor derives a signature quality metric, the ratio of the placed volume to the theoretical column volume, which flags over-consumption or a possible neck. Figures 3-29 and 3-30 show the same information in two proprietary forms, a Menard per-column installation log and a Keller in-cab data-acquisition display. The same acquisition system presents these quantities to the operator in real time in the cab of the rig as each column is formed (Figure 3-31; Menard 2022), and the diameters those rigs install are shown in Figure 3-32.

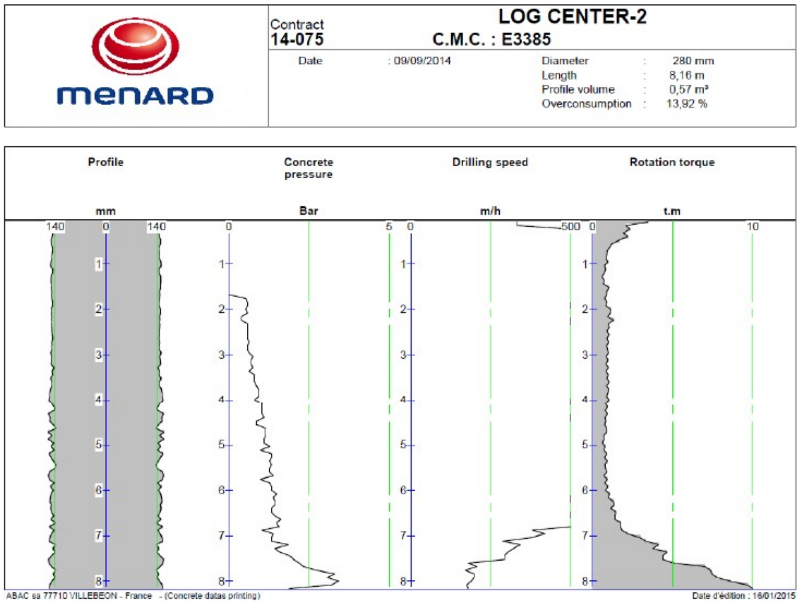
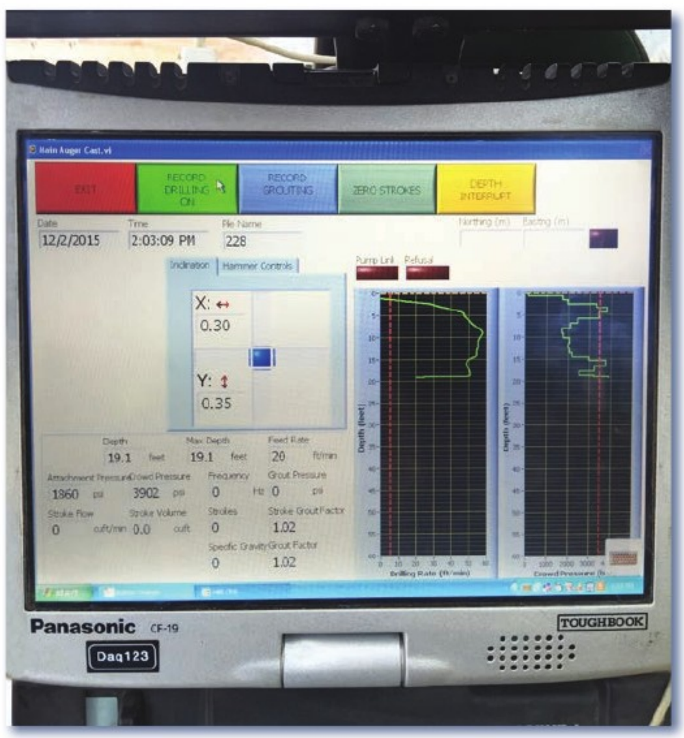



**Figure 3-29.** A Menard controlled-modulus-column production record (one "LOG CENTER" log per column), plotting the column profile, concrete pressure, drilling speed and rotation torque against depth and reporting the grout over-consumption ratio that is its signature quality metric. *reproduced from Menard CMC documentation (Menard 2022).*




*In-cab DAQ monitor displaying real-time quality control feedback during RI*

**Figure 3-30.** A Keller / Hayward Baker in-cab data-acquisition unit displaying real-time quality-control feedback during rigid-inclusion installation: depth, feed rate, crowd and grout pressure, stroke count and grout factor. *reproduced from Keller / Hayward Baker documentation (Blackburn and Franz 2018).*

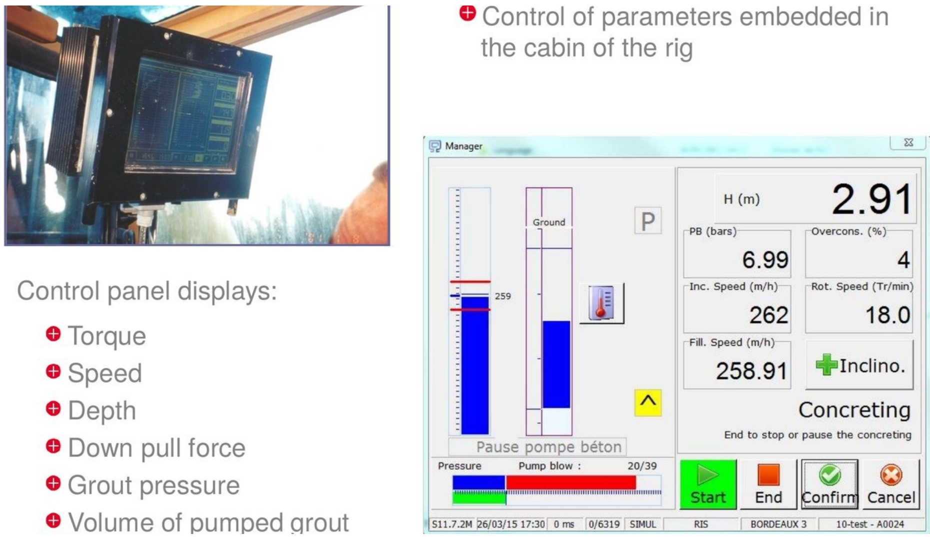



**Figure 3-31.** In-cab quality monitoring on a controlled-modulus-column rig: the control panel embedded in the cab (left) and the real-time acquisition display (right), which reports installed depth (H), grout pressure (PB), penetration, rotation and fill speeds, and grout over-consumption as each column is formed. *(Menard 2022)*

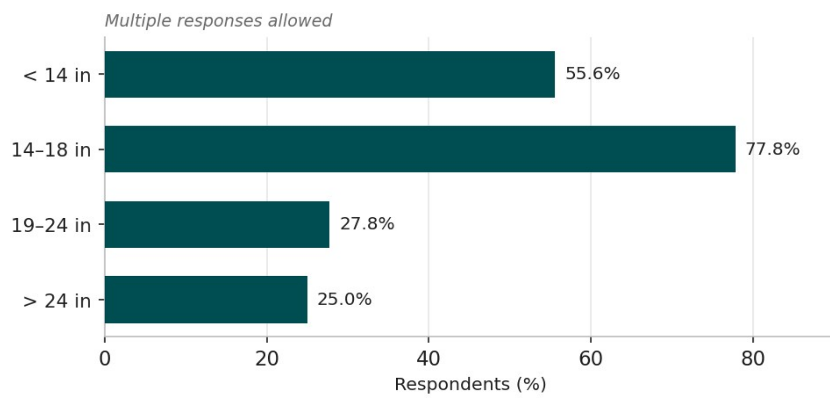



**Figure 3-32.** Common rigid-inclusion diameters installed in United States practice; diameters of 14–18 in and below 14 in dominate. Respondents could select more than one range. *redrawn from Han et al. (2025), Figure 3.9.*

### 3.5.4 Installation effects, specifications and contract practice

The very mechanism that gives displacement installation its advantages, the lateral movement of the soil, also produces effects that must be managed and recorded. Full-scale and numerical studies put the disturbance at two to four inclusion diameters (Suleiman et al. 2016; Samy et al. 2023), and lateral soil movements of several inches have been measured near test arrays (Lamb et al. 2022); the lateral displacement imposes bending moments on columns already in the ground, and unreinforced columns are prone to cracking (King et al. 2018), while the installation sequence governs the uplift and bending induced in neighbours, and a column cured for twenty-eight days suffers less tensile yielding than one cured for a single day (Nguyen et al. 2019; Larisch et al. 2015). Two consequences follow: the order of installation is itself a parameter worth recording, and production-time monitoring of the surrounding ground and structures becomes part of the installation quality record.

Construction specifications for rigid inclusions in the United States are still limited and have not been standardized, but several state transportation agencies have issued special provisions for individual projects. Table 3-11 sets the provisions of four agencies, Kansas, Minnesota, Iowa and Pennsylvania, side by side (Han et al. 2025). They agree on the essentials, that the columns must reach a competent bearing stratum, that the load-transfer platform is a compacted granular fill reinforced with geogrid, that acceptance rests on individual static load tests to about twice the design load, on low-strain integrity testing and on settlement monitoring, and that a global factor of safety of at least 1.5 applies, but they differ in the inclusion types they permit, the diameters and spacings they allow, the testing frequencies they require and the settlement limits they set. The ongoing NCHRP 10-121 project is expected to provide performance-based national guidelines, and the DFI two-part commentary (Smith et al. 2023a, 2023b) is the closest the United States has to a consensus statement of good practice. Consistent with this fragmentation, the survey shows that the governing specification is most often the owner's or agency's own provisions, that contracts are most often let design-build (Figures 3-34 and 3-35).

**Table 3-11.** Construction specifications for rigid inclusions issued as special provisions by four state transportation agencies (after Han et al. 2025, Tables 2.3 to 2.6).

| Provision | KDOT (Kansas) | MnDOT (Minnesota) | IaDOT (Iowa) | PennDOT (Pennsylvania) |
|---|---|---|---|---|
| **Inclusion types** | Auger-cast, steel / prestressed / cast-in-place concrete piles, grout (compaction) columns | CMC, CFA, VCC or DDC | Grout (CMC, APGD, ORTD) or concrete (VCC, GCC) | Drilled-displacement piles, CMC, DDP, VCC or CFA |
| **Diameter** | Per FHWA-NHI-16-028 | Minimum 12 in | Contractor's design | 12, 16 or 18 in (contractor) |
| **Spacing** | Per FHWA-NHI-16-028 | Maximum 10 ft | Contractor's design (load-test optimized) | Maximum 10 to 14 ft, square pattern |
| **Load testing** | Tied to performance (payment) | ≥ 2 static tests at 200% design (ASTM D1143) | Static load test on production columns | Static test on ≥ 1/250 columns at 200% design (ASTM D1143) |
| **Integrity testing** | No explicit provision | ≥ 2% of production, after ≥ 10 days (ASTM D5882) | Installation monitoring of each column | On ≥ 100 production columns |
| **Instrumentation** | Weekly settlement to 60 days; piezometers / inclinometers if required | Settlement plates, shape array, vibrating-wire piezometers | Torque, down-pressure, advance rate, grout pressure and volume | Vibration monitoring; strain gauges at top, tip and mid-length |
| **Settlement criteria** | Total ≤ 0.75 in (unit cell); ≤ 1.5 in per 100 ft; FoS ≥ 1.5 | Total ≤ 1.5 in; differential ≤ 1 in per 100 ft in 50 yr; FoS ≥ 1.5 | Contractor design to project criteria | Long-term settlement ≤ 1 in; FoS ≥ 1.5 |

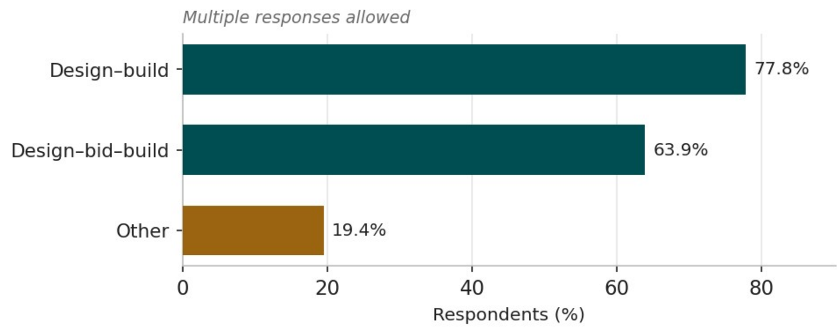



**Figure 3-34.** Common contract methods for rigid-inclusion projects in United States practice. Respondents could select more than one method. *redrawn from Han et al. (2025), Figure 3.23.*

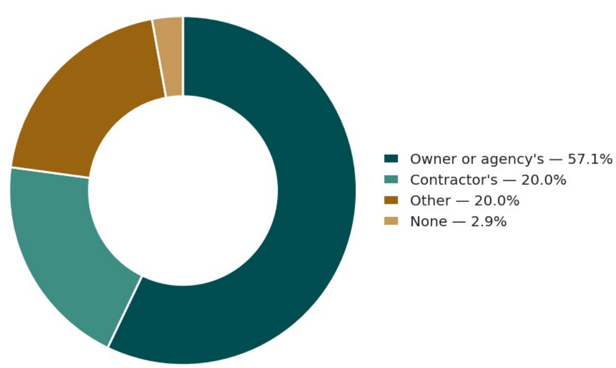



**Figure 3-35.** Common specifications or standards governing rigid-inclusion projects in United States practice. *redrawn from Han et al. (2025), Figure 3.24.*

### 3.5.5 Construction of the load-transfer platform

Once the inclusions are installed and their heads trimmed to the design cut-off level, the load-transfer platform is constructed over them. It is the element through which arching develops and load is shared between the inclusion heads and the soft soil, as Section 3.3.4 set out, so its construction is as consequential as that of the columns themselves. In United States practice the platform is most often a compacted granular fill, a gravel or graded-aggregate base (Figure 3-33), placed in controlled lifts over the trimmed heads; FHWA GEC-13 requires it to be a select structural fill with an effective friction angle of at least 35 degrees (FHWA GEC-13 2017). Where the working platform built to carry the installation plant survives in place, it commonly becomes the lowest layer of the load-transfer platform, so that its material, thickness and level pass directly into the platform record (Section 3.3).

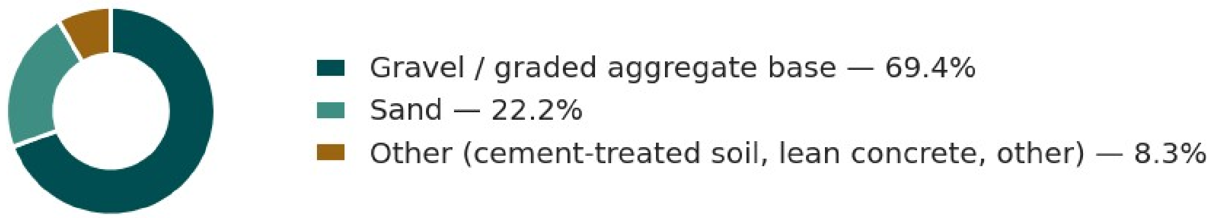



**Figure 3-33.** Common load-transfer-platform fill materials in United States practice; gravel or graded aggregate base dominates.
*redrawn from Han et al. (2025), Figure 3.19.*

Where the platform is reinforced, one or more layers of geosynthetic, most often a geogrid and less often a woven geotextile, are placed within the fill at the elevations and in the orientations the design specifies, with the principal reinforcement direction and the overlap or connection of adjacent rolls controlled so that the design tension can be developed across the full treated area (FHWA GEC-13 2017; van Eekelen 2015). The state transportation special provisions reviewed in Section 3.5.4 are consistent on this point, specifying a compacted granular fill reinforced with geogrid, placed and compacted in lifts to a target density (Han et al. 2025). A distinctive feature of contractor practice is that the specialty contractor frequently designs the load-transfer platform but does not construct it, the fill and reinforcement being placed by the earthworks contractor; the clear transfer of the platform specification between those parties is therefore itself part of the data a standardized record must carry (Blackburn and Franz 2018).

Construction quality is governed by the placement and compaction of the fill and the correct positioning of the reinforcement. The DFI Rigid Inclusion commentary on construction and verification treats the platform as a controlled structural element, calling for inspection of the fill placement, compaction and density testing at a frequency set by the supported foundation, and confirmation of the reinforcement type, strength and elevation before it is covered (Smith et al. 2023b). ASIRI similarly requires the platform to be verified for thickness, grading and compaction, including a plate-load (EV2) deformation modulus (ASIRI 2012). Because these checks are fixed at the design stage, the as-built platform record, its plan extent, thickness, material, compaction and reinforcement, closes forward into the post-construction verification of Section 3.6 and is precisely the body of platform data a standardized record must capture.

### 3.5.6 Geosynthetic reinforcement of the load-transfer platform

Where the load-transfer platform must span between widely spaced inclusion heads, or where the design relies on a tensile membrane to carry part of the applied load onto the heads, the granular platform is reinforced with one or more layers of geosynthetic, most often a biaxial or multi-axial geogrid and sometimes a high-strength geotextile, laid horizontally at or near the base of the platform (Figure 3-36). The reinforcement performs two related functions. In combination with soil arching it completes the transfer of load from the soft soil to the inclusion heads: the fraction of load not carried directly by arching is picked up by the reinforcement in tension and delivered to the heads, which reduces the vertical stress reaching the soft soil and limits both total and differential settlement at the surface. It also ties the heads together horizontally, resisting the lateral thrust generated at the base of an embankment and improving the stability of the side slopes.

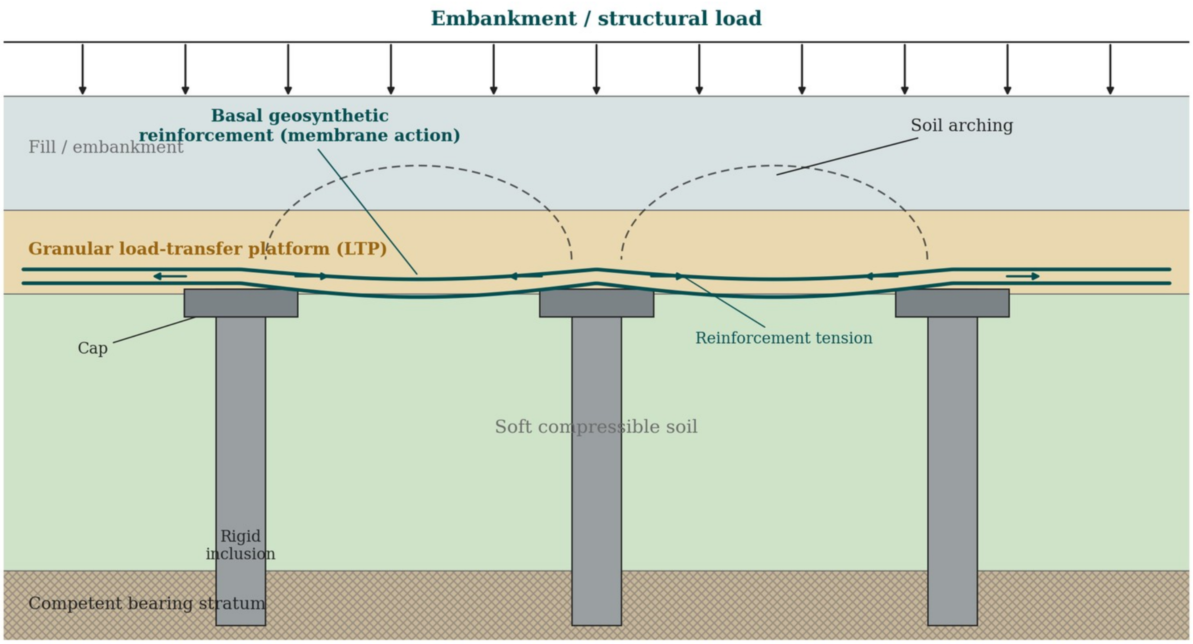



**Figure 3-36.** Geosynthetic reinforcement of the load-transfer platform. One or more layers of geogrid at the base of the granular platform carry, in tension, the part of the load not transferred to the inclusion heads by soil arching, and tie the heads together against lateral thrust. *(original)*

The four design standards reviewed in Section 3.4 treat this reinforcement differently. BS 8006-1 (2010) and EBGEO (2011) size the geosynthetic explicitly as a load-carrying membrane, combining the arching model with a tensioned-membrane calculation to obtain the required short-term and long-term tensile stiffness and strength, and they limit the strain in the reinforcement, typically to a few per cent, so that the platform deflection between heads stays within tolerance. ASIRI (2012) and FHWA GEC-13 (2017) place greater emphasis on the granular platform and the load-transfer mechanism itself, treating basal reinforcement as a complement rather than the primary load path, although both recognize its role in controlling differential settlement and edge stability. In all four the geosynthetic is a serviceability element as much as a strength element, and its stiffness, creep behaviour and connection detailing at the heads govern the deflection that ultimately reaches the structure (van Eekelen et al. 2015).

In United States practice the load-transfer platform is most often a compacted granular fill reinforced with geogrid, and the state transportation special provisions reviewed in Section 3.5.4 are consistent in specifying geogrid reinforcement within the platform (Han et al. 2025). The number of layers, their vertical spacing within the platform and their tensile stiffness are the principal variables and, because the reinforcement is placed within the platform lift sequence during construction, its specification, placement and overlap belong to the construction stage as much as to the design.

## 3.6 Post-Construction and Verification

The final phase asks whether the constructed system meets the design intent, and its parameters come from the acceptance ranges the design standards and codes publish (Table 3-9) and the special provisions of the owners (Table 3-11). Because those limits are fixed at the design stage, the post-installation record closes the loop back to the design intent, and the standardized set stores each acceptance criterion alongside the value it judges. Verification is organized into a small number of categories. The as-built survey records the position and verticality of every column against tolerance; integrity testing, by low-strain reflection or impedance and, increasingly, by thermal integrity profiling, checks the continuity and cross-section of the column; material-strength testing of cylinders or cores confirms the concrete or grout; and load testing, static maintained-load, plate and group or composite tests, taken in United States practice to prescribed multiples of the design load, confirms the load-settlement response. The load-transfer platform is verified for thickness, compaction and fill strength, and the system is monitored for settlement, differential settlement, pore-water pressure, load on heads and soil, reinforcement strain and lateral movement.

Practice weights these categories unevenly. As Figure 3-37 shows, load testing on individual columns is by far the most common verification method, used by more than eighty percent of respondents, followed by pile integrity testing at about fifty-six percent, with thermal integrity profiling, composite-ground load tests and tests on the surrounding soil used less often (Han et al. 2025). Instrumentation and monitoring, by contrast, are not routinely used: only a small fraction of respondents always instrument their projects, and a large fraction do so only sometimes or rarely (Figure 3-38). This pattern, heavy reliance on individual-column load testing and integrity testing and under-use of longer-term monitoring, guides which verification datasets a standardized record must treat as first-class, and it explains why documented case studies so often report predicted rather than measured settlement. Full-scale instrumented load tests, though comparatively rare, provide the most complete verification datasets and are the natural benchmark against which the standardized verification record should be shaped.

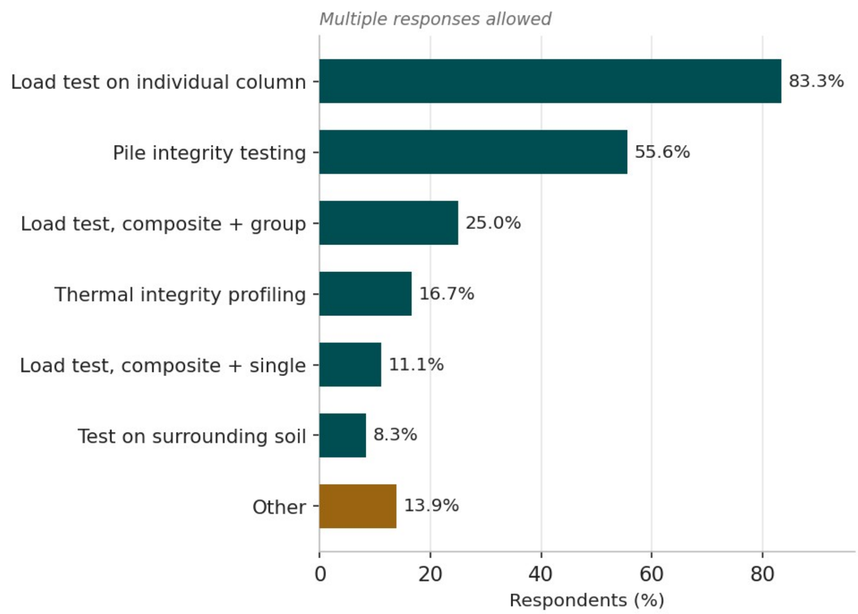



**Figure 3-37.** Methods used in practice to assess the quality and performance of rigid inclusions; individual-column load testing and pile integrity testing dominate. Respondents could select more than one method. *redrawn from Han et al. (2025), Figure 3.20.*

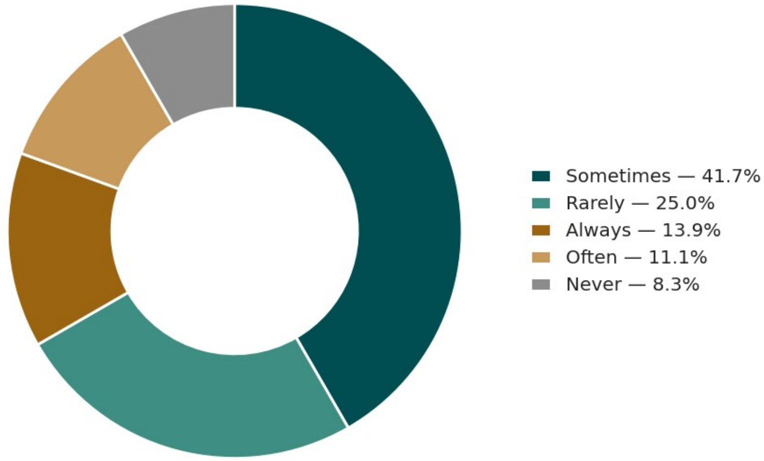



**Figure 3-38.** Frequency of instrumentation and monitoring of rigid-inclusion projects in United States practice; routine monitoring is uncommon. *redrawn from Han et al. (2025), Figure 3.21.*

Beyond the survey overview, the practice of an experienced specialty contractor illustrates how the quality and performance of installed rigid inclusions are verified in the field. Controlled-modulus-column programmes combine shaft control tests on the hardened columns with load-bearing verification. The shaft is checked for integrity by low-strain testing, for diameter by exhumation, and for material quality by compressive-strength and slump testing of the placed concrete (Figure 3-39). Load-bearing behaviour is verified by plate, or static, load tests carried to the design load or beyond, using a reaction-beam arrangement over a single column (Figure 3-40), and the measured load-settlement response is compared with the finite-element prediction used in design as a check on the assumed column and soil stiffness (Figure 3-41; Menard 2022).


**Figure 3-39.** Shaft control tests on installed controlled modulus columns: (from left) low-strain integrity testing, diameter verification of an exhumed column, compressive-strength testing of the placed concrete, and a concrete slump test. *(Menard 2022)*

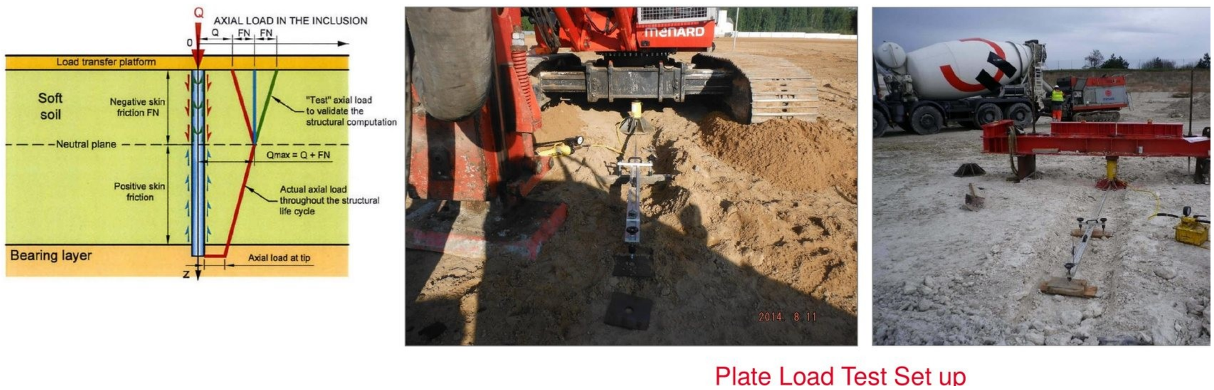



**Figure 3-40.** Plate load test on a single column to verify load-bearing behaviour: the axial load-transfer principle against which the test validates the design (left) and the reaction-beam test set-up in the field (centre and right). *(Menard 2022)*

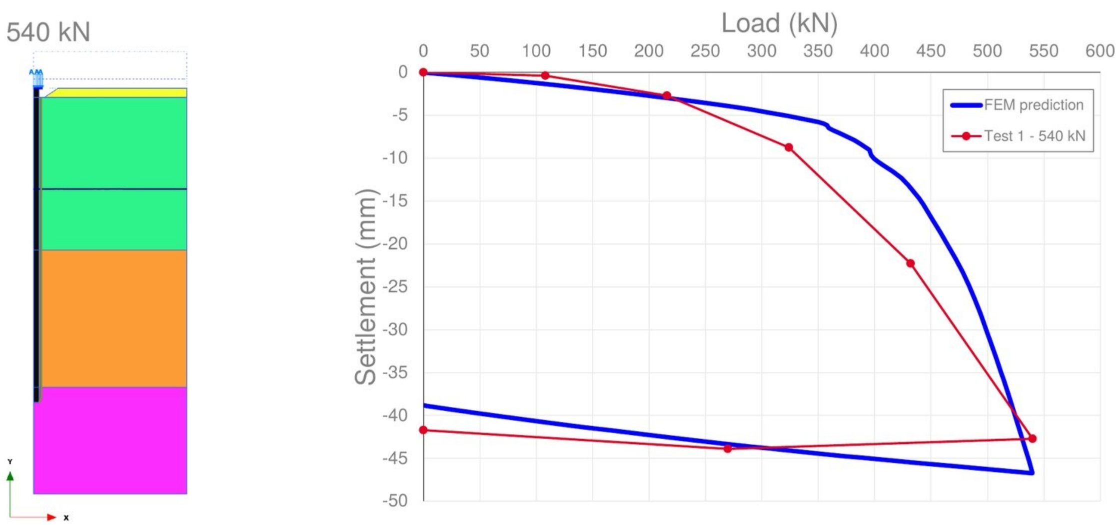



**Figure 3-41.** Measured load-settlement response from a plate load test carried to 540 kN compared with the finite-element (FEM) prediction used in design. *(Menard 2022)*

## 3.7 Summary

Rigid inclusions are a mature, widely used ground-improvement technology installed to control the settlement of soft, compressible soils, whose behaviour is governed by the composite action of stiff columns, soft soil and a load-transfer platform, and whose performance is sensitive to a small set of parameters. Followed through the life cycle of a project, the review shows a consistent picture: the geotechnical investigation must quantify the strength and compressibility of the soft ground with its provenance; the system is described by a compact set of physical components and a load-sharing mechanism common to all methods; the four design standards share that physical picture but implement it through different arching models, reinforcement assumptions, safety formats and acceptance limits, and demonstrably diverge when checked against measured behaviour; the specialty contractors build the same kind of column and record the same physical quantities but in incompatible proprietary formats; and the state-of-the-practice survey shows a wide range of named options for every project attribute, with drilled displacement columns, individual-column load testing and owner-written specifications dominant and routine monitoring uncommon. This literature review therefore establishes both the physical system to be described and the fragmentation, across standards and contractors, that motivates the standardized, method-agnostic and contractor-neutral data set developed in the chapters that follow.

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**CHAPTER 4**

# Essential Elements

*The methodology of Chapter 2 produced a standardized set of 235 rigid-inclusion parameters and mapped them onto the DIGGS schema. This chapter has two parts. It first explains how the two main physical elements of a rigid-inclusion system, the column and the load-transfer platform, are represented as objects in DIGGS, as sampling features, the constructed-ground family to which boreholes and soundings belong, and not as driven piles. It then presents the full set of essential data elements phase by phase, and for each phase it explains, group by group, what the essential elements are, why they matter to the rigid-inclusion system, and how they sit in DIGGS, closing with a summary table that gives, for every element, its symbol, its unit, the source that requires it, the stakeholder who uses it, and its DIGGS status. Throughout, the focus is the drilled displacement column. The element names used here follow the rigid-inclusion module being developed in the DIGGS 3.0 Extension, described in full in Chapter 6.*

## 4.1 The DIGGS Representation of a Rigid-Inclusion Element

Before enumerating the parameters, it is worth establishing how DIGGS represents a rigid-inclusion system, because the column and the load-transfer platform are built on the same pattern. The key point is what kind of thing a rigid inclusion is. It is not a foundation element supplied and driven from above; it is a column constructed in place, in the ground, by a displacement tool, and then surveyed, logged and load-tested. That is exactly what DIGGS already models as a sampling feature: a borehole, a sounding and a trial pit are all constructed linear features in the ground that carry a geometry and a set of observations. The extension therefore represents the column as a new linear sampling feature, RigidInclusion, deriving from AbstractLinearSamplingFeature, the same head as the borehole and the sounding; the load-transfer platform and the whole improved block are planar sampling features. Each inherits a stable set of properties from its head, an identifier and name, a project reference, spatial referencing and its geometry, and each is contained by the feature above it in the system: the columns and the geosynthetics sit within a LoadTransferPlatform, and the platforms sit within an RIFoundationSystem that represents the whole treated block. The schema describes the element and its construction, not the design calculation.

This is a displacement column and not a pile, and the distinction matters throughout. A pile is placed and its construction is a driving activity, with a hammer and a blow count; a rigid inclusion is constructed in place and its installation is captured as properties of the column itself, a construction method and a construction event, with no driving activity anywhere. The two sections that follow set out the column and the platform in turn; the remaining sections then return to the full, phase-by-phase list of essential parameters. The complete element model is given in Chapter 6.

## 4.2 The Rigid-Inclusion Column Element

The rigid-inclusion column is represented by a new RigidInclusion object that derives from the existing AbstractLinearSamplingFeature head, the same head from which the borehole and the sounding derive. Figure 4-1 shows the object model. From the sampling-feature head the column inherits its identity and, above all, its geometry: a referencePoint at the head, a centerLine running from the head to the toe, and a linear spatial reference system that lets every measurement be located by its depth from the head. The column is contained by the LoadTransferPlatform of its zone, which is in turn contained by the RIFoundationSystem for the whole improved block, so that the column, its platform and the block form one self-contained hierarchy.

To these inherited properties the RigidInclusion object adds the as-built attributes that describe the finished column: its as-built diameter, its cut-off and tip elevations and total length, its toe embedment into the bearing stratum, its as-built verticality (the tip deviation), the material and its characteristic strength and modulus, and a flag marking whether it is a production or a test column. Figure 4-2 illustrates these geometric quantities on a vertical section through the column.

**Figure 4-1 model – the rigid-inclusion column as a linear sampling feature**

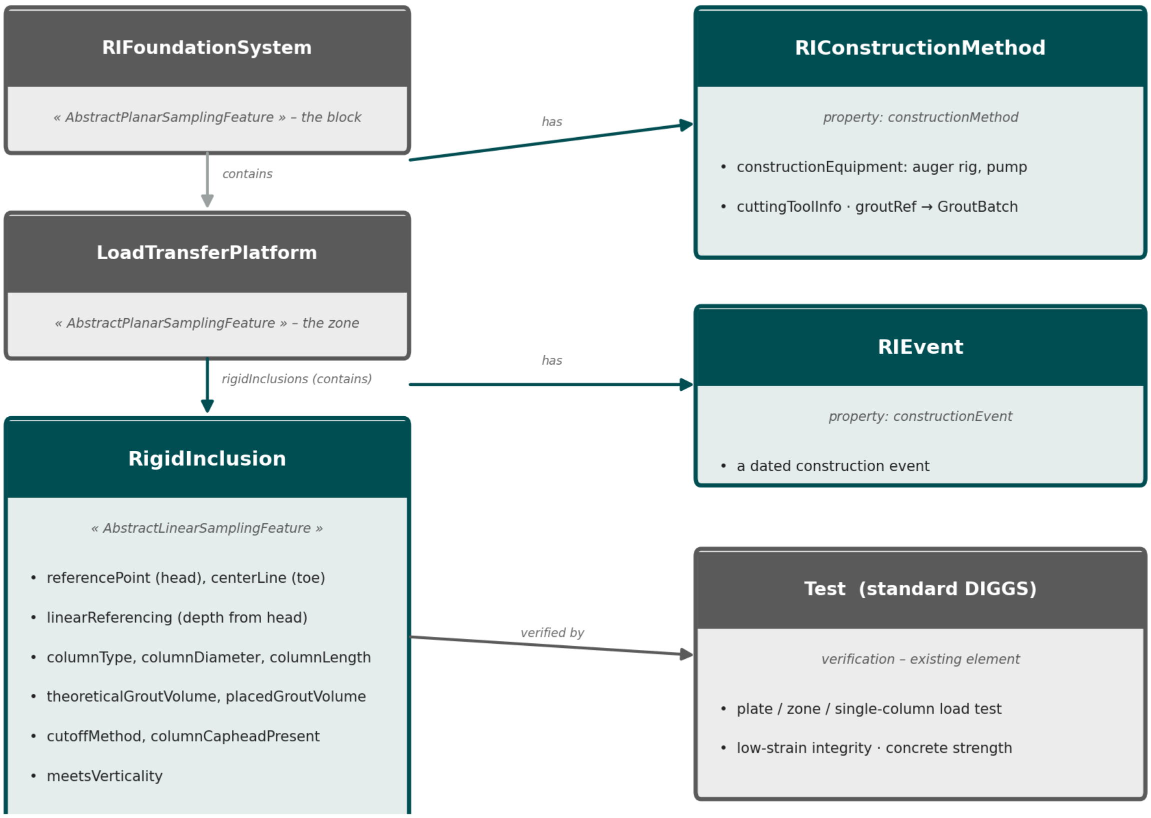


**Figure 4-1.** The rigid-inclusion column in DIGGS. The RigidInclusion object (teal) derives from the existing AbstractLinearSamplingFeature head (gray), the borehole and sounding head; it is contained by its LoadTransferPlatform within the RIFoundationSystem, carries its installation as a construction method and a construction event, and is verified by standard load and integrity tests that reference it. *(original, after the DIGGS 3.0 Extension rigid-inclusion model)*

**To be revised by Dan**

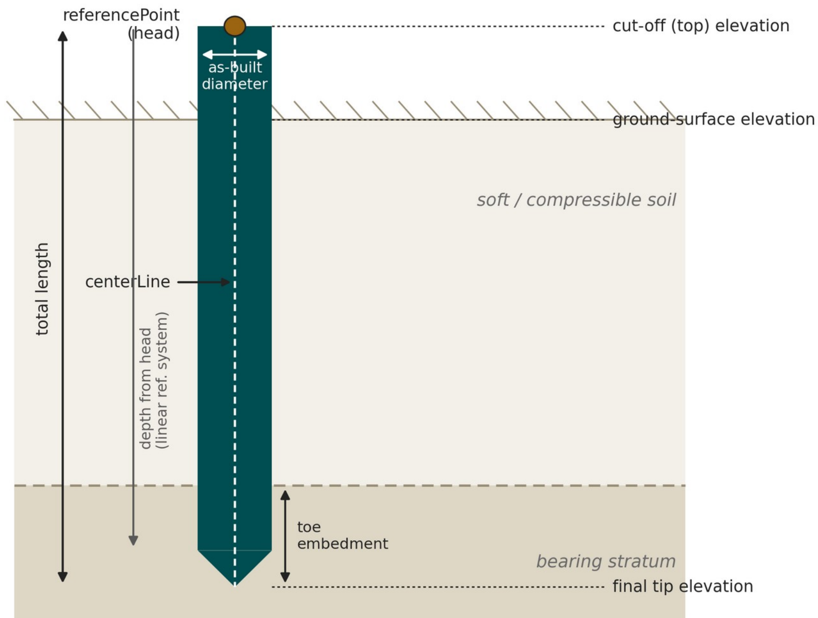


**Figure 4-2.** As-built geometry of the column, inherited from the sampling-feature head: a referencePoint at the head and a centerLine to the toe, with the cut-off, ground-surface and final-tip elevations, the total length, the toe embedment and the as-built diameter located by depth from the head. *(original)*

How the column was built is carried on the column itself, as two property objects rather than as a separate driving activity. An RIConstructionMethod, attached through the column's construction-method property, records the installation method, the equipment, the displacement auger rig and the concrete or mortar pump, the cutting tool, and a reference to the grout batch; because a whole zone is usually built by the same rig with the same method, one construction-method object is defined and every column references it. An RIEvent, attached through the column's construction-event property, records a dated event such as reaching the termination criterion. The as-built column then carries its own attributes directly: the column type, the as-built diameter and length, the theoretical and the placed grout volume, whose difference signals soft ground, the cut-off method, and the flags recording the caphead and the verticality. The load and integrity tests reference the column from the post-installation phase using the schema's existing Test element. A rigid-inclusion column therefore reuses the sampling-feature framework, the geometry, the referencing and the test links, and adds only the column-specific object with its attributes and its construction method and event.

## 4.3 The Load-Transfer-Platform Element

The load-transfer platform is areal rather than linear, and it is represented by a new LoadTransferPlatform object that derives from the AbstractPlanarSamplingFeature head, the planar counterpart of the head used for the column. Figure 4-3 shows the object model. From the planar head the platform inherits an areal footprint geometry, a surface rather than a line, and a referencePoint, and it is contained by the RIFoundationSystem for the improved block, in the same family of sampling features as the column. It is not merely contained by the block; it is itself the container for the columns of its zone, holding them and its geosynthetic layers directly.

The LoadTransferPlatform object adds the attributes that describe the layer: its areal extent, its total area, its fill described as a material, its top elevation and its implied thickness. The material attribute is what lets a single object cover the full range the platform takes in practice, from an unreinforced granular fill, through a granular fill carrying one or more layers of geogrid, to a soil-mixed or cemented raft or, in the most demanding case, a structural concrete slab; and because whether a platform is present at all is itself a design decision, the same object accommodates the uncommon case in which none is used. Because a platform commonly carries one or more geosynthetic layers, each layer is a Geosynthetic sub-object that records its type and its elevation within the platform, so that a platform with any number of layers, or none, is described without changing the structure. Its lowest layer is frequently the working platform placed to carry the installation plant, whose level is captured with the as-built record of the installation phase. Figure 4-4 shows the platform in cross-section, the fill and its geosynthetic layers spanning the column heads over the soft soil.

**Figure 4-3 model – the load-transfer platform and the features it contains**

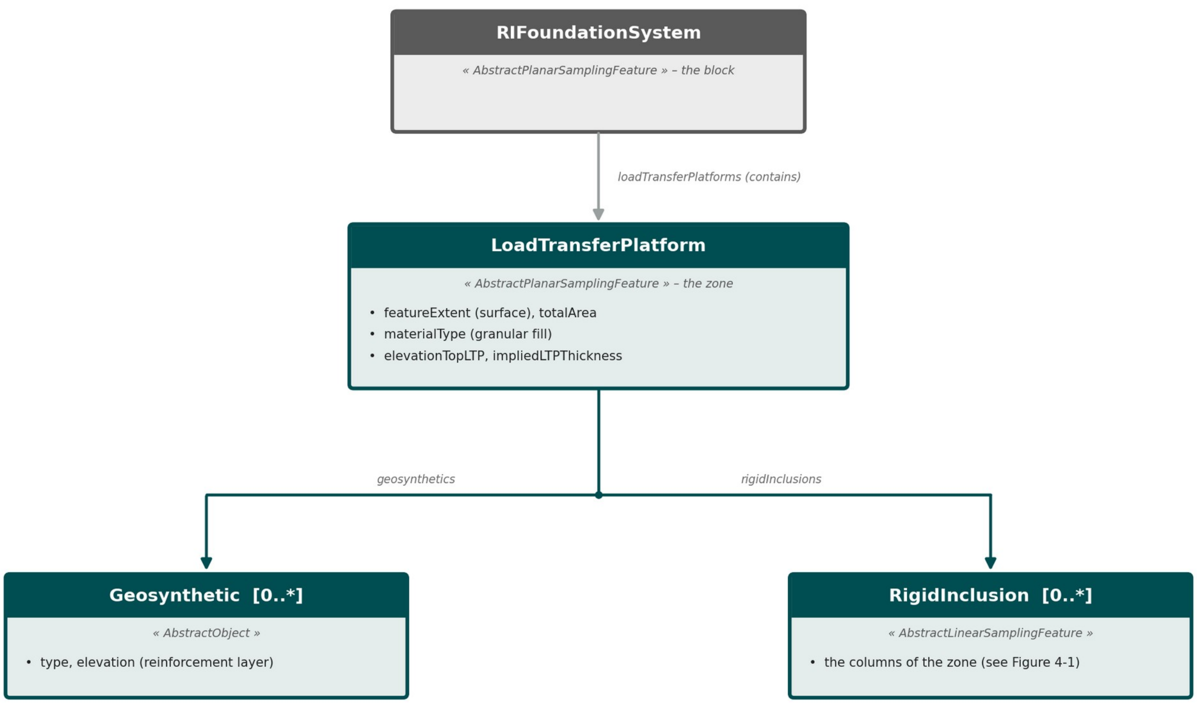


**Figure 4-3.** The load-transfer platform in DIGGS. The LoadTransferPlatform object (teal) derives from the existing AbstractPlanarSamplingFeature head (gray), is contained by the RIFoundationSystem, and is itself the container for its Geosynthetic layers and its RigidInclusion columns. *(original, after the DIGGS 3.0 Extension rigid-inclusion model)*

**To be revised by Dan**

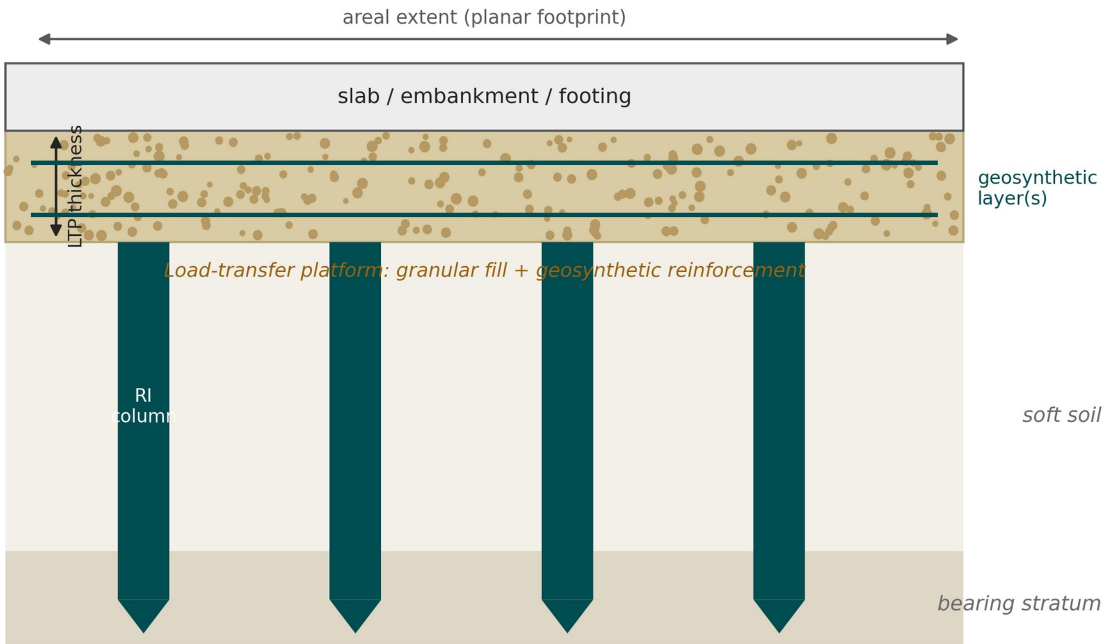


**Figure 4-4.** Cross-section of the load-transfer platform: a granular fill of defined thickness, carrying one or more geosynthetic layers, spanning the rigid-inclusion column heads over the soft soil across the planar footprint. *(original)*

The platform therefore mirrors the block above it and the columns below it: one sampling feature deriving from a planar head, with its own attributes, containing its geosynthetic layers and its columns. Its as-built thickness and compaction and its verified fill strength are recorded on the platform and its associated tests, and the whole improved block, its platforms and its columns form a single, navigable hierarchy of features under the RIFoundationSystem. The block's own planar geometry carries the support area, the plan extent actually treated with inclusions, recorded on the RIFoundationSystem in its own right because, as the practitioner interviews stressed, it is frequently made wider than the footprint of the structure or embankment above rather than sharing its outline. The platform differs from the column only in that its geometry is a surface rather than a line.

## 4.4 How the Essential Elements Are Presented

Each of the following four sections treats one project phase and ends with a summary table. The tables share a common format, and reading them requires three conventions.

**Required by (source).** This column names the standard or document that requires the element. For the design phase these are chiefly the four design standards, ASIRI, BS 8006-1, EBGEO and FHWA GEC No. 13, which is why many design rows read “all four standards” or name the subset that requires the parameter. For the other phases the requiring sources are the ground-investigation and testing standards (Eurocode 7 and the ASTM or ISO test methods), the execution standards (EN 12699 and EN 14679 for displacement and grouted elements), and the project-specific documents (the specification, the method statement and the inspection-and-test plan, ITP).

**Used by.** This column names the stakeholder who principally uses the element: the owner, the geotechnical engineer who characterizes the ground, the designer who sizes the system, the specialty contractor who installs it, and the quality-assurance or quality-control engineer (QA/QC) who verifies it. Most elements are produced by one party and consumed by another, and the column names the primary consumer in the rigid-inclusion workflow.

**DIGGS status.** This column records how the element maps onto the schema, in one of three states. “Existing” means a feature already defined in DIGGS provides the home, and the feature is named; the rigid-inclusion load and integrity tests fall here, because they are carried by the schema's standard Test element referencing the column. “Bridge” means an existing feature can host the element with a minor, backward-compatible addition, typically a soil value that is measured by an existing test feature but must be re-stated in a design context. “New” means no suitable feature exists and a rigid-inclusion element is required; the element is named together with the DIGGS abstract head it derives from. The new elements are those of the rigid-inclusion module developed in the DIGGS 3.0 Extension and described in Chapter 6: the RIFoundationSystem, LoadTransferPlatform and Geosynthetic that build the improved block; the RigidInclusion column with its RIConstructionMethod and RIEvent; and the staged design under RigidInclusionProgram, with its RIDesignSpec, ColumnSpec, RIPerformanceSpec and RISystemDesign. The element and head names are those of the 3.0 Extension rigid-inclusion model.

Across the four phases the 235 elements divide into 98 that an existing DIGGS feature already carries, 15 that bridge onto an existing feature, and 122 that require one of the rigid-inclusion elements. The balance shifts sharply from phase to phase, and each section states its own split.

## 4.5 Pre-installation Phase: the Ground Model

The pre-installation phase captures the geotechnical ground model against which the system is designed and its performance later judged. This is where DIGGS is strongest: of the forty-nine essential elements, forty-eight already have a home in the existing schema and only one is new, because site characterization, boreholes, soundings, sampling and the full suite of laboratory and in-situ tests, is precisely what DIGGS was first built to carry. The elements fall into eight groups.

**A, Project and investigation context.** The root records that bind the four phases into one dataset and locate the works: the project identity, the site location and coordinate reference system, the investigation programme and its governing standard, and each investigation point (borehole, sounding or trial pit) with its type, coordinates and date, together with the investigation density and depth and any survey of existing fill or obstructions. All reuse existing features, the Project and DocumentInformation records, the GML coordinate reference system, and the Borehole, Sounding and TrialPit features.

**B, Stratigraphy, profile and groundwater.** The soil and rock profile that defines the compressible zone to be reinforced and the bearing stratum for the toe: the lithological profile, the lateral variability of soft-layer thickness and depth-to-bearing across the footprint, the soil description and classification, the groundwater or piezometric level, and the founding-stratum level and description. These reuse the schema's LithologySystem, StratigraphySystem and GeoUnitSystem for the ground model and the Well, WaterLevelMonitoring and Monitor features for groundwater.

**C, Index and classification properties.** The laboratory index tests that classify the soft soil and feed the phase relations: natural water content, Atterberg limits, particle-size distribution and fines content, bulk and dry unit weight, specific gravity, initial void ratio and organic content. Each reuses its own established test feature, WaterContentTest, AtterbergLimitsTest, ParticleSizeTest, LabDensityTest, SpecificGravityTest and LossOnIgnitionTest, carried on the Sample.

**D, Strength properties.** The strengths that govern soft-soil bearing, negative skin friction, stability and end bearing: undrained shear strength, effective cohesion and friction angle, the constant-volume friction angle, sensitivity, the in-situ friction angle and relative density of the granular and bearing strata, and the at-rest earth-pressure coefficient. These reuse the TriaxialTest, DirectShearTest, InsituVaneTest and UnconfinedCompressiveStrengthTest features, and the cone and penetration soundings, with a few values derived from them.

**E, Compressibility and consolidation.** The oedometer parameters that quantify the settlement the rigid-inclusion system is meant to reduce: the compression, recompression and secondary-compression indices, the preconsolidation pressure and overconsolidation ratio, the coefficient of volume compressibility, the constrained modulus, and the vertical and horizontal coefficients of consolidation and permeability. All but the permeability route reuse the single ConsolidationTest feature, with pore-pressure dissipation and permeability tests carrying the rest.

**F, In-situ test data.** The soundings that profile strength and stiffness and set the anchorage target: the cone resistance, sleeve friction and pore pressure of the CPT and CPTu, the SPT blow count, the Ménard pressuremeter modulus and limit pressure, the field vane strength, and the shear-wave velocity and small-strain modulus. These reuse the Sounding feature with the StaticConePenetrationTest, the DrivenPenetrationTest (the schema's feature for the SPT), the PressuremeterTest, the InsituVaneTest and the GeophysicalFieldSurvey.

**G, Environmental and durability inputs.** The chemistry that sets the exposure class and the durability provisions for the concrete or grout: the sulfate, chloride and pH of soil and groundwater and the resulting aggressivity classification, carried on the schema's LabChemicalTest and environmental-screening features.

**H, Seismic and special inputs (conditional).** The seismic and liquefaction inputs used where they apply: the seismic site class or ground type and the liquefaction-susceptibility inputs, both reusing the geophysical survey and the penetration soundings, and the design ground acceleration, which is the one new element of this phase, and is in fact a design input carried with the design record. Table 4-1 summarizes the pre-installation elements.

**Table 4-1.** Essential elements of the pre-installation phase (49 elements; 48 carried on existing DIGGS features, 1 new).

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) To be revised by Dan |
|---|---|---|---|---|---|
| **A · Project & investigation context** | | | | | |
| **Project / job identification** | – | – | All four standards; EC7 | Owner; geotech engineer | Existing: Project |
| **Site location & coordinate system** | – | – | All four standards; EC7 | Owner; geotech engineer | Existing: Project + location (GML CRS / PointLocation) |
| **Investigation program & standard** | – | – | ASIRI; FHWA; EC7 | Owner; geotech engineer | Existing: DocumentInformation + SamplingActivity |
| **Investigation point (borehole / sounding / pit) ID, type, coordinates, date** | – | – | EC7; ASTM | Owner; geotech engineer | Existing: Borehole / Sounding / TrialPit |
| **Investigation density / depth** | – | m | ASIRI | Owner; geotech engineer | Existing: Borehole / Sounding (feature geometry) |
| **Existing fill / made-ground / obstruction survey** | – | m | ASIRI | Owner; geotech engineer | Existing: TrialPit + LithologySystem |
| **B · Stratigraphy, profile & groundwater** | | | | | |
| **Stratigraphic / lithological profile** | – | m | All four standards; EC7 | Designer | Existing: LithologySystem / StratigraphySystem / GeoUnitSystem |
| **Soft-layer thickness heterogeneity / depth-to-bearing variability** | – | m | ASIRI | Designer | Existing: GeoUnitSystem / StratigraphySystem |
| **Soil description & classification** | – | – | EC7; ASTM | Designer | Existing: LithologySystem (+ ConstituentSystem / ColorSystem) |
| **Groundwater / piezometric level** | – | m | All four standards | Designer | Existing: Well + WaterLevelMonitoring + Monitor |
| **Bearing (founding) stratum, level & description** | – | m | All four standards | Designer | Existing: GeoUnitSystem / StratigraphySystem |

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) To be revised by Dan |
|---|---|---|---|---|---|
| **C · Index & classification properties (laboratory)** | | | | | |
| **Natural water content** | w | % | EC7; ASTM | Designer; geotech engineer | Existing: WaterContentTest |
| **Atterberg limits (LL, PL, PI)** | $w_L$, $w_P$, $I_P$ | % | EC7; ASTM | Designer; geotech engineer | Existing: AtterbergLimitsTest |
| **Particle-size distribution / fines content** | – | % | ASTM | Designer; geotech engineer | Existing: ParticleSizeTest |
| **Bulk & dry unit weight** | $\gamma$, $\gamma_d$ | kN/m³ | All four standards; ASTM | Designer; geotech engineer | Existing: LabDensityTest |
| **Specific gravity of solids** | $G_s$ | – | ASTM | Designer; geotech engineer | Existing: SpecificGravityTest |
| **Initial void ratio** | $e_0$ | – | ASIRI | Designer; geotech engineer | Existing: LabDensityTest / SpecificGravityTest (derived) |
| **Organic / peat content** | – | % | ASIRI; ASTM | Designer; geotech engineer | Existing: LossOnIgnitionTest |
| **D · Strength properties** | | | | | |
| **Undrained shear strength** | $c_u$ ($s_u$) | kPa | All four standards | Designer | Existing: TriaxialTest / InsituVaneTest / UnconfinedCompressiveStrengthTest |
| **Effective strength parameters** | c', φ' | kPa, ° | All four standards | Designer | Existing: TriaxialTest / DirectShearTest |
| **Critical-state / constant-volume friction angle** | $\varphi'_{cv}$ | ° | BS 8006-1; FHWA | Designer | Existing: TriaxialTest / DirectShearTest (derived) |
| **Sensitivity** | $S_t$ | – | EC7 | Designer | Existing: InsituVaneTest / LabVaneTest |
| **In-situ friction angle & relative density of granular / bearing strata** | φ', $D_r$ | °, % | ASIRI; FHWA | Designer | Existing: StaticConePenetrationTest / DrivenPenetrationTest / RelativeDensityTest |
| **At-rest earth-pressure coefficient** | $K_0$ | – | ASIRI; EC7 | Designer | Existing: FlatPlateDilatometerTest / derived |
| **E · Compressibility & consolidation (oedometer)** | | | | | |
| **Compression index** | $C_c$ | – | ASIRI; FHWA; ASTM | Designer | Existing: ConsolidationTest |
| **Recompression / swelling index** | $C_s$ ($C_r$) | – | FHWA; ASTM | Designer | Existing: ConsolidationTest |
| **Secondary-compression (creep) index** | $C_\alpha$ | – | ASIRI; EC7 | Designer | Existing: ConsolidationTest |
| **Preconsolidation pressure** | $\sigma'_p$ | kPa | ASIRI; FHWA | Designer | Existing: ConsolidationTest |
| **Overconsolidation ratio** | OCR | – | ASIRI | Designer | Existing: ConsolidationTest (derived) |
| **Coefficient of volume compressibility** | $m_v$ | MPa⁻¹ (m²/MN) | FHWA; EC7 | Designer | Existing: ConsolidationTest |
| **Constrained / oedometric modulus** | $E_{oed}$ | MPa | ASIRI | Designer | Existing: ConsolidationTest |
| **Coefficient of consolidation** | $c_v$ | m²/yr | ASIRI; EC7 | Designer | Existing: ConsolidationTest |
| **Coefficient of horizontal consolidation** | $c_h$ ($k_h$) | m²/yr (m/s) | EC7 | Designer | Existing: PorePressureDissipationTest / ConsolidationTest |
| **Permeability** | k ($k_v$) | m/s | ASIRI | Designer | Existing: InsituPermeabilityTest / LabPermeabilityTest / PumpingTest / LugeonTest |
| **F · In-situ test data** | | | | | |
| **CPT/CPTu cone resistance** | $q_c$ ($q_t$) | MPa | ASIRI; ASTM; Project spec/ITP | Designer; geotech engineer | Existing: Sounding + StaticConePenetrationTest |
| **CPT sleeve friction & friction ratio** | $f_s$, $R_f$ | kPa, % | ASTM | Designer; geotech engineer | Existing: StaticConePenetrationTest |
| **CPTu pore pressure** | $u_2$ | kPa | ASTM | Designer; geotech engineer | Existing: StaticConePenetrationTest / PorePressureDissipationTest |
| **SPT blow count** | N ($N_{60}$) | blows/300 mm | EC7; ASTM | Designer; geotech engineer | Existing: Borehole + DrivenPenetrationTest |
| **Pressuremeter (Ménard) modulus** | $E_M$ | MPa | ASIRI; ISO | Designer; geotech engineer | Existing: PressuremeterTest |
| **Pressuremeter limit pressure** | $p_l$ ($p_l$*) | kPa | ASIRI | Designer; geotech engineer | Existing: PressuremeterTest |
| **Field vane shear strength** | $c_{u,vane}$ | kPa | ASIRI; ASTM | Designer; geotech engineer | Existing: InsituVaneTest |
| **Shear-wave velocity / small-** | $V_s$, $G_{max}$ | m/s, MPa | ASIRI | Designer; geotech | Existing: GeophysicalFieldSurvey / |

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) To be revised by Dan |
|---|---|---|---|---|---|
| **strain modulus [I]** | | | | engineer | LabVelocityTest |
| **G · Environmental / durability inputs** | | | | | |
| **Sulfate content (soil & groundwater)** | $SO_4^{2-}$ | mg/kg, mg/L | ASIRI; Project spec/ITP | Designer; owner | Existing: LabChemicalTest / EnvironmentalScreeningTest |
| **Chloride content** | $Cl^-$ | mg/kg, mg/L | ASIRI | Designer; owner | Existing: LabChemicalTest |
| **pH of soil / groundwater** | pH | – | ASIRI | Designer; owner | Existing: LabChemicalTest / RedoxTest |
| **Aggressivity / exposure classification** | – | – | ASIRI | Designer; owner | Existing: Derived (LabChemicalTest) + code list |
| **H · Seismic & special inputs (conditional)** | | | | | |
| **Seismic site class / ground type** | – | – | ASIRI | Designer | Existing: GeophysicalFieldSurvey (Vs30) + code list |
| **Design ground acceleration / hazard** | $a_g$ (PGA) | g | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (seismic design action) |
| **Liquefaction-susceptibility inputs** | – | – | ASIRI | Designer | Existing: StaticConePenetrationTest / DrivenPenetrationTest / ParticleSizeTest |

## 4.6 Design Phase: the Design Record

The design phase is where DIGGS is weakest and where the extension is largest. Of the ninety-seven essential elements, eighty-two require a new rigid-inclusion element and eleven more must be bridged onto an existing feature, because no feature in the current schema models a rigid-inclusion design. The proposed home is a new RigidInclusionDesign feature for the design quantities and a RigidInclusionSpecification feature for the tolerances and acceptance criteria the design fixes; only four elements reuse existing features. Because the standards disagree, the set is the union of what all four require, so a design reported in it is complete and comparable whichever method produced it. The elements fall into ten groups.

**A, Design basis and method.** The self-describing header of the design: the governing standard and edition, the design and arching method, the supported structure type, the rigid-inclusion type, the safety format, the design working life, the consequence or reliability class, the partial or global factor set, the parameter provenance and the method-applicability gate. These make a design reproducible; most are new on RigidInclusionDesign, while the parameter provenance links to the existing soundings and the contracting method reuses the Project and its business associates.

**B, Geometry and layout.** The geometry that fixes the composite system: the support (treated) area, which may extend beyond the loaded footprint, the grid pattern and centre-to-centre spacing, the inclusion diameter and length, the tributary diameter, the head or cap, the area replacement ratio, the critical full-arching height, the embankment batter, and the position and verticality tolerances. All are new, the geometry on RigidInclusionDesign and the tolerances on RigidInclusionSpecification.

**C, Inclusion, material and structural.** The inclusion as a structural element: its material and characteristic compressive strength (reusing the existing DesignGroutMix), its design strength and mean-stress limit, its elastic modulus (bridged from the mix to the design), its axial capacity and utilization, the buckling and shear checks, the column reinforcement where tension or bending occurs, and the lateral or bending demand. Apart from the material, these are new design quantities.

**D, Shaft, soil interface and neutral plane.** The load-transfer chain along the shaft, chiefly ASIRI's: the pressuremeter modulus as a design input (bridged from the PressuremeterTest), the shaft and tip load-transfer moduli, the negative skin-friction limit, the neutral-plane depth and the maximum axial force and downdrag at that plane. These are new, being design results rather than measurements.

**E, Toe and base.** The end bearing: the bearing-stratum level (bridged from the ground model), the toe and shaft resistances, the minimum embedment criterion, the limit tip resistance or limit pressure (bridged from the pressuremeter), and the design axial load per column. Mostly new design quantities.

**F, Load-transfer platform.** The platform and its arching, where the standards diverge most: whether an LTP is present, its type or material (granular, geogrid-reinforced, soil-mixed or cemented, or a concrete slab) and its thickness (a new LoadTransferPlatform feature), the fill friction angle and unit weight (bridged from the shear and density tests), and the method-specific arching quantities, the BS arching coefficient and cap-stress ratio, the Hewlett-Randolph efficiencies, the EBGEO critical stress ratio and the stresses on soil and element, the adapted-Terzaghi coefficient, and the ASIRI head-punching factor and head stress. The standard-specific quantities are all new.

**G, Geosynthetic reinforcement (conditional).** The basal reinforcement where provided: whether it is present and how many layers, the tensile stiffness, the long-term design strength and its reduction factors, the design tension, the anchorage length, the allowable and creep strains and the minimum load fraction. All are new.

**H, Soft soil, ground model and loading.** The design values of the soil and the loads: the design undrained and effective strengths, the compressibility parameters and the groundwater level, all bridged from the pre-installation tests but re-stated as design inputs, together with the subgrade-reaction modulus, the embankment height, the surcharge and the structure or slab load, which are new. This group shows the bridge concept most clearly: the value is measured by an existing test feature, but the design context that selects and factors it is new.

**I, Load transfer and verification.** The system checks at the ultimate limit state: the load efficacy and settlement efficacy, the active earth-pressure coefficient, the global-stability method and factor of safety, the lateral-spreading demand and factor of safety, the pile-group lateral extent and the seismic or liquefaction outcome. All are new design results.

**J, Serviceability and design-set acceptance.** The serviceability limits and the acceptance regime fixed at design: the total and differential settlement, the serviceability criterion, the reinforcement mid-span deflection and the durability provisions, and the load-test, integrity-test and monitoring regimes the design sets for later verification. The acceptance items are carried on RigidInclusionSpecification, and all are new. Table 4-2 summarizes the design elements.

**Table 4-2.** Essential elements of the design phase (97 elements; 4 existing, 11 bridged, 82 new). The union of the four standards' requirements; a dash in the symbol or unit column marks a coded or descriptive element.

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) To be revised by Dan |
|---|---|---|---|---|---|
| **A · Design basis & method (make the design self-describing)** | | | | | |
| **Governing design standard(s) & edition** | – | – | All four standards | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Design method (coded)** | – | – | All four standards | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Supported structure type (load source)** | – | – | ASIRI; FHWA; Han (2025) | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (structureType code) |
| **RI type / installation-technique family (coded)** | – | – | FHWA; Han (2025) | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (rigidInclusionType code) |
| **Project delivery / contracting method** | – | – | Han (2025) | Designer | Existing: Project + BusinessAssociate (role) |
| **Arching model (coded)** | – | – | All four standards | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Safety format / design approach** | – | – | All four standards | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Design working life** | $t_L$ | yr | BS 8006-1; EBGEO; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Consequence / reliability class** | – | – | ASIRI; BS 8006-1; | Designer | New: RIDesignSpec / RISystemDesign → |

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) To be revised by Dan |
|---|---|---|---|---|---|
| | | | EBGEO | | RigidInclusionProgram |
| **Partial / global factor set** | γ, f | – | All four standards | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Parameter provenance (PMT / CPT / SPT / lab)** | – | – | ASIRI; EBGEO; FHWA | Designer | Existing: samplingFeatureRef → Sounding/Borehole + test procedures |
| **Method-applicability gate & geometry envelope** | $k_{s,T}/k_s$ | – | EBGEO | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **B · Geometry & layout** | | | | | |
| **Support (treated) area** | $A_{RI}$ | m² | ASIRI; EBGEO; FHWA | Designer; Contractor | New: RIFoundationSystem → AbstractPlanarSamplingFeature (featureExtent) |
| **Grid pattern (square / triangular)** | – | – | ASIRI; EBGEO; FHWA | Designer; contractor | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (layout) |
| **Spacing (centre-to-centre)** | s ($s_x$, $s_y$) | m | BS 8006-1; FHWA | Designer; contractor | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (geometry) |
| **Inclusion diameter** | D (d) | m | ASIRI; EBGEO; FHWA | Designer; contractor | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (geometry) |
| **Inclusion length / toe depth** | L | m | ASIRI | Designer; contractor | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (geometry) |
| **Equivalent / tributary diameter** | $d_{Ers}$, $D_e$ | m | EBGEO; FHWA | Designer; contractor | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (derived geometry) |
| **Head / cap present & size** | a | m | BS 8006-1; EBGEO; FHWA | Designer; contractor | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Area replacement ratio** | α ($a_s$) | % | ASIRI; EBGEO; FHWA | Designer; contractor | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (derived) |
| **Critical (full-arching) height** | $H_{crit}$ | m | BS 8006-1; EBGEO; FHWA | Designer; contractor | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (derived) |
| **Embankment side-slope / batter** | n (β) | – (or °) | BS 8006-1; FHWA | Designer; contractor | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (loading geometry) |
| **Position tolerance** | – | mm | ASIRI | Designer; contractor | New: RIPerformanceSpec / RIDesignSpec → RigidInclusionProgram |
| **Verticality / inclination tolerance** | – | % | ASIRI | Designer; contractor | New: RIPerformanceSpec / RIDesignSpec → RigidInclusionProgram |
| **C · Inclusion, material & structural** | | | | | |
| **Material (concrete / grout / mortar)** | – | – | ASIRI; EBGEO; FHWA | Designer | Existing: DesignGroutMix |
| **Characteristic compressive strength** | $f_{ck}$ ($f_{ck}$*) | MPa | ASIRI; FHWA | Designer | Existing: DesignGroutMix |
| **Design compressive strength** | $f_{cd}$ | MPa | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (structural) |
| **Mean-stress limit (fixed cap)** | $\bar{\sigma}_{lim}$ | MPa | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (structural limit) |
| **Elastic modulus** | E | GPa | ASIRI; FHWA | Designer | Bridge: DesignGroutMix / new RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Column axial capacity (structural)** | $R_c$ | kN | ASIRI; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (structural) |
| **Column axial utilization** | $N/R_c$ | – | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (structural check) |
| **Buckling check (soft-soil)** | – | – | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (structural check) |
| **Shear check** | $\tau_{cp}$ | MPa | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (structural check) |
| **Column reinforcement (Domain 1 / tension zones)** | $\rho_s$, $A_s$, $L_{reinf}$ | %, mm², m | ASIRI; Han (2025) | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (reinforcement) |
| **Lateral / bending demand on RI (eccentric · lateral · seismic)** | M, $H_{lat}$ | kN·m, kN | ASIRI; Han (2025) | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (structural check) |
| **D · Shaft, soil interface & neutral plane** | | | | | |
| **Pressuremeter (Ménard) modulus (design input)** | $E_M$ | MPa | ASIRI | Designer | Bridge: PressuremeterTest (data) → new RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Shaft load-transfer modulus** | $k_\tau$ | kN/m³ | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (t–z) |
| **Tip load-transfer modulus** | $k_q$ | kN/m³ | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → |

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) To be revised by Dan |
|---|---|---|---|---|---|
| | | | | | RigidInclusionProgram (t–z) |
| **Negative skin-friction limit** | $q_s$ | kPa | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Neutral-plane depth** | $h_c$ ($z_n$) | m | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Max axial force / downdrag at neutral plane** | $Q_{max}$ | kN | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **E · Toe / base (end bearing)** | | | | | |
| **Bearing-stratum level** | – | m | ASIRI; FHWA | Designer | Bridge: GeoUnitSystem (data) → new RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Toe / base resistance** | $R_b$ | kN | All four standards | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Shaft (positive) resistance** | $R_s$ | kN | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Minimum bearing-stratum embedment (design criterion)** | – | × D / m | ASIRI; Han (2025) | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (criterion) |
| **Limit unit tip resistance / limit pressure** | $q_p$, $p_l$ | kPa | ASIRI | Designer | Bridge: PressuremeterTest (data) → new RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Design axial load per column (demand)** | $Q_r$ (N, $Q_p(0)$) | kN | ASIRI; EBGEO; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (demand) |
| **F · Load-transfer platform (LTP)** | | | | | |
| **LTP present** | – | – | ASIRI; EBGEO; FHWA | Designer | New: LoadTransferPlatform → AbstractPlanarSamplingFeature |
| **LTP type / material** | – | – | ASIRI; EBGEO; FHWA | Designer | New: LoadTransferPlatform → AbstractPlanarSamplingFeature (materialType) |
| **LTP thickness** | $H_M$ (h, h*) | m | ASIRI; EBGEO; FHWA | Designer | New: LoadTransferPlatform → AbstractPlanarSamplingFeature |
| **LTP fill friction angle** | φ' | ° | All four standards | Designer | Bridge: DirectShearTest / TriaxialTest (data) → new RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **LTP fill unit weight** | γ | kN/m³ | BS 8006-1; FHWA | Designer | Bridge: LabDensityTest (data) → new RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Arching coefficient (BS)** | $C_{c,arch}$ | – | BS 8006-1 | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (arching) |
| **Cap-stress ratio (BS)** | $p'_c/\sigma'_v$ | – | BS 8006-1 | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (arching) |
| **Hewlett–Randolph arching efficiency & passive coefficient (BS)** | $E_{crown}$, $E_{cap}$, $E_{min}$, $K_p$ | – | BS 8006-1; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (arching) |
| **Critical stress ratio (EBGEO)** | $K_{crit}$ | – | EBGEO | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (arching) |
| **Stress on soft soil between elements** | $\sigma_{zo,k}$ | kPa | EBGEO | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (arching) |
| **Stress on element (head)** | $\sigma_{zs,k}$ | kPa | EBGEO | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (arching) |
| **Adapted-Terzaghi earth-pressure coefficient** | K | – | FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (arching) |
| **Head-punching bearing factor (ASIRI)** | $N_q$ | – | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (arching) |
| **Head stress on inclusion** | $q_p^+$ | kPa | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (arching) |
| **G · Geosynthetic reinforcement (conditional)** | | | | | |
| **Geosynthetic present** | – | – | All four standards | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (reinforcement) |
| **Number of reinforcement layers** | $n_{geo}$ | – | EBGEO; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (reinforcement) |
| **Tensile stiffness** | J ($J_k$) | kN/m | BS 8006-1; EBGEO; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (reinforcement); cf. no geosynthetic feature |
| **Long-term design strength** | $T_D$ ($R_{B,d}$) | kN/m | BS 8006-1; EBGEO; | Designer | New: RIDesignSpec / RISystemDesign → |

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) To be revised by Dan |
|---|---|---|---|---|---|
| | | | FHWA | | RigidInclusionProgram (reinforcement) |
| **Reduction factors (creep, damage, durability)** | A1–A5 / $f_m$ | – | BS 8006-1; EBGEO | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (reinforcement) |
| **Reinforcement design tension** | $T_{rp}$ · $E_{M,geo}$ · T | kN/m | BS 8006-1; EBGEO; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (reinforcement) |
| **Reinforcement anchorage / bond length** | $L_e$, $L_b$ ($L_A$) | m | BS 8006-1; EBGEO; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (reinforcement) |
| **Allowable / design strain** | ε | % | BS 8006-1; EBGEO; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (reinforcement) |
| **Creep-strain limit over design life** | $\Delta\varepsilon_{kr}$ | % | BS 8006-1; EBGEO | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (reinforcement) |
| **Minimum reinforcement load fraction** | – | % | BS 8006-1 | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (reinforcement) |
| **H · Soft soil, ground model & loading** | | | | | |
| **Undrained shear strength (design)** | $c_u$ | kPa | ASIRI; FHWA | Designer | Bridge: TriaxialTest/InsituVaneTest (data) → new RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Effective strength (design)** | c', φ' | kPa, ° | ASIRI; BS 8006-1 | Designer | Bridge: TriaxialTest/DirectShearTest (data) → new RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Compressibility parameters (design)** | $E_{oed}$, $C_c$, $C_s$, $\sigma'_p$, OCR | – | ASIRI; FHWA | Designer | Bridge: ConsolidationTest (data) → new RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Subgrade reaction modulus (Winkler)** | $k_s$ | kN/m³ | EBGEO | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (derived) |
| **Groundwater level (design)** | – | m | ASIRI | Designer | Bridge: Well/WaterLevelMonitoring (data) → new RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Embankment height** | H | m | BS 8006-1; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (loading) |
| **Fill / embankment unit weight** | γ | kN/m³ | BS 8006-1 | Designer | Bridge: LabDensityTest (data) → new RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Surcharge / live load** | $w_s$ (q, $p_k$) | kPa | BS 8006-1; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (loading action) |
| **Structure / slab load** | q | kPa | ASIRI; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (loading action) |
| **I · Load transfer & verification (ULS / stability)** | | | | | |
| **Load (stress) efficacy** | E | – | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (verification) |
| **Settlement efficacy / settlement-reduction ratio** | $E_{tass}$ | – | ASIRI | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (verification) |
| **Active earth-pressure coefficient (fill)** | $K_a$ | – | BS 8006-1; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (verification) |
| **Global-stability analysis method (coded)** | – | – | All four standards; Han (2025) | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (stability method) |
| **Global / slope stability factor** | FS | – | BS 8006-1; EBGEO; FHWA; Han (2025) | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (verification) |
| **Lateral spreading demand** | $T_{ds}$ / $P_{Lat}$ | kN/m | BS 8006-1; EBGEO; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (verification) |
| **Lateral-spreading factor of safety** | $FS_{ls}$ | – | FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (verification) |
| **Pile-group lateral extent** | $L_p$ | m | BS 8006-1; FHWA | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (verification) |
| **Seismic / liquefaction outcome** | – | – | ASIRI; Han (2025) | Designer | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (verification) |
| **J · Serviceability & design-set acceptance** | | | | | |
| **Total settlement** | $s_{tot}$ | mm | ASIRI; FHWA | Designer; QA/QC | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (SLS) |
| **Differential settlement** | Δs | mm | ASIRI; FHWA | Designer; QA/QC | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (SLS) |
| **Serviceability criterion** | – | – | ASIRI; BS 8006-1; Han | Designer; QA/QC | New: RIPerformanceSpec / RIDesignSpec → |

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) To be revised by Dan |
|---|---|---|---|---|---|
| | | | (2025) | | RigidInclusionProgram (acceptance) |
| **Reinforcement mid-span deflection** | y | mm | BS 8006-1 | Designer; QA/QC | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram (SLS) |
| **Durability provisions** | – | – | ASIRI | Designer; QA/QC | New: RIDesignSpec / RISystemDesign → RigidInclusionProgram / DesignGroutMix |
| **Load-test regime & acceptance (set at design)** | – | – | ASIRI; FHWA | Designer; QA/QC | New: RIPerformanceSpec / RIDesignSpec → RigidInclusionProgram → Post load tests |
| **Integrity-testing regime** | – | – | ASIRI | Designer; QA/QC | New: RIPerformanceSpec / RIDesignSpec → RigidInclusionProgram → Post integrity |
| **Monitoring / instrumentation plan** | – | – | ASIRI; EBGEO; FHWA; Han (2025) | Designer; QA/QC | New: RIPerformanceSpec / RIDesignSpec → RigidInclusionProgram → Monitor |

## 4.7 Installation Phase: the As-built Production Record

The installation phase records what was actually built, column by column. Here DIGGS is mixed: of the fifty-two essential elements, seventeen reuse existing features, one is bridged and thirty-four are new. The physical inclusion and its construction record are new (a RigidInclusion feature for the column, with an RIConstructionMethod for the method and equipment and an RIEvent for the dated construction events), but the fresh concrete or grout and its tests reuse the schema's existing grout-batch and material features almost in full. The elements fall into seven groups.

**A, Column identity and installation context.** What, where, when and by what the column was built: the column identifier (new), the project, the rig or machine, the operator or crew and the tool diameter (all reusing the Project, Equipment and business-associate features), and the installation date, method and sequence (new on the installation activity).

**B, As-built geometry.** The geometry as constructed: the as-built diameter and length, the maximum depth reached, the working-platform reference level (the level of the granular or cemented mat built to carry the plant and retained as the lowest layer of the load-transfer platform, which sets the datum for every recorded depth), the working-platform material and thickness, the cut-off (top) and toe elevations, the as-built head position (reusing PointLocation) and the as-built verticality. Apart from the head position these are new on the RigidInclusion and its record.

**C, Depth-indexed process record.** The heart of the installation record: the penetration and extraction rate, the rotation speed and torque, the crowd or down-thrust, the grout or concrete pressure and volume, the pump strokes and the volume over-consumption, logged against depth and carried with the column's construction method and events, together with the adjacent-element or ground-movement check that reuses the Monitor feature. This is the depth-indexed channel set that every displacement contractor records and that the standardized record captures once.

**D, Concrete or grout as-placed.** The material actually placed, where DIGGS grouting support is strong: the total volume placed, the grout-level monitoring of uncovered columns and the wet-squeeze or grout-return volume that flag grout drop or necking, the mix and strength class, the cement type, the fresh slump, the transit time, the fresh grout density or flow, the temperature and air content, the batch-ticket data, the cylinders cast and the pump and plant calibration. Almost all of the material tests reuse existing features, DesignGroutMix, SlumpTest, GroutBatch, the mud-balance, flow-cone and Marsh-funnel tests, the grout pump and plant, and the Sample and grout specimen; the total placed volume is bridged to the installation activity, and the grout-level and grout-return checks are new on the construction record.

**E, Termination and anchorage.** How the drilling was ended and the founding confirmed, beginning with the indicator or pre-production programme that correlates the drilling response to the stratigraphy and establishes the termination criteria: the coded refusal or anchorage criterion, the achieved termination outcome, the as-built toe embedment into the competent stratum, and the drilling time in the compressible versus the competent soil. All are new on the installation activity, because they describe a rigid-inclusion-specific decision.

**F, Reinforcement (conditional).** Where the column is reinforced: whether reinforcement is present and of what type, and its length and section, both new attributes of the RigidInclusion.

**G, Durations and production.** The production record: the drilling and concreting durations, the remarks and downtime, and the attached native rig-log file, which reuses the AssociatedFile feature while the durations are new. Table 4-3 summarizes the installation elements.

**Table 4-3.** Essential elements of the installation phase (52 elements; 17 existing, 1 bridged, 34 new). Focused on the drilled displacement column.

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) To be revised by Dan |
|---|---|---|---|---|---|
| **A · Column identity & installation context** | | | | | |
| **Column / element ID** | – | – | EN 12699/14679; Project spec/ITP | Contractor | New: RigidInclusion → AbstractLinearSamplingFeature (gml:name) |
| **Project / contract** | – | – | Project spec/ITP | Contractor | Existing: Project + projectRef |
| **Rig / machine ID** | – | – | Project spec/ITP | Contractor | Existing: Equipment / DrillRig (constructionEquipment) |
| **Operator / crew** | – | – | Project spec/ITP | Contractor | Existing: BusinessAssociate (role) |
| **Installation date & start/end time** | – | date-time | EN 12699/14679; Project spec/ITP | Contractor | New: RIConstructionMethod (constructionMethod of RigidInclusion) |
| **Installation method (coded)** | – | – | ASIRI; EN 12699/14679; Project spec/ITP | Contractor | New: RIConstructionMethod (constructionMethod of RigidInclusion) (method code) |
| **Installation sequence / order** | – | – | EN 12699/14679; Han (2025) | Contractor | New: RIConstructionMethod (constructionMethod of RigidInclusion) |
| **Tool / auger diameter** | – | mm | Project spec/ITP | Contractor | Existing: Equipment (tool) |
| **B · As-built geometry** | | | | | |
| **As-built diameter** | D | mm | ASIRI; Project spec/ITP | Contractor; QA/QC | New: RigidInclusion → AbstractLinearSamplingFeature |
| **As-built length** | L | m | EN 12699/14679; Project spec/ITP | Contractor; QA/QC | New: RigidInclusion → AbstractLinearSamplingFeature |
| **Maximum depth reached** | $z_{max}$ | m | Project spec/ITP | Contractor; QA/QC | New: RigidInclusion → AbstractLinearSamplingFeature / Require new: RIEvent / installation record (on RigidInclusion) |
| **Working-platform / rig-reference level (depth datum)** | – | m | EBGEO; Project spec/ITP | Contractor; QA/QC | New: RIConstructionMethod (constructionMethod of RigidInclusion) (datum) |
| **Working-platform material** | – | – | EBGEO; Project spec/ITP | Contractor; QA/QC | New: RIConstructionMethod (constructionMethod of RigidInclusion) (working platform) |
| **Working-platform thickness** | $t_{wp}$ | m | EBGEO; Project spec/ITP | Contractor; QA/QC | New: RIConstructionMethod (constructionMethod of RigidInclusion) (working platform) |
| **Top (cut-off) elevation** | – | m | ASIRI; Project spec/ITP | Contractor; QA/QC | New: RigidInclusion → AbstractLinearSamplingFeature (geometry) |
| **Toe elevation** | – | m | ASIRI; FHWA | Contractor; QA/QC | New: RigidInclusion → AbstractLinearSamplingFeature (geometry) |
| **As-built head position (x, y)** | – | m | ASIRI | Contractor; QA/QC | Existing: PointLocation (on RigidInclusion) |
| **Verticality / inclination (as-built)** | – | ° or % | ASIRI | Contractor; QA/QC | New: RIEvent / installation record (on RigidInclusion) |
| **C · Depth-indexed process record (the coverage)** | | | | | |
| **Penetration rate vs depth** | $v_{pen}$ | m/h (cm/min) | Project spec/ITP | Contractor | New: RIEvent / installation record (on RigidInclusion) (cf. PileDrivingRecord / MeasurementWhileDrilling) |
| **Extraction rate vs depth** | $v_{ext}$ | m/h | Project spec/ITP | Contractor | New: RIEvent / installation record (on RigidInclusion) |
| **Rotation speed** | $n_{rot}$ | rev/min | Menard panel (GIW p.41) | Contractor | New: RIEvent / installation record (on RigidInclusion) |

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) To be revised by Dan |
|---|---|---|---|---|---|
| **Rotation torque vs depth** | C ($M_t$) | kN·m (t·m) | Menard anchorage V<250 m/h & C>5 t·m/0.5 | Contractor | New: RIEvent / installation record (on RigidInclusion) (cf. MeasurementWhileDrilling) |
| **Crowd / down-thrust force** | $F_{crowd}$ | kN (bar) | HBI DAQ (crowd/torque); Menard down-pull | Contractor | New: RIEvent / installation record (on RigidInclusion) |
| **Grout / concrete pressure vs depth** | $P_B$ | bar (kPa) | Menard CMC brochure (controlled pressure | Contractor | New: RIEvent / installation record (on RigidInclusion) |
| **Grout / concrete volume vs depth** | V | m³ | HBI neat & actual grout volume | Contractor | New: RIEvent / installation record (on RigidInclusion) |
| **Pump strokes** | – | count | HBI pump stroke count; Menard pump blows | Contractor | New: RIEvent / installation record (on RigidInclusion) |
| **Volume overconsumption** | – | % | Menard CMC record (GIW p.42) | Contractor | New: RIEvent / installation record (on RigidInclusion) (derived) |
| **Adjacent-element / ground-movement check during install** | – | mm / obs | Keller HBI QC (monitor adjacent location | Contractor | Existing: Monitor + MonitorResult |
| **D · Concrete / grout as-placed** | | | | | |
| **Total concrete / grout volume placed** | $V_{tot}$ | m³ | EN 12699/14679; Project spec/ITP | Contractor; QA/QC | Bridge: GroutBatch / new RIConstructionMethod (constructionMethod of RigidInclusion) |
| **Grout-level monitoring / grout drop (uncovered column)** | – | – | Project spec/ITP; DFI (2023) | Contractor; QA/QC | New: RIConstructionMethod (constructionMethod of RigidInclusion) (grout-level check) |
| **Wet-squeeze / grout-return volume** | $V_{ws}$ | m³ | Project spec/ITP; DFI (2023) | Contractor; QA/QC | New: RIConstructionMethod (constructionMethod of RigidInclusion) (grout-return check) |
| **Concrete mix / strength class** | – | – | Project spec/ITP | Contractor; QA/QC | Existing: DesignGroutMix |
| **Cement type** | – | – | Project spec/ITP | Contractor; QA/QC | Existing: DesignGroutMix (component) |
| **Fresh-concrete slump** | – | mm | ASTM; Project spec/ITP | Contractor; QA/QC | Existing: SlumpTest |
| **Delivery-to-placement elapsed time (transport window)** | – | min | Project spec/ITP | Contractor; QA/QC | Existing: GroutBatch (timing) |
| **Fresh grout density / flow (grout-based RI)** | – | kg/m³ (s) | ASTM | Contractor; QA/QC | Existing: MudBalanceTest / FlowConeTest / MarshFunnelTest |
| **Concrete temperature** | – | °C | ASTM; Project spec/ITP | Contractor; QA/QC | Existing: MaterialTest / GroutBatch |
| **Air content** | – | % | ASTM | Contractor; QA/QC | Existing: MaterialTest |
| **Batch-ticket data** | – | – | IR A-1 | Contractor; QA/QC | Existing: GroutBatch + AssociatedFile |
| **Fresh-concrete sampling (cylinders cast)** | – | – | ASTM; Project spec/ITP | Contractor; QA/QC | Existing: Sample / GroutSpecimen (from GroutBatch) |
| **Grout-pump / plant calibration** | – | – | EN 12699/14679 | Contractor; QA/QC | Existing: GroutPump / GroutPlant |
| **E · Termination & anchorage** | | | | | |
| **Indicator / pre-production test programme** | – | – | Project spec/ITP; DFI (2023) | Contractor; QA/QC | New: RIConstructionMethod (constructionMethod of RigidInclusion) (indicator programme) |
| **Refusal / anchorage criterion (coded)** | – | – | Project spec/ITP | Contractor; QA/QC | New: RIConstructionMethod (constructionMethod of RigidInclusion) (refusalCriterionType) |
| **Achieved termination outcome** | – | – | Project spec/ITP | Contractor; QA/QC | New: RIConstructionMethod (constructionMethod of RigidInclusion) |
| **As-built toe embedment into competent stratum** | – | m (× D) | ASIRI; Han (2025) | Contractor; QA/QC | New: RigidInclusion → AbstractLinearSamplingFeature (derived geometry) |
| **Drilling time, compressible vs competent layer** | – | min | Project spec/ITP | Contractor; QA/QC | New: RIConstructionMethod (constructionMethod of RigidInclusion) / Require new: RIEvent / installation record (on RigidInclusion) |
| **F · Reinforcement (conditional)** | | | | | |
| **Reinforcement present & type** | – | – | ASIRI | Contractor | New: RigidInclusion → AbstractLinearSamplingFeature (reinforcement) |
| **Reinforcement length / section** | – | m | ASIRI | Contractor | New: RigidInclusion → AbstractLinearSamplingFeature (reinforcement) |
| **G · Durations & production** | | | | | |
| **Drilling duration** | – | min | Project spec/ITP | Contractor; owner | New: RIConstructionMethod (constructionMethod of RigidInclusion) |
| **Concreting duration** | – | min | Project spec/ITP | Contractor; owner | New: RIConstructionMethod (constructionMethod of RigidInclusion) |

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) To be revised by Dan |
|---|---|---|---|---|---|
| **Remarks / downtime (breakdown, standby, move)** | – | – | Project spec/ITP | Contractor; owner | New: RIConstructionMethod (constructionMethod of RigidInclusion) (note) |
| **Attached rig-log file (PDF)** | – | – | Project spec/ITP | Contractor; owner | Existing: AssociatedFile |

## 4.8 Post-installation Phase: Verification and Monitoring

The post-installation phase confirms the constructed system against the design intent. DIGGS covers this phase well: the monitoring reuses the Monitor feature, the material verification reuses the material-test features, and the load and integrity tests reuse the standard Test feature, referencing the RigidInclusion column. Of the thirty-seven essential elements, twenty-nine reuse existing features, three are bridged and five are new. The elements fall into six groups.

**A, As-built verification survey.** The survey that checks the column against tolerance: the post-construction head coordinates and the verticality check (reusing PointLocation and a derived survey value), the position deviation against tolerance (bridged to the specification that sets the limit), and the post-construction diameter and trim-level verification (new).

**B, Integrity and material verification.** The integrity checks: the low-strain integrity test and the thermal integrity profiling (TIP), each carried by a standard Test referencing the column, and the compressive strength of cylinders or cores, which reuses the existing material-test features.

**C, Load testing.** The performance verification that United States practice relies on most: the test-element class and load level, the static maintained-load test on a single column and its load-settlement curve, the setup and procedure, the critical creep load, the plate load test on the head and the zone load test on the platform, any dynamic test, and the measured column stiffness or capacity. In the rigid-inclusion model this whole group is carried by the schema's standard Test element, each test referencing the RigidInclusion column it was performed on, so that individual-column load testing, the dominant quality-control method, is expressed without any rigid-inclusion-specific test element.

**D, Load-transfer platform verification.** The platform as built: the as-built thickness (new on the LoadTransferPlatform), the compaction and the verified fill friction angle (reusing the in-situ density and shear tests), the geosynthetic installation record (bridged with an attached file), and the LTP special-inspection regime that sets the visual and density-testing frequency by foundation type and area (new on the specification).

**E, Instrumentation and monitoring.** The performance monitoring, which reuses the schema in full: settlement and differential settlement, the measured settlement reduction, pore-water pressure, the load on the head and soil, the reinforcement strain, the lateral movement and the adjacent-structure deformation, the shear deformation of the platform measured by a shape array, and the composite shear-wave velocity measured to confirm the seismic Site Class after improvement, all carried on the Monitor and its results or the existing geophysical survey. DIGGS monitoring support is complete, so none of these is new.

**F, Acceptance and records.** The acceptance trail: the acceptance criteria (new on the specification), the inspection sign-off and non-conformance records, the measured-versus-predicted settlement (bridged to the design), the achieved test frequency and coverage, the as-built and verification report, and the monitoring duration and frequency, most of which reuse the DocumentInformation, AssociatedFile and Monitor features. Table 4-4 summarizes the post-installation elements.

**Table 4-4.** Essential elements of the post-installation phase (37 elements; 29 existing, 3 bridged, 5 new). The load and integrity tests reuse the standard Test element, referencing the column.

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) |
|---|---|---|---|---|---|
| **A · As-built verification survey** | | | | | |
| **Post-construction head coordinates** | – | m | ASIRI; Project spec/ITP | QA/QC; surveyor | Existing: PointLocation (on RigidInclusion) |
| **Position deviation vs tolerance** | – | mm | ASIRI; Project spec/ITP | QA/QC; surveyor | Bridge: Derived (survey) vs new RIPerformanceSpec / RIDesignSpec → RigidInclusionProgram |
| **Verticality check** | – | ° or % | ASIRI | QA/QC; surveyor | Existing: Derived (survey) |
| **Post-construction diameter verification** | – | m | Project spec/ITP | QA/QC; surveyor | New: RigidInclusion → AbstractLinearSamplingFeature (as-verified) / MaterialTest |
| **Cut-off / trim level (verified)** | – | m | ASIRI; Project spec/ITP | QA/QC; surveyor | New: RigidInclusion → AbstractLinearSamplingFeature (geometry) |
| **B · Integrity & material verification** | | | | | |
| **Low-strain integrity test** | – | – | ASIRI; ASTM; Han (2025) | QA/QC | Existing: Test (integrity test on the RigidInclusion column) |
| **Thermal integrity profiling (TIP)** | – | – | DFI (2023); ASTM D7949 | QA/QC | Existing: Test (thermal integrity test on the RigidInclusion column) |
| **Concrete cylinder / core compressive strength** | $f_c$ (7 d / 28 d) | MPa | ASTM; Project spec/ITP | QA/QC | Existing: UnconfinedCompressiveStrengthTest / MaterialTest (on GroutSpecimen) |
| **C · Load testing (performance verification)** | | | | | |
| **Test-element class & test-load level** | – | – / % design load | ASIRI; EC7 | QA/QC; designer | Existing: Test (load test on the RigidInclusion column) |
| **Static load test on single column** | Q–s | kN, mm | ASIRI; FHWA; ASTM; Han (2025) | QA/QC; designer | Existing: Test (load test on the RigidInclusion column) |
| **Load-test setup / reaction & procedure** | – | – | BS 8006-1; ASTM | QA/QC; designer | Existing: Test (load test on the RigidInclusion column) |
| **Critical creep load** | $R_{c,cr}$ | kN | ASIRI | QA/QC; designer | Existing: Test (load test on the RigidInclusion column) |
| **Plate Load Test on column head (PLT)** | – | kN, mm | ASTM; Project spec/ITP | QA/QC; designer | Existing: Test (load test on the RigidInclusion column) |
| **Zone Load Test on LTP (ZLT)** | – | kN, mm | BS 8006-1; Project spec/ITP | QA/QC; designer | Existing: Test (load test on the RigidInclusion column) |
| **Dynamic load test (if used)** | – | kN, mm | FHWA; ASTM | QA/QC; designer | Existing: Test (load test on the RigidInclusion column) |
| **Measured column stiffness / capacity** | – | kN, kN/mm | ASIRI; FHWA | QA/QC; designer | Existing: Test (load test on the RigidInclusion column) |
| **D · Load-transfer platform verification** | | | | | |
| **LTP thickness (as-built)** | – | m | ASIRI; EBGEO; FHWA | QA/QC | New: LoadTransferPlatform → AbstractPlanarSamplingFeature (as-built) |
| **LTP compaction / degree of compaction** | $D_{Pr}$ | % | EBGEO; FHWA | QA/QC | Existing: InsituDensityTest / LabCompactionTest |
| **LTP fill friction angle (verified)** | φ' | ° | EBGEO; FHWA | QA/QC | Existing: DirectShearTest / TriaxialTest |
| **Geosynthetic installation record** | – | – | EBGEO; FHWA | QA/QC | Bridge: AssociatedFile + new LoadTransferPlatform → AbstractPlanarSamplingFeature |
| **LTP special-inspection frequency / regime** | – | – | DFI (2023); Project spec/ITP | QA/QC; owner | New: RIPerformanceSpec → AbstractProgramSpecificaton (LTP inspection regime) |
| **E · Instrumentation & monitoring** | | | | | |
| **Settlement monitoring** | $s_{meas}$ | mm | ASIRI; EBGEO; FHWA | Owner; designer | Existing: Monitor + MonitorResult |
| **Differential settlement (measured)** | $\Delta s_{meas}$ | mm | ASIRI; FHWA | Owner; designer | Existing: Monitor + MonitorResult |
| **Measured settlement reduction (treated vs untreated)** | – | – / mm | ASIRI | Owner; designer | Existing: Monitor + MonitorResult (derived) |
| **Pore-water pressure** | u | kPa | ASIRI | Owner; designer | Existing: Monitor + MonitorResult (piezometer) |
| **Load on head / soil (earth-pressure cells)** | – | kPa | ASIRI; EBGEO | Owner; designer | Existing: Monitor + MonitorResult |

| Parameter (essential element) | Symbol | Unit | Required by (source) | Used by | DIGGS status (existing / new element) |
|---|---|---|---|---|---|
| **Reinforcement / geosynthetic strain** | $\varepsilon_{meas}$ | % | BS 8006-1; EBGEO | Owner; designer | Existing: Monitor + MonitorResult |
| **Lateral movement (inclinometer)** | – | mm | EBGEO; FHWA | Owner; designer | Existing: Monitor + MonitorResult |
| **Adjacent-structure / adjacent-RI deformation (installation effect)** | – | mm | Han (2025) | Owner; designer | Existing: Monitor + MonitorResult |
| **Composite shear-wave velocity / seismic Site Class (post-improvement)** | $V_{s,comp}$ | m/s | DFI (2023); IBC (Site Class) | Designer; geotech engineer | Existing: GeophysicalFieldSurvey (post-improvement composite Vs) |
| **LTP shear-deformation monitoring (shape array)** | – | mm | DFI (2023) | Owner; designer | Existing: Monitor + MonitorResult |
| **F · Acceptance & records** | | | | | |
| **Acceptance criteria (settlement / differential / load test)** | – | – | ASIRI; FHWA; Project spec/ITP | QA/QC; owner | New: RIPerformanceSpec / RIDesignSpec → RigidInclusionProgram (acceptance) |
| **Inspection sign-off & non-conformance (NCR) records** | – | – | ASIRI; Project spec/ITP | QA/QC; owner | Existing: DocumentInformation + AssociatedFile |
| **Measured vs predicted settlement** | – | mm | ASIRI; FHWA | QA/QC; owner | Bridge: Monitor (derived) vs new RIDesignSpec / RISystemDesign → RigidInclusionProgram |
| **Test frequency / coverage achieved** | – | – | ASIRI; Project spec/ITP | QA/QC; owner | Existing: DocumentInformation (procedure metadata) |
| **As-built / verification report** | – | – | ASIRI; EC7 | QA/QC; owner | Existing: DocumentInformation + AssociatedFile |
| **Monitoring duration & frequency** | – | – | ASIRI; EBGEO; FHWA | QA/QC; owner | Existing: Monitor (metadata) |

## 4.9 Summary: the Shape of the DIGGS Extension

Taken together, the four phases define 235 essential elements, forty-nine for pre-installation, ninety-seven for design, fifty-two for installation and thirty-seven for post-installation. Of these, ninety-eight already have a home in DIGGS, fifteen can be bridged onto an existing feature, and 122 require one of the rigid-inclusion elements (Table 4-5). The distribution is not uniform, and its shape is the chapter's central result. Where the schema is strong, the ground investigation of the pre-installation phase, the fresh-material tests of the installation phase, and the monitoring and load, integrity and material testing of the post-installation phase, the essential elements reuse the existing features almost entirely. Where the schema had no rigid-inclusion object, the design record and the physical system of the improved block, its platforms and its columns with their construction, the new elements are concentrated.

**Table 4-5.** The four phases of the standardization framework: the source basis for each phase, the number of standardized parameters, and how they map onto DIGGS (carried on an existing feature, bridged onto a related feature, or requiring a new rigid-inclusion feature).

| Phase | Basis for parameter selection | Parameters | Existing | Bridged | New |
|---|---|---|---|---|---|
| **Pre-installation** | Geotechnical investigation: ground model, soil strength and compressibility, in-situ and laboratory tests | 49 | 48 | 0 | 1 |
| **Design** | Four standards compared (ASIRI, BS 8006-1, EBGEO, FHWA); union of all requirements | 97 | 4 | 11 | 82 |
| **Installation** | Contractor production records compared (Menard, Keller and others); displacement tools and effects | 52 | 17 | 1 | 34 |
| **Post-installation** | Acceptance ranges published in the standards; integrity and load tests, monitoring | 37 | 29 | 3 | 5 |
| **Total** | | **235** | **98** | **15** | **122** |

The extension those gaps require is small and orderly, and it is the model developed in the DIGGS 3.0 Extension and set out in full in Chapter 6. On the physical side it adds the RIFoundationSystem for the improved block, the LoadTransferPlatform and Geosynthetic for the platform, and the RigidInclusion column with its RIConstructionMethod and RIEvent; on the design side it adds the staged RigidInclusionProgram with its performance, design and system specifications. Every one of these derives from an existing DIGGS abstract head, chiefly the sampling-feature and program heads, and everything else, the investigation, the grout, and the load, integrity and material tests, is reused from the schema as it stands. This confirms the methodology's finding that bringing rigid inclusions into DIGGS is a concentrated, backward-compatible extension that describes the rigid-inclusion system rather than re-implementing any design method.

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**CHAPTER 5**

# Case Study

*This chapter presents, in detail, a real controlled-modulus-column (CMC) rigid-inclusion project used to validate the standardized data framework of this report: the ground improvement of a 110/13.8 kV electrical substation in the Kingdom of Saudi Arabia. It sets out the project and its ground conditions, the geotechnical investigation and its field and laboratory tests, the reasoning that made CMC the appropriate solution, the design of the CMC and load-transfer-platform system, the as-built layout and quantities, the installation method and its monitoring record, and the post-installation load and integrity testing. All figures are reproduced from the project's own documents and have been anonymized; the project, the owner and the contractors are not named.*

## 5.1 Project Overview

The project is the construction of a 110/13.8 kV electrical substation in the Kingdom of Saudi Arabia, comprising a substation building, transformer (TR) tunnels, three power transformers, an auxiliary transformer, a shunt reactor, outdoor electrical equipment, a boundary wall and ancillary structures, over a broadly square site of about 70 m by 60 m (4,200 m²). The plot is flat and was vacant at the time of the works. Three parties are referred to throughout: the owner (a national electricity authority), the main (civil) contractor, and the specialty ground-improvement contractor who designed and installed the rigid inclusions.

Figure 5-1 shows the plan of the foundation system and the location of each structure across the site: the substation building on a raft with isolated footings, the transformer tunnels, the three power transformers, the auxiliary transformer, the shunt reactor, the outdoor equipment, the boundary wall and the septic tank, every one carrying an average applied load of about 150 kPa.

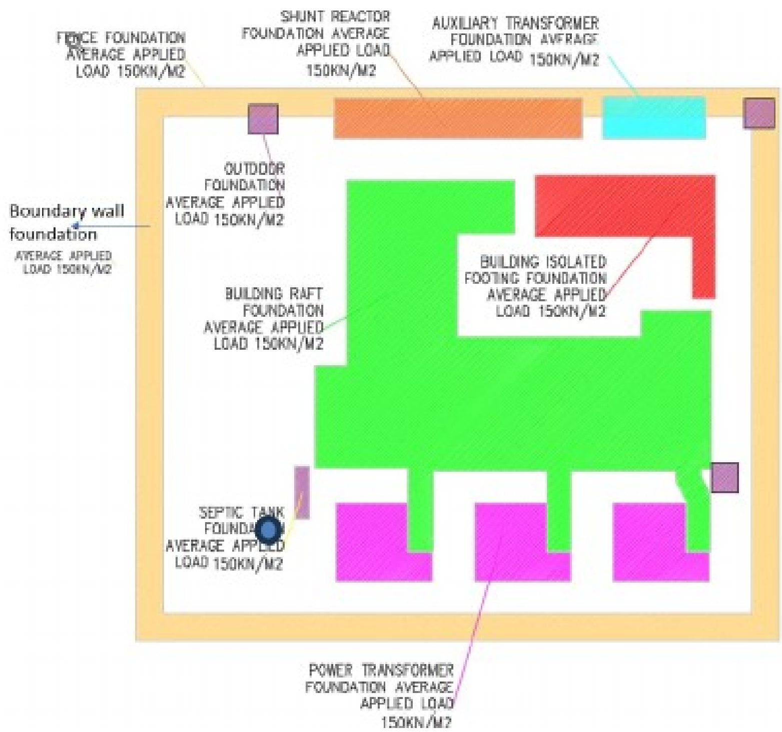



**Figure 5-1.** Plan of the foundation system and the location of each structure across the site; each foundation carries an average applied load of about 150 kPa. *reproduced from the project CMC report, Project Information section (anonymized).*

The performance requirements set for the improved ground were a net allowable bearing pressure of 150 kPa under an applied structural load of the order of 150 kPa, with total settlement limited to 50 mm under raft foundations and 25 mm under isolated footings. The works were governed by the Saudi Building Code (SBC 303-CR, 2018) together with the French rigid-inclusion recommendations ASIRI (2012) and the reinforced-fill code BS 8006, and verified by ASTM and BS in-situ test methods. Table 5-1 collects the key project parameters and reference levels.

**Table 5-1.** Key project parameters and design requirements.

| Item | Value |
|---|---|
| **Structure** | 110/13.8 kV electrical substation (building, TR tunnels, transformers, shunt reactor, outdoor equipment, boundary wall) |
| **Location** | Kingdom of Saudi Arabia |
| **Site area** | 70 m × 60 m ≈ 4,200 m² |
| **Ground-improvement solution** | Controlled modulus columns (CMC), a drilled-displacement rigid inclusion |
| **Applied structural load** | 150 kPa |
| **Required net allowable bearing capacity** | 150 kN/m² |
| **Allowable settlement (raft / isolated)** | 50 mm / 25 mm |
| **Working-platform level / existing ground level** | +6.25 m / +5.90 m |
| **Average water-table level** | +5.25 m (groundwater 0.65 to 0.75 m below ground) |
| **Governing standards** | SBC 303-CR (2018); ASIRI (2012); BS 8006; ASTM / BS test methods |

## 5.2 Geotechnical Investigation

The improved-ground design rests on a geotechnical investigation carried out for the substation in mid-2024. This section presents the fundamental data of that investigation: its scope, the subsurface profile it revealed, the field and laboratory tests performed, the chemistry of the ground, and the soil design parameters derived from it.

### 5.2.1 Scope of the investigation

The investigation comprised seven boreholes, five within the substation footprint and two just outside it, drilled to depths of 10 to 15 m using a rotary rig with the wash-boring technique and a bentonite slurry. Standard penetration tests (SPT) were performed continuously, at 0.75 m intervals to 3 m depth and at 1.5 m intervals thereafter, with a split-spoon sampler, in accordance with ASTM D-1586. The boreholes were supplemented by geophysical and thermal field testing: four electrical-resistivity tests (ERT) for corrosivity screening and four thermal-resistivity tests (TRT), the latter relevant to the buried power cables of a substation. The specialty contractor additionally carried out cone penetration tests (CPT) across the site before installation, which were used directly in the rigid-inclusion design.

The seven boreholes were positioned to cover the substation footprint, five within the site boundary and two just outside it, as shown in Figure 5-2.

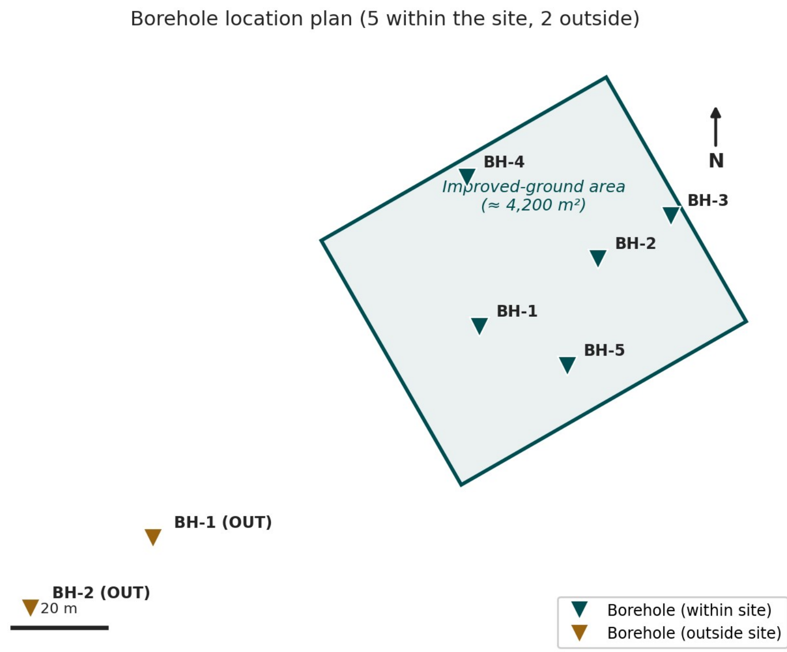



**Figure 5-2.** Borehole location plan: five boreholes within the substation footprint (BH-1 to BH-5) and two just outside it (BH-1 OUT, BH-2 OUT), on the improved-ground boundary. *borehole layout drawn from the project survey coordinates (anonymized).*

### 5.2.2 Subsurface profile and groundwater

The ground is a classic soft, aggressive coastal profile. Beneath a thin surface fill lies a thick, very loose sabkha, then a loose to medium-dense silty sand, over a stiff lean clay. The most important feature is the sabkha layer between about 0.75 m and 4.5 m depth, in which the SPT blow count is essentially N = 1, indicating a very loose, highly compressible and collapsible soil. Groundwater stands very high, only 0.65 to 0.75 m below the surface. Figure 5-3 reproduces a representative borehole log, Table 5-2 summarizes the profile, and Table 5-3 sets out the complete standard-penetration-test record for all seven boreholes.

## LOG OF BORING — BOREHOLE BH-1

**Electrical substation project, Kingdom of Saudi Arabia**

**Ground investigation (representative borehole); total depth 15.0 m; GWL ~0.65 m**

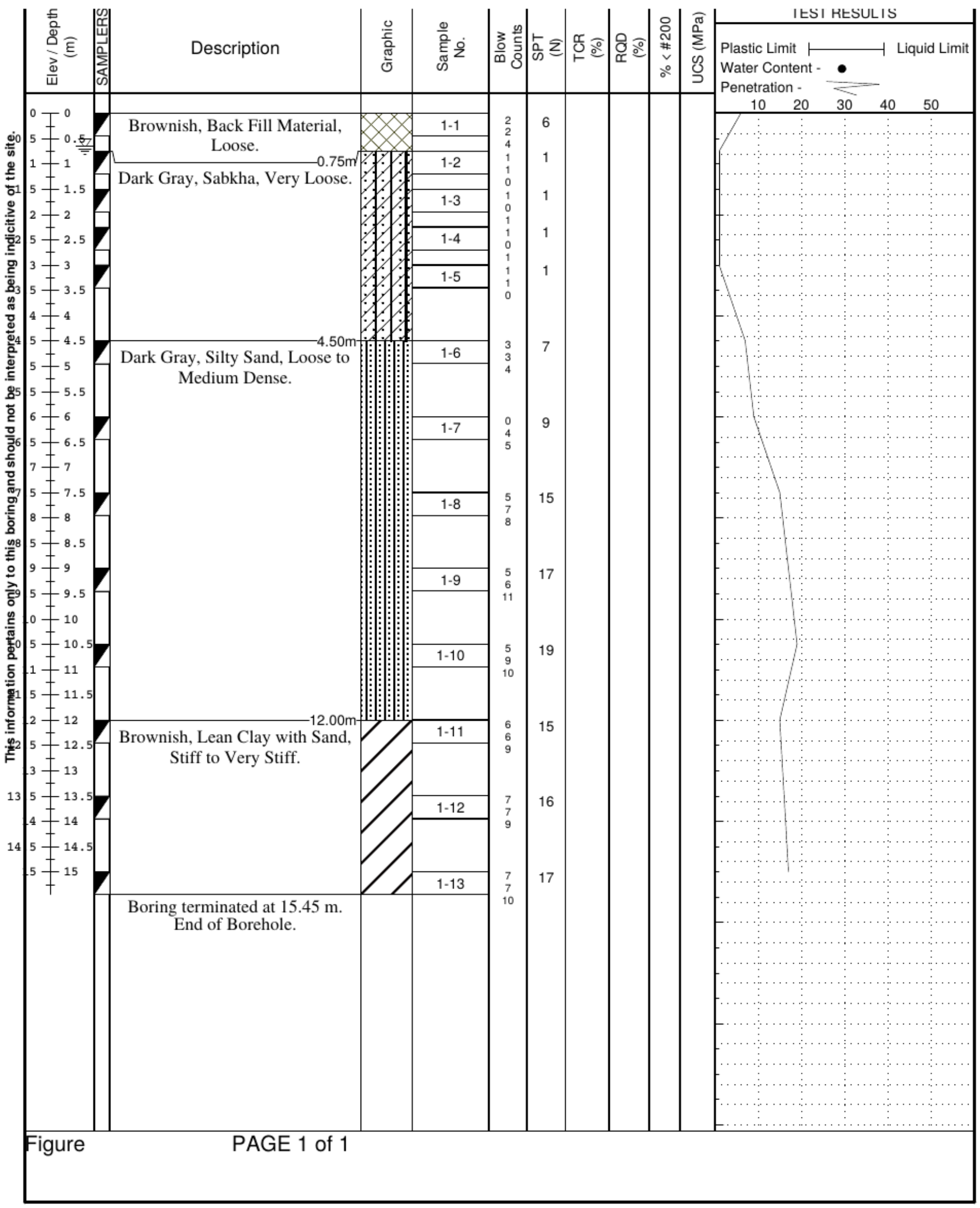



**Figure 5-3.** Representative borehole log (BH-1) from the site investigation, showing the fill, the very loose sabkha (SPT N ≈ 1), the loose to medium-dense silty sand and the stiff lean clay, with the water table about 0.65 m below ground. *reproduced from the project geotechnical report (anonymized).*

**Table 5-2.** Subsurface stratigraphy and representative SPT resistance.

| Layer | Depth (m) | Description | SPT N (blows/300 mm) |
|---|---|---|---|
| **Fill** | 0.0 – 0.75 | Brownish, loose to medium-dense backfill | 6 – 17 |
| **Sabkha** | 0.75 – 4.5 | Dark gray, very loose sabkha (collapsible, aggressive) | ≈ 1 |
| **Silty sand** | 4.5 – 12.0 | Dark gray, very loose to medium-dense silty sand | 8 – 21 |
| **Lean clay with sand** | 12.0 – 15.0 | Brownish, stiff to very stiff lean clay | 15 – 20 |

The very loose sabkha is not a local feature but is present across the whole site. Table 5-3 gives the standard-penetration-test blow count at every test depth in each of the seven boreholes: the SPT N-value collapses to about one blow per 300 mm through the 1.5 m to 3.75 m interval in every borehole without exception, and recovers only in the silty sand below about 4.5 m. This site-wide consistency is what justified treating the sabkha as a single, continuous compressible layer in the improved-ground design.

**Table 5-3.** Standard-penetration-test blow count N (blows/300 mm) at each test depth in the seven boreholes (dash indicates the borehole terminated above that depth).

| Depth (m) | BH-1 | BH-2 | BH-3 | BH-4 | BH-5 | BH-1 (OUT) | BH-2 (OUT) |
|---|---|---|---|---|---|---|---|
| **0.75** | 6 | 17 | 6 | 8 | 6 | 12 | 17 |
| **1.50** | 1 | 1 | 1 | 1 | 1 | 1 | 1 |
| **2.25** | 1 | 1 | 1 | 1 | 1 | 2 | 1 |
| **3.00** | 1 | 1 | 1 | 1 | 1 | 1 | 5 |
| **3.75** | 1 | 1 | 1 | 1 | 1 | 1 | 2 |
| **4.50** | 7 | 9 | 5 | 7 | 6 | 4 | 10 |
| **6.00** | 9 | 14 | 14 | 12 | 10 | 8 | 11 |
| **7.50** | 15 | 11 | 11 | 18 | 11 | 9 | 8 |
| **9.00** | 17 | 18 | 16 | 21 | 12 | 17 | 18 |
| **10.00** | – | – | 15 | 15 | – | – | – |
| **10.50** | 19 | 16 | – | – | 13 | 18 | 19 |
| **12.00** | 15 | 16 | – | – | 14 | 17 | 16 |
| **13.50** | 16 | 16 | – | – | – | 20 | 17 |
| **15.00** | 17 | 18 | – | – | – | 18 | 18 |

The five borehole logs were combined into a generalized geological section across the substation footprint (Figure 5-4). The same four-layer sequence recurs at every location — a thin surface fill over sabkha, passing into silty sand and, at depth, into sandy clay — confirming that the ground is laterally uniform, with only minor variation in layer thickness and blow count between boreholes.

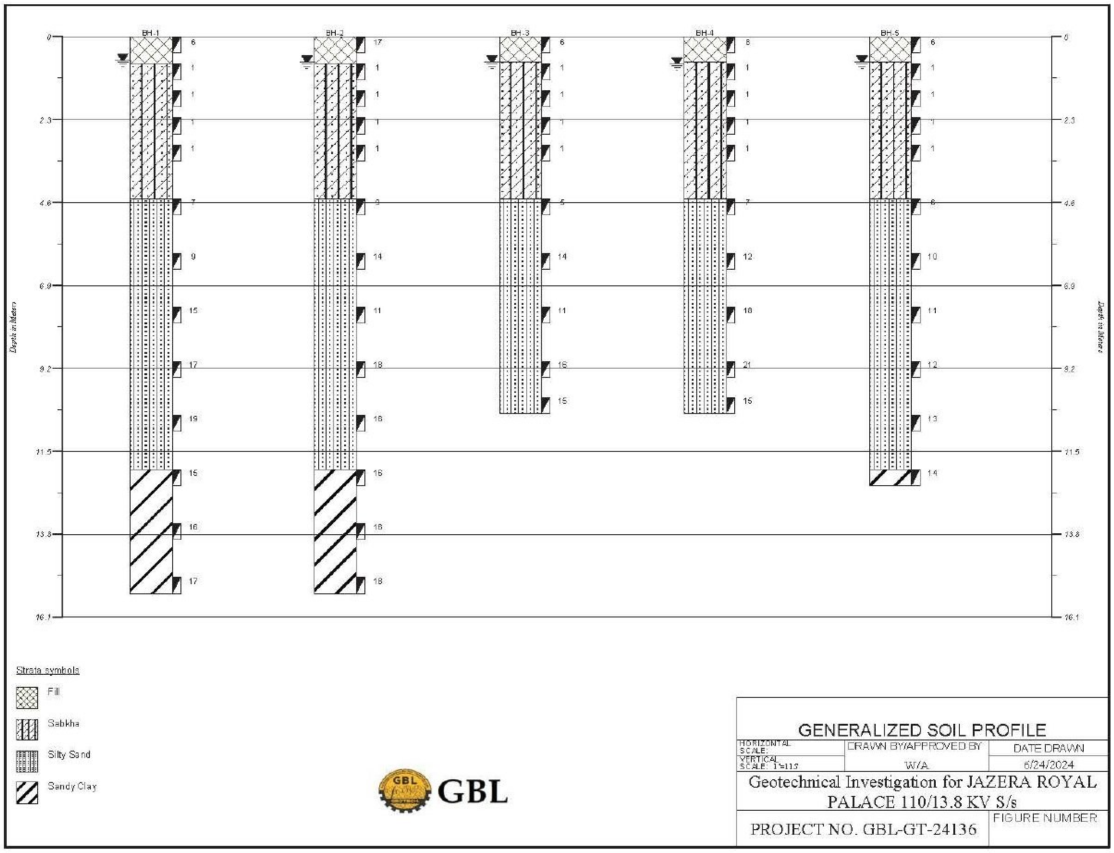



**Figure 5-4.** Generalized geological section across the five boreholes (BH-1 to BH-5), showing the fill, sabkha, silty-sand and sandy-clay sequence with standard-penetration blow counts. *(GBL geotechnical investigation report GBL-GT-24136)*

### 5.2.3 Field and laboratory tests

The investigation combined in-situ field testing with a laboratory programme on recovered samples. The field tests were the SPT in every borehole, the four ERT and the four TRT arrays, and the contractor's pre-installation CPT. The laboratory suite classified and characterized the soils and, critically, measured the aggressivity of the ground. Table 5-4 lists the tests and the standards under which they were performed.

**Table 5-4.** Field and laboratory tests performed in the investigation.

| Test | Type | Number | Standard |
|---|---|---|---|
| **Standard penetration test (SPT)** | Field (in borehole) | 7 boreholes, continuous | ASTM D-1586 |
| **Cone penetration test (CPT)** | Field (pre-installation) | Site coverage | ASTM D-5778 |
| **Electrical resistivity test (ERT)** | Field (geophysical) | 4 | (corrosivity screening) |
| **Thermal resistivity test (TRT)** | Field (geophysical) | 4 | (cable thermal design) |
| **Sieve / particle-size analysis** | Laboratory | 11 | ASTM D-422 |
| **Atterberg limits** | Laboratory | 11 | ASTM D-4318 |
| **Engineering soil classification (USCS)** | Laboratory | 11 | ASTM D-2487 |
| **Chemical analysis (sulfate, chloride, pH, Mg)** | Laboratory | 3 | BS 1377 Part 3 |
| **Modified Proctor compaction and CBR** | Laboratory | 2 | ASTM D-1557 / D-1883 |

### 5.2.4 Ground chemistry and aggressivity

The chemical tests confirmed a highly aggressive ground, consistent with a sabkha. The pore fluid carried a sulfate content of 5,810 to 8,515 ppm, a chloride content of 0.51 to 0.71 percent and a magnesium content of 571 to 954 ppm, at a pH of 7.9 to 8.2, and the site was accordingly classified as a very severe exposure. The corresponding concrete requirement is severe: a minimum characteristic strength of the order of 38 MPa, a sulfate-resisting cement with a pozzolanic addition (fly ash or silica fume), a maximum water-to-binder ratio of about 0.36 to 0.40, and a minimum binder content near 390 kg/m³. The electrical-resistivity survey independently returned a very severe corrosivity class. This aggressivity is a first-order design driver, because any ground-improvement element left in the ground must survive it, and it strongly influences the choice of solution discussed in Section 5.3.

### 5.2.5 Soil design parameters

From the field and laboratory data the investigation derived the soil design parameters summarized in Table 5-5, which give, for each layer, the unit weight, the representative SPT resistance and the strength parameters used in the foundation and improved-ground analyses.

**Table 5-5.** Soil design parameters by layer (from the geotechnical report).

| Depth below NSL (m) | Material | Unit weight (kN/m³) | Avg. SPT N | Cohesion (kPa) | Friction angle φ (°) |
|---|---|---|---|---|---|
| **0.0 – 0.75** | Fill | 16.0 | 10 | – | 29 |
| **0.75 – 4.5** | Sabkha | – | 1 | – | – |
| **4.5 – 12.0** | Silty sand | 16.5 | 13 | – | 30 |
| **12.0 – 15.0** | Lean clay with sand | 16.5 | 17 | 20 | – |

## 5.3 Why CMC Was the Appropriate Solution

The combination of a very loose, collapsible and aggressive sabkha, a shallow water table and settlement-sensitive substation structures is exactly the situation for which rigid inclusions are best suited, and it is worth setting out why the controlled modulus column was preferred over the alternatives the investigation itself proposed. The geotechnical report offered two conventional options: to excavate the site to about 5 m and replace the soft ground with compacted granular fill and rockfill beneath a geotextile, or to install stone columns to about 12 m. Both are workable in principle, but both are poorly matched to this site.

**Excavation and replacement** would require a deep excavation carried well below a water table only 0.65 m down, with the heavy dewatering, support and spoil-disposal that implies in an aggressive sabkha, and it treats only the upper soil rather than transferring load to the competent stratum. It is slow, costly and difficult to control at this groundwater level.

**Stone columns** derive their capacity from the lateral confinement the surrounding soil provides, and in a very loose sabkha with SPT N of about 1 that confinement is largely absent, so the columns are prone to bulging and give limited settlement control; the aggressive, fine sabkha is also an unfavourable host for a granular column.

The controlled modulus column overcomes both objections. As a rigid, cast concrete inclusion it does not rely on lateral confinement and cannot bulge, so it performs in the very loose sabkha where a stone column struggles; it carries the structural load down through the sabkha and the silty sand to the competent bearing stratum, controlling settlement to the required tolerance through the composite action of the columns, the soil and the load-transfer platform. Being installed by displacement it generates virtually no spoil and needs no mass excavation or dewatering, an important advantage at this water table, and it is quiet and vibration-free. Its concrete can be designed as a sulfate-resisting mix to survive the very severe exposure. For a settlement-sensitive electrical substation on this ground, the CMC rigid inclusion is therefore the technically and economically appropriate solution, and it is the one that was designed and built.

## 5.4 Design of the CMC and Load-Transfer-Platform System

### 5.4.1 Design basis and method

The rigid inclusions were designed to the French recommendations ASIRI (2012) and the associated standards NF P94-261 and NF P94-262, using the site CPT as the primary ground model. In this framework the soil-inclusion interaction is expressed through load-transfer laws: the base resistance is taken as $q_b = K_c \times q_c$ from the cone resistance below the toe, with the Frank and Zhao end-bearing factor $K_q$ (taken as 4.8 for the coarse-grained bearing soils of this site), the limit skin friction $q_s$ is a function of the cone resistance with the Frank and Zhao friction factor $K_T$ (0.8 for the coarse soils), and the negative skin friction developed in the settling sabkha is included through a K·tanδ term. The pressuremeter modulus used for settlement is related to the deformation modulus through a rheological factor. Figure 5-5 shows the design soil profile, the CPT cone-resistance and SPT profiles from which the layer parameters were taken, and Table 5-6 gives the design soil layers and their moduli.

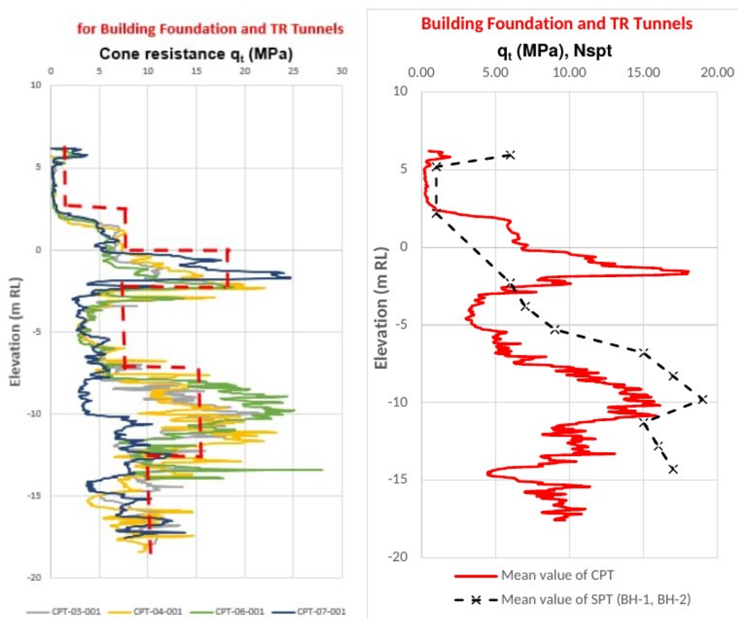



Figure 4: Soil Profile and parameters for Building foundation and TR Tunnels.

**Figure 5-5.** Design subsurface profile for the building and TR-tunnel zone: cone resistance $q_t$ from the site CPTs (left) and the mean CPT and SPT profiles against elevation (right), from which the design soil layering and moduli were derived. *reproduced from the project CMC design report (anonymized).*

Comparable design profiles were derived for the other foundation zones. Figure 5-6 and Figure 5-7 present the profiles for the power-transformer foundations and for the boundary-wall, auxiliary-transformer, shunt-reactor and outdoor-equipment areas; in each, the mean cone resistance from the pre-treatment CPTs is plotted against elevation together with the SPT blow counts of the corresponding boreholes.

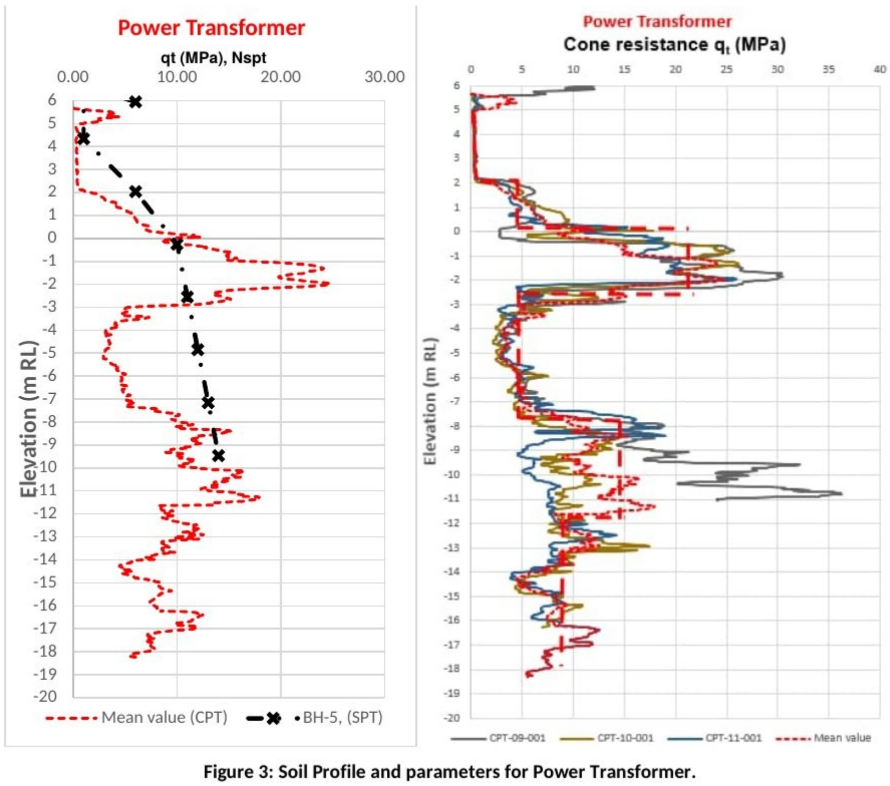



**Figure 5-6.** CPT-derived design soil profile for the power-transformer foundations: cone resistance $q_t$ and SPT N against elevation (mean CPT and borehole BH-5). *(Menard FMSA-24047 report, Figure 3)*

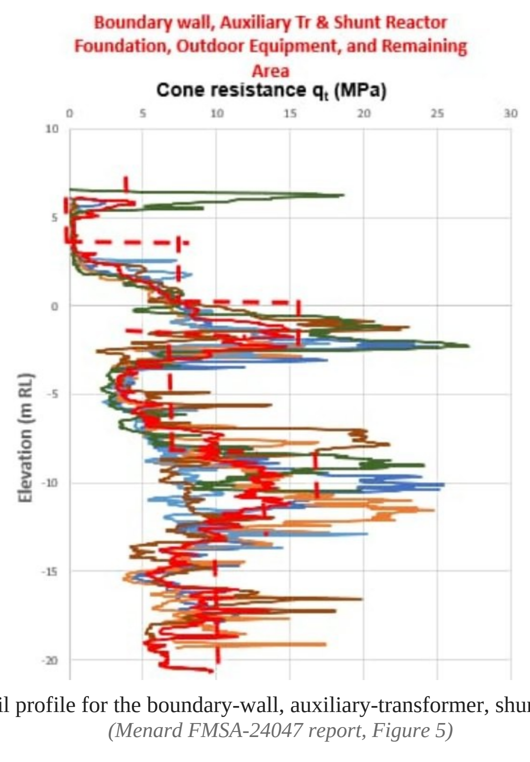



**Figure 5-7.** CPT-derived design soil profile for the boundary-wall, auxiliary-transformer, shunt-reactor and outdoor-equipment areas. *(Menard FMSA-24047 report, Figure 5)*

**Table 5-6.** Design soil layers and deformation moduli used in the CMC analysis.

| Layer | Young's modulus E (kPa) | Unit weight (kN/m³) | c′ (kPa) | φ′ (°) |
|---|---|---|---|---|
| **Load-transfer platform** | 60,000 | 16 | 0 | 38 |
| **Fill (working platform)** | 30,000 | 16 | 0 | 28 |
| **Fill** | 3,000 | 16 | 0 | 29 |
| **Sabkha (drained)** | 1,320 | 17 | 0 | 27 |
| **Clean to silty sand** | 24,000 | 17 | 0 | 27 |
| **Clayey silt to silty clay** | 18,000 | 17 | 20 | 0 |
| **Silty sand to sandy silt** | 36,000 | 17 | 0 | 27 |
| **Clayey silt to silty clay (lower)** | 24,000 | 17 | 20 | 0 |

The design soil profile divides the ground into layers, each assigned a Young’s modulus ($E_Y$), unit weight ($\gamma$), effective cohesion ($c'$), effective friction angle ($\varphi'$) and Poisson’s ratio ($\upsilon$). Because the CPTs showed the three foundation zones to differ mainly in the stiffness of the granular layers, a separate set of layer moduli was adopted for each zone; the full set is given in Table 5-7.

**Table 5-7.** Soil-profile design parameters by layer for the three foundation zones (power transformer; building and TR tunnels; boundary wall and other areas). *(Menard FMSA-24047 report, Tables 1–3)*

| No. | Description | $E_Y$, PT (kPa) | $E_Y$, Bldg/TR (kPa) | $E_Y$, Bound. (kPa) | γ (kN/m³) | c′ (kPa) | φ′ (°) | υ |
|---|---|---|---|---|---|---|---|---|
| 1 | **LTP** | 60000 | 60000 | 60000 | 16 | 0 | 38 | 0.3 |
| 2 | **Fill material (working platform)** | 30000 | 30000 | 30000 | 16 | 0 | 28 | 0.3 |
| 3 | **Fill material** | 7500 | 3000 | 6250 | 16 | 0 | 29 | 0.3 |
| 4 | **Sabkha (drained)** | 1260 | 1320 | 2925 | 17 | 0 | 27 | 0.3 |
| 5 | **Clean sand to silty sand** | 45000 | 24000 | 23250 | 17 | 0 | 27 | 0.3 |
| 6 | **Clayey silt to silty clay** | 15000 | 18000 | 20000 | 17 | 20 | 0 | 0.3 |
| 7 | **Silty sand †** | 21000 | 36000 | — | 17 | 0 | 27 | 0.3 |
| 8 | **Clayey silt to silty clay** | 30000 | 24000 | — | 17 | 20 | 0 | 0.3 |
| 9 | **Silty sand to sandy silt** | 36000 | — | — | 17 | 0 | 27 | 0.3 |
| 10 | **Clayey silt to silty clay** | 24000 | — | — | 17 | 20 | 0 | 0.3 |

The layer moduli were derived from the CPT cone resistance $q_c$ using the correlations of Lunne and Christofferson (1983), Lunne et al. (1997) and Bowles (2001), applied according to the friction ratio $R_f$:

$$\textit{Sandy soils } (R_f \le 1.5\%): \quad E_s = (2 \text{ to } 4)\, q_c$$

$$\textit{Silty soils } (1.5\% < R_f < 3\%): \quad E_s = 2\, q_c \; (q_c \le 2.5 \text{ MPa}); \quad E_s = 4\, q_c + 5 \; (2.5 < q_c \le 12.5); \quad E_s = 2\, q_c + 20 \; (q_c > 12.5)$$

$$\textit{Clays } (R_f \ge 3\%): \quad E_s = (3 \text{ to } 8)\, q_c$$

The modulus of the controlled modulus columns themselves was taken from ASIRI (2012), Clause 3.1.1.4, as a function of the 28-day concrete strength $f_{ck}$:

$$E_Y = 3700 \cdot f_{ck}^{1/3}$$

The settlement analysis followed the ASIRI (2012) pressuremeter approach. The Young’s moduli were converted to equivalent Ménard pressuremeter moduli through $E_m = \alpha_M \cdot E_Y$ ($\alpha_M$ the rheological factor), and the axial capacity of each inclusion was verified from the CPT: end-bearing $q_b = K_c \cdot q_c$ at the toe, mobilized through the Frank–Zhao factor, and limit skin friction $q_s$ for each layer, both to NF P 94-262. The design targets — a maximum allowable bearing pressure of 150 kN/m², with settlement within 50 mm for rafts and 25 mm for isolated footings — governed the grid spacing selected for each zone (Table 5-8).

### 5.4.2 CMC geometry by zone

The columns are 600 mm in diameter throughout, installed on a square grid at a centre-to-centre spacing of 2.0 m over most of the site and 1.8 m beneath the more heavily loaded power transformers, and taken to lengths that vary from 10 m to 18.5 m according to the load and the required bearing level of each structure. The concrete has a minimum characteristic compressive strength of 25 MPa, and the installation refusal criterion is a penetration of no more than 5 cm over two consecutive minutes, with a 600 mm displacement auger closed at its base by a lost cap. Table 5-8 sets out the design geometry by zone.

**Table 5-8.** CMC design geometry by structure / zone (diameter 0.6 m; square grid throughout).

| Zone / foundation | Total length (m) | Grid spacing (m) |
|---|---|---|
| **Building raft and TR tunnels** | 18.5 | 2.0 × 2.0 |
| **Building isolated footing** | 17.0 | 2.0 × 2.0 |
| **Power transformer** | 18.5 | 1.8 × 1.8 |
| **Auxiliary transformer** | 14.0 | 2.0 × 2.0 |
| **Shunt reactor** | 14.5 | 2.0 × 2.0 |
| **Outdoor equipment** | 12.0 | 2.0 × 2.0 |
| **Boundary wall** | 16.0 | 2.0 × 2.0 |
| **Septic tank / remaining area** | 10.0 | 2.0 × 2.0 |

### 5.4.3 Load-transfer platform

On top of the column network a load-transfer platform of well-compacted granular fill spreads the structural load onto the columns and produces the global, quasi-elastic behaviour of the composite soil-inclusion-platform system. The platform was designed with a deformation modulus of 60 MPa and an acceptance requirement of more than 65 MPa measured by plate load test, with an effective friction angle of 38 degrees and zero cohesion. Its thickness follows the difference between the CMC cut-off level (the working platform at +6.25 m) and the bottom of each foundation, and therefore varies from 0.3 m under the boundary wall to 1.75 m under the auxiliary transformer and shunt reactor, as set out in Table 5-9. In line with normal practice, and as noted in the schema discussion of Chapter 4, the specialty contractor designed the platform but the main contractor constructed it.

**Table 5-9.** Load-transfer-platform thickness by structure (CMC cut-off at +6.25 m).

| Structure | Bottom of foundation (m RL) | LTP thickness (m) |
|---|---|---|
| **Boundary wall** | 6.55 | 0.30 |
| **Building foundation and TR tunnels** | 6.85 | 0.60 |
| **Power transformer** | 7.25 | 1.00 |
| **Outdoor equipment** | 7.75 | 1.50 |
| **Auxiliary transformer** | 8.00 | 1.75 |
| **Shunt reactor** | 8.00 | 1.75 |

## 5.5 Foundation Allocation, Layout and Quantities

The foundation system for the whole substation is the same in principle everywhere: each structure bears on a load-transfer platform that sits on a grid of rigid inclusions, sized and spaced for that structure's load and settlement tolerance as set out above. The building raft and TR tunnels, the isolated footings, the three power transformers, the auxiliary transformer, the shunt reactor, the outdoor-equipment pillars, the boundary wall and the septic tank each have their own CMC grid and platform thickness, so that a single, uniform improved-ground solution serves the entire site while adapting locally to each load.

Figure 5-8 reproduces the as-built layout, which records every installed column, the four plate-load-test locations and the single zone-load-test location, within the improved-ground boundary of 4,200 m². In total the project comprises 1,075 controlled modulus columns of 600 mm diameter, ranging from 10 m to 18.5 m in length by zone, for about 18,221 linear metres of drilling and some 5,150 m³ of placed concrete, which at an area of about 4,200 m² corresponds to an area replacement ratio of roughly seven percent for the 2.0 m grid. This is the as-built population of columns that a standardized installation record, of the kind developed in this report, must be able to carry, one record per column, each tied to its coordinates, its geometry and its production data.

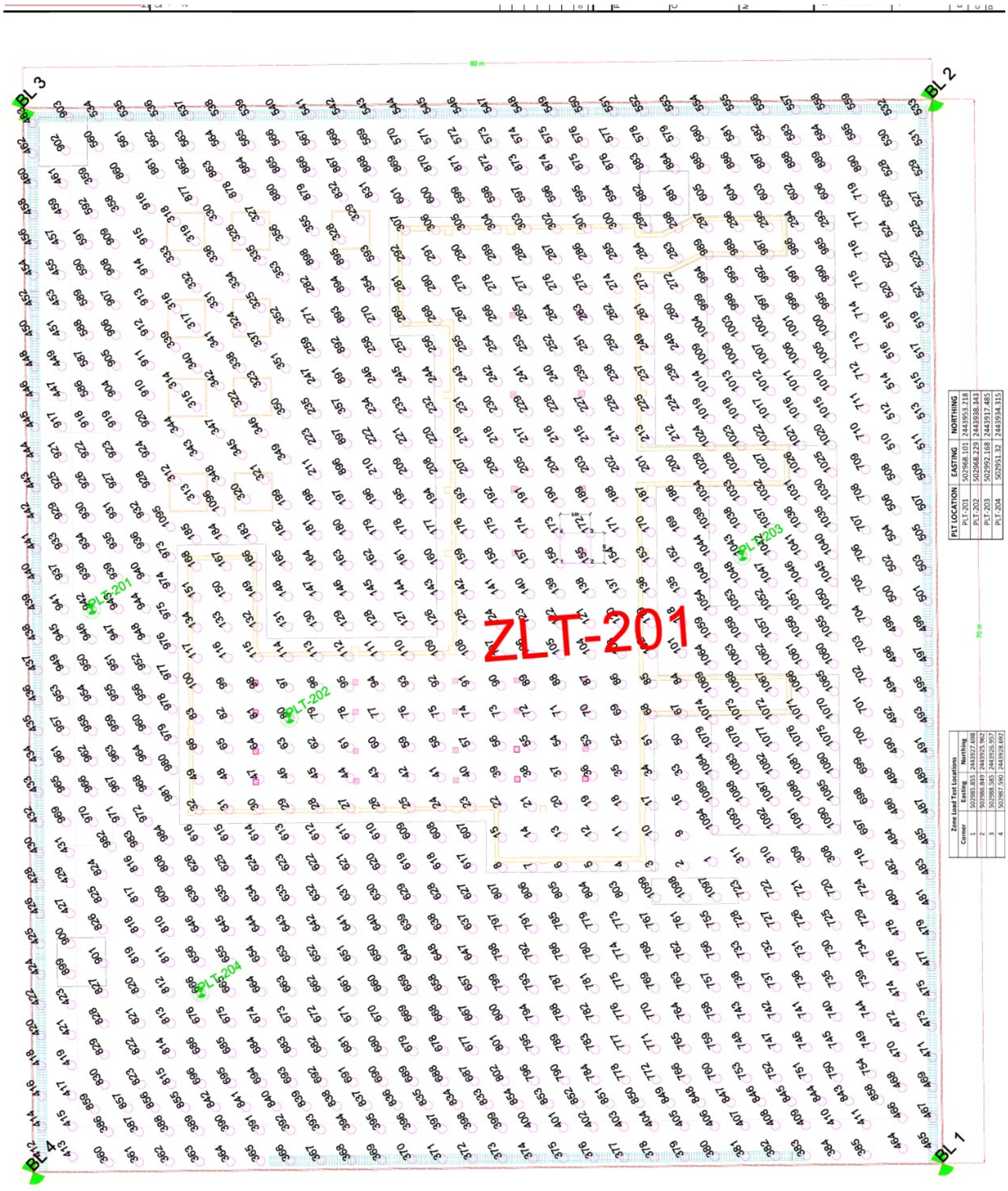



**Figure 5-8.** As-built layout of the rigid-inclusion works: the 1,075 controlled modulus columns within the 4,200 m² improved-ground boundary, the building and equipment foundation footprints, and the plate-load-test (PLT-201 to 204) and zone-load-test (ZLT-201) locations. *reproduced from the project as-built drawing (anonymized).*

The improved ground is organized as thirteen distinct load-transfer zones, one for each structure or foundation, and every installed column belongs to exactly one zone. Table 5-10 gives the as-built quantities zone by zone: the number of columns, their length, the resulting drilling metreage and the placed-concrete volume. The boundary wall and the building raft together account for the great majority of the columns, while the outdoor equipment and the septic tank need only a handful each, and the totals reconcile to the 1,075 columns recorded in the instance file.

**Table 5-10.** As-built CMC quantities by zone (diameter 0.6 m throughout; from the DIGGS instance file).

| Zone / foundation | No. of columns | Column length (m) | Total drilling (m) | Placed concrete ($m^3$) |
|---|---|---|---|---|
| **Boundary wall (perimeter)** | 503 | 16.0 | 8,048 | 2,277 |
| **Building raft and TR tunnels** | 352 | 18.5 | 6,512 | 1,842 |
| **Building isolated footings** | 70 | 17.0 | 1,190 | 338 |
| **Power transformer #1 (West)** | 25 | 18.5 | 462 | 130 |
| **Power transformer #2 (Centre)** | 30 | 18.5 | 555 | 157 |
| **Power transformer #3 (East)** | 28 | 18.5 | 518 | 146 |
| **Shunt reactor** | 39 | 14.5 | 566 | 160 |
| **Auxiliary transformer** | 18 | 14.0 | 252 | 72 |
| **Outdoor equipment (four pillars)** | 9 | 12.0 | 108 | 30 |
| **Septic tank** | 1 | 10.0 | 10 | 3 |
| **Total** | 1,075 | – | 18,221 | 5,150 |

## 5.6 Installation and Monitoring

The controlled modulus column is a semi-rigid inclusion formed by drilled displacement, without spoil and with very little vibration, so that it can be installed close to sensitive structures. A specially shaped hollow-stem auger is screwed into the ground to the design depth, displacing the soil laterally rather than removing it and thereby densifying the surrounding ground; the operation is monitored continuously from the rig cab through the torque and the rate of penetration per revolution.

Each 0.6 m-diameter column is set out and then formed in one continuous operation. When the auger reaches the design depth — or a refusal criterion of 5 cm penetration in two consecutive minutes — a highly workable cement mixture (or concrete where required) is pumped through the hollow stem, displacing a sacrificial end cap, and the auger is withdrawn at a rate matched to the grout flow so that the column forms from the toe upward. The reverse-flight auger keeps the soil above compacted and the column is bonded to the ground along its full length. A compacted granular load-transfer platform is then placed over the column network to spread the structural load (Figure 5-9).

The columns were installed by the drilled-displacement method that defines the controlled modulus column. For each column the sequence is: set out the column position; drive a 600 mm displacement auger, closed at its base by a lost cap, into the ground by simultaneous pushing and rotation, displacing the soil laterally rather than removing it, while the drilling parameters (depth, penetration rate and torque) are monitored in real time from the rig cabin; begin feeding the concrete through the hollow stem before the design depth is reached; stop driving when the design depth is reached or the refusal criterion (5 cm penetration over two consecutive minutes) is met; and extract the auger without unscrewing while concrete is pumped under pressure through the stem, the extraction rate being controlled against the concrete flow so that a continuous, full column is formed from the toe upward. The rig then moves to the next column. Because the soil is displaced and not excavated, the technique produces virtually no spoil and is quiet and vibration-free, which suited the aggressive sabkha and the shallow water table. Figure 5-9 shows the three-stage installation schematic.

icinity to sensitive structures.

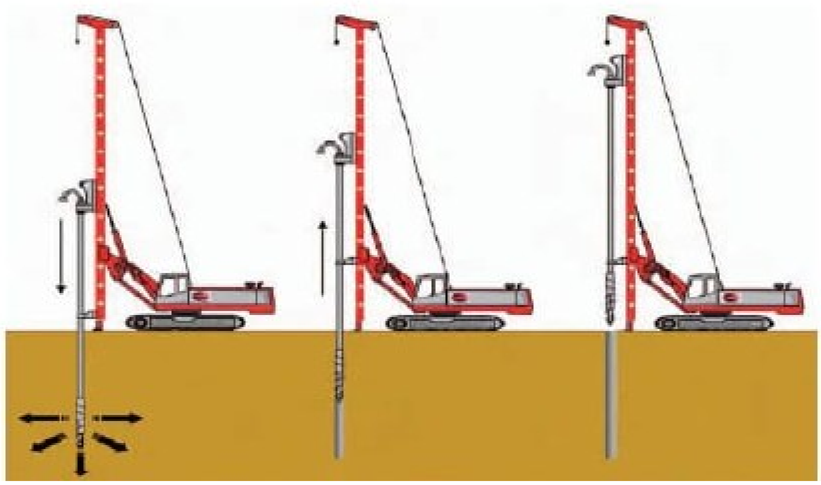


**Figure 5-9.** The drilled-displacement installation sequence of a controlled modulus column: the displacement auger is driven to depth (left), concrete is pumped through the hollow stem (centre), and the auger is extracted while concrete is placed under pressure to form the column (right). *reproduced from the project CMC report (anonymized).*

Every column produced a depth-indexed installation record. Figure 5-10 shows a representative record of the kind captured by the contractor's rig instrumentation: the column profile and the torque pressure, the grout (concrete) pressure and the drilling and extraction speed, all logged continuously against depth, together with the column identifier, the machine, the diameter, the achieved depth and the drilling and concreting durations. This per-column record is precisely the depth-indexed installation coverage that the DIGGS rigid-inclusion extension of Chapter 4 represents through the proposed RI-InstallationRecord object; the case study therefore both exercises and validates that part of the data model. The placed concrete was verified by cylinder compressive tests, with 41 batches sampled over the works.

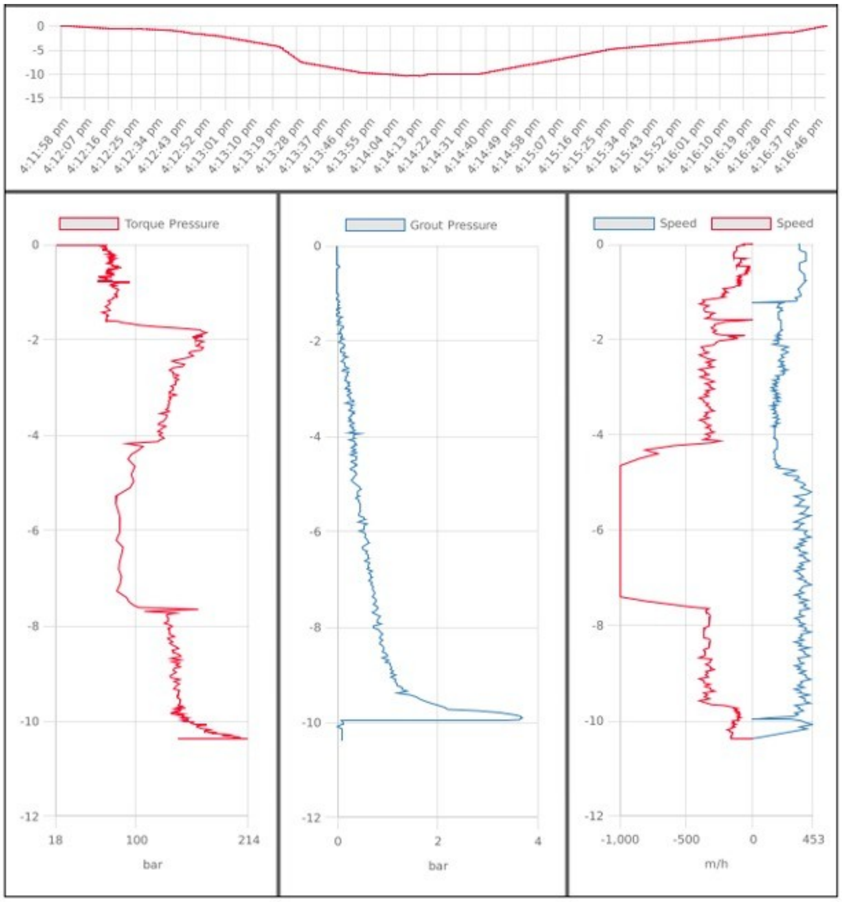



**Figure 5-10.** Representative per-column CMC installation record: the column profile, torque pressure, grout / concrete pressure and drilling / extraction speed logged against depth, with the column identifier, machine, diameter and durations. This is the depth-indexed data captured for every column during installation. *representative CMC installation record (anonymized); the record is a sample of the monitoring data, not from the case-study columns.*

## 5.7 Post-installation Verification and Testing

The completed works were verified by an extensive post-installation testing programme, summarized on the as-built testing layout (Figure 5-11) and in Table 5-13. Four kinds of test were performed: static plate load tests on single column heads, a full-scale zone load test over a group of four columns, low-strain pile-integrity testing on a sample of columns, and compressive-strength testing of the column concrete. The plan positions of the plate, zone and integrity tests, and the standard set-ups for the plate and zone tests, are shown on the layout.

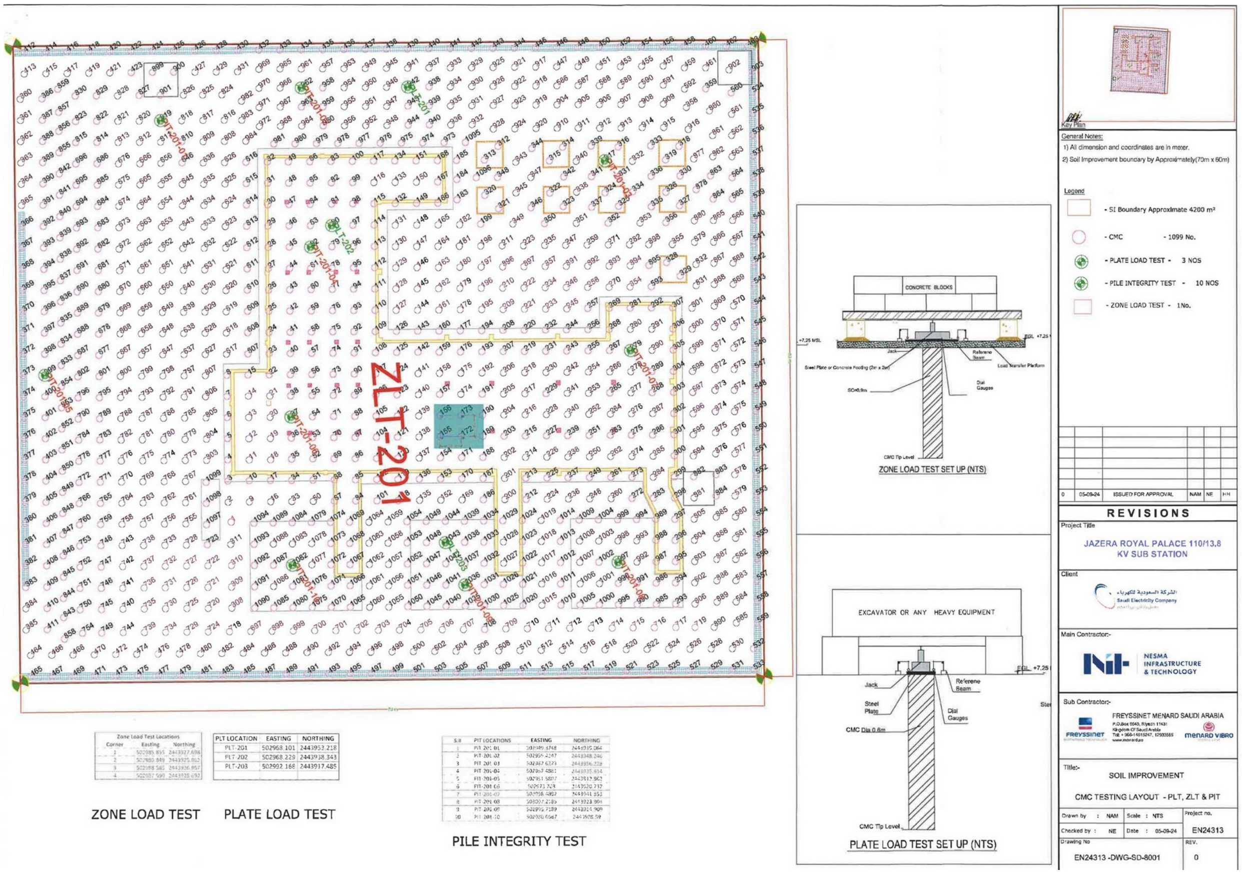



**Figure 5-11.** As-built CMC testing layout over the column grid: locations of the plate load tests (PLT-201 to 204), the zone load test (ZLT-201, over columns 155, 156, 172 and 173) and the pile-integrity tests, with the plate- and zone-load-test set-ups. *(Menard FMSA-24047 report)*

The performance of the improved ground was verified by a programme of load and integrity testing, whose results confirm that the CMC system meets the design requirements with a very wide margin. The programme comprised four plate load tests on single column heads, one zone load test over a group of columns, low-strain integrity testing of a sample of columns, and the concrete cylinder tests noted above.

### 5.7.1 Plate load tests on single columns

Four plate load tests (PLT-201 to 204) were carried out on individual column heads, on columns CMC-1043, CMC-448, CMC-666 and CMC-942, each loaded through a 457 mm plate in seven increments to 225 kPa, that is, to 150 percent of the 150 kPa design pressure, following the static plate-load-test method (ASTM D-1194 / D-1196). The measured settlements at the full test load were 0.45, 0.61, 0.36 and 0.25 mm respectively, all far below the acceptance criteria, and every column rebounded to a residual of 0.10 mm or less on unloading. Figure 5-12 shows the load-settlement curves for all four tests, and Table 5-11 lists the settlements at the design and maximum loads; the near-linear, low-settlement response is characteristic of a stiff rigid inclusion carrying its load to the competent stratum.

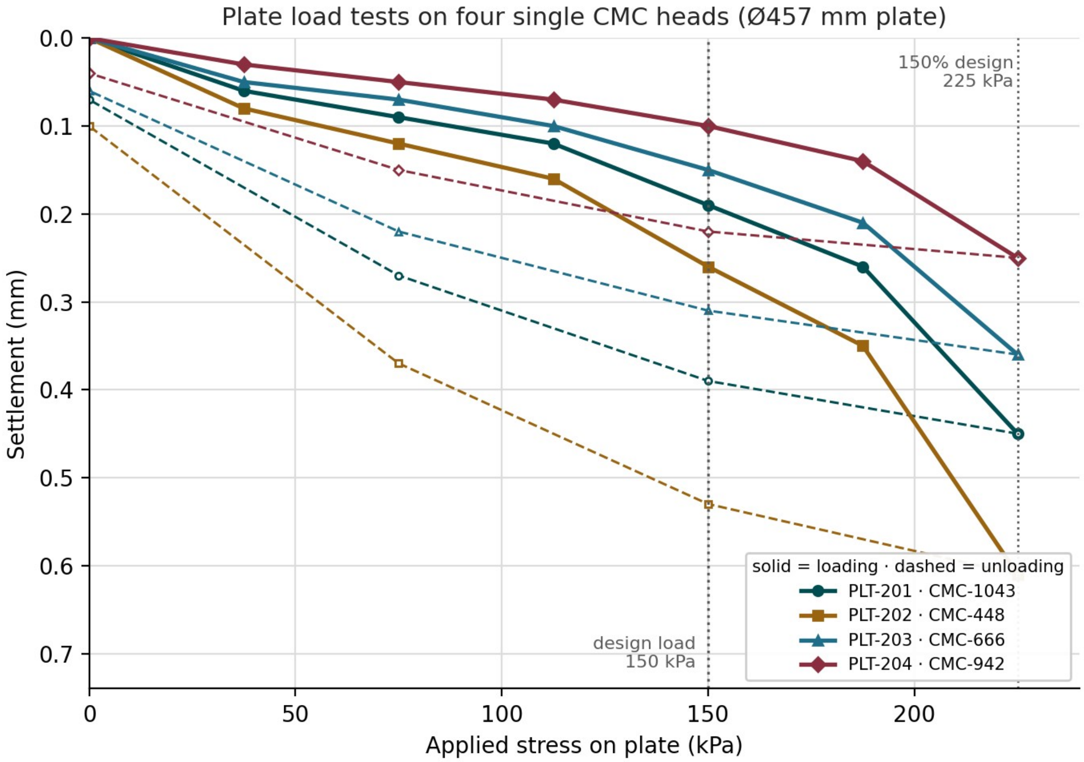



**Figure 5-12.** Plate load tests on four single CMC heads (PLT-201 to 204): applied stress versus settlement, each loaded to 150 percent of the design pressure (225 kPa). Peak settlements ranged from 0.25 to 0.61 mm with almost full rebound on unloading. *measured data plotted from the project plate-load-test records (anonymized).*

**Table 5-11.** Plate-load-test results: settlement of each column head at the design and maximum test loads.

| Test | Column | Plate (mm) | Settlement at 150 kPa (mm) | Settlement at 225 kPa (mm) | Residual after unload (mm) |
|---|---|---|---|---|---|
| **PLT-201** | CMC-1043 | 457 | 0.19 | 0.45 | 0.07 |
| **PLT-202** | CMC-448 | 457 | 0.26 | 0.61 | 0.10 |
| **PLT-203** | CMC-666 | 457 | 0.15 | 0.36 | 0.06 |
| **PLT-204** | CMC-942 | 457 | 0.10 | 0.25 | 0.04 |

Each plate load test was carried on a single column head to 150% of the design pressure. The four tests — PLT-201 on column CMC-1043, PLT-202 on CMC-80, PLT-203 on CMC-666 and PLT-204 on CMC-942 — recorded head settlements of only 0.45, 0.61, 0.36 and 0.25 mm at the 150 kN/m² design pressure, far inside the acceptance limit. The measured pressure–settlement response of PLT-201 is shown in Figure 5-13.

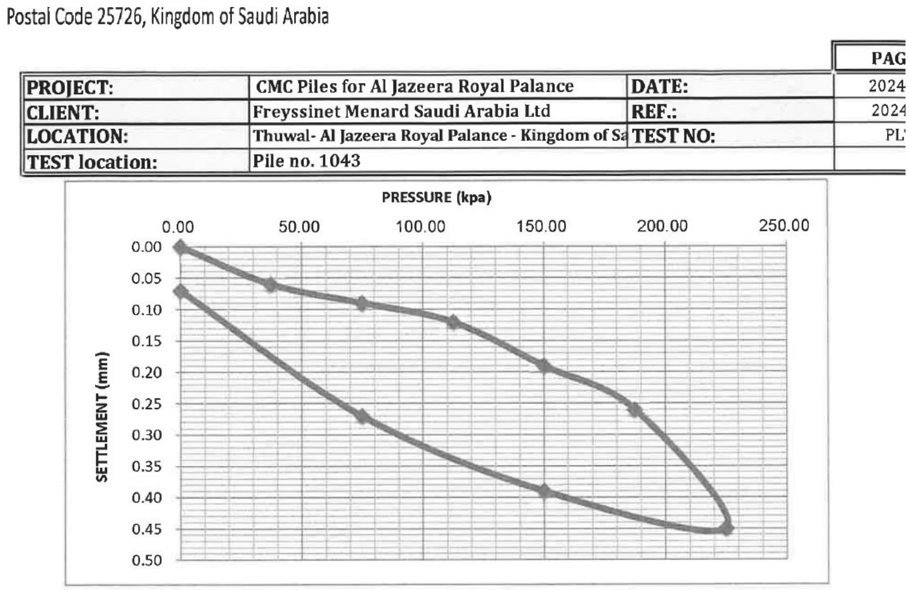



**Figure 5-13.** Plate load test PLT-201 on column head CMC-1043: measured pressure–settlement response to 225 kN/m² (150% of the 150 kN/m² design pressure). *(Applus / Menard FMSA-24047 report)*

### 5.7.2 Zone load test over a column group

A single zone load test (ZLT-201) was performed over a 2.0 m by 2.0 m steel plate bearing on a group of four columns, loaded in six stages to 225 kN/m², again 150 percent of the design pressure, in accordance with the in-situ zone-load-test method (BS 1377 Part 9). The average settlement was 3.36 mm at the design load and 6.29 mm at the maximum test pressure, with a residual settlement of 4.51 mm after unloading, as set out stage by stage in Table 5-12 and shown in Figure 5-14. The zone test loads the composite soil-column-platform system as a whole, rather than a single head, and its modest settlement confirms the global behaviour the design intends.

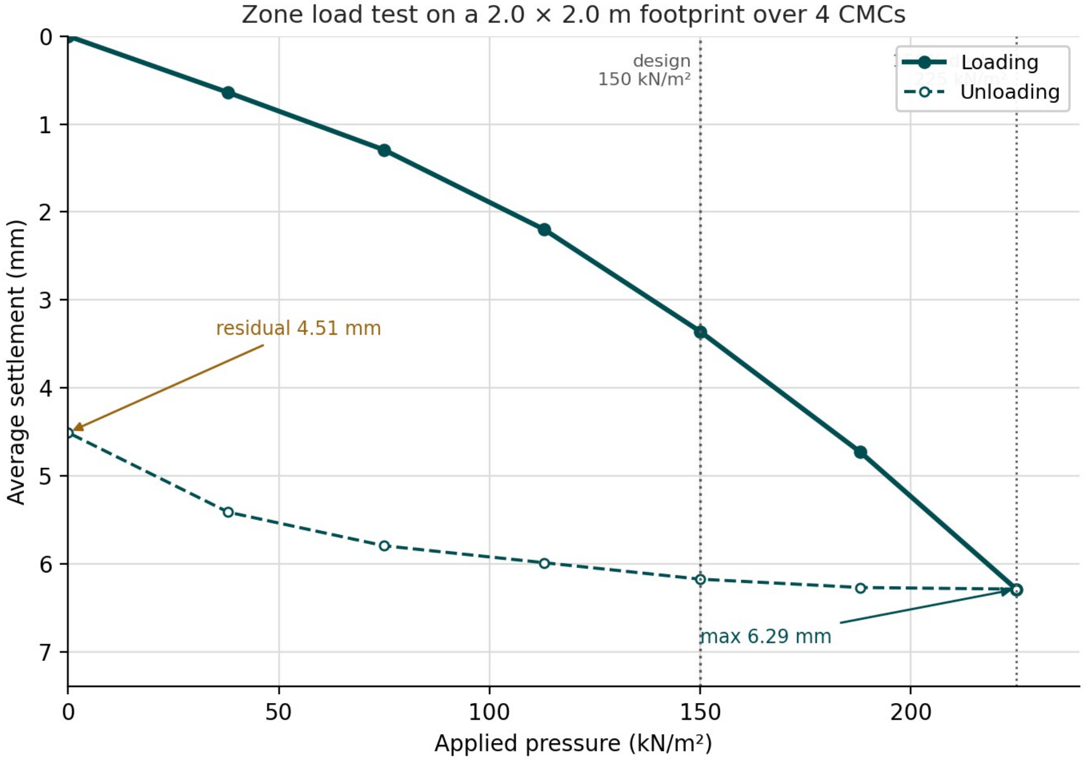



**Figure 5-14.** Zone load test (ZLT-201) on a 2.0 × 2.0 m footprint over four columns: applied pressure versus average settlement, loaded to 150 percent of the design pressure. The composite system settled 3.36 mm at the design load and 6.29 mm at the maximum test pressure. *measured data plotted from the project zone-load-test record (anonymized).*

**Table 5-12.** Zone-load-test (ZLT-201) average settlement at each load stage, loading and unloading.

| Applied pressure (kN/m²) | Settlement, loading (mm) | Settlement, unloading (mm) |
|---|---|---|
| **0** | 0.00 | 4.51 |
| **38** | 0.64 | 5.42 |
| **75** | 1.30 | 5.80 |
| **113** | 2.20 | 5.99 |
| **150 (design)** | 3.36 | 6.18 |
| **188** | 4.73 | 6.27 |
| **225 (max)** | 6.29 | 6.29 |

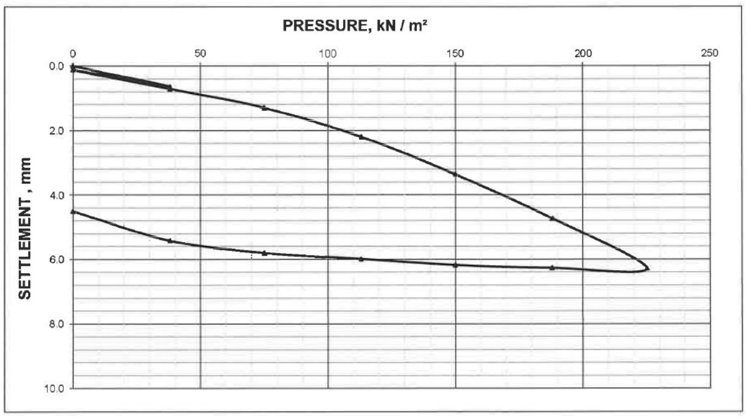
The zone load test ZLT-201 loaded a 2.0 × 2.0 m concrete footing bearing on four columns (CMC 155, 156, 172 and 173) in seven increments to 225 kN/m² — 150% of the design pressure — with holding times of up to two hours per stage (BS 1377, Part 9). The footing settled 3.36 mm at the 150 kN/m² design pressure and 6.29 mm at the 225 kN/m² maximum, recovering to a residual 4.51 mm on unloading, well within the 50 mm serviceability limit. The full pressure–settlement loop is given in Figure 5-15.

GULF CONSULT For Soil & Material Testing

**ZONE LOAD TEST**

**APPLIED PRESSURE VS SETTLEMENT GRAPH**

| | | | |
|---|---|---|---|
| **Client:** | : SAUDI WATER AUTHORITY (SWA) | | |
| **Project** | : AL JAZEERA PALACE SUB STATION PROJECT | | |
| **Location** | : Jeddah, Saudi Arabia | **Steel Test Plate** | : 2.0m x 2.0m |
| **Soil Description** | : CMC#155,156,172,173 | **Job No.** | : 17823/24 |
| | | **Design Load** | : 150 kN/m² |
| **Date of Test** | : 22-Dec-24 | **Max. Test Pressure** | : 225 kN/m² |




| % of Applied Load | Test Pressure, ($kN/m^2$) | Settlement at Applied Test pressure (mm) | Residual Settlement (mm) |
|---|---|---|---|
| 100 | 150 | 3.360 | 4.510 |
| 150 | 225 | 6.290 | |

*Gulf Consult:P.O.Box 684- Al khobar 31952- Tel.8944468/8955036/8949872 - Fax:8949015.*
*Jubail Support Industry: P.O.Box 10113- Jubail 31961- Tel.3410018 - Fax:3410017*

**Figure 5-15.** Zone load test ZLT-201 on a 2.0 × 2.0 m footing over four columns (CMC 155, 156, 172, 173): applied-pressure–settlement loop to 225 kN/m², giving 3.36 mm at the design pressure and 6.29 mm at 150%. *(Gulf Consult / Menard FMSA-24047 report)*

### 5.7.3 Integrity and material testing

The integrity of the columns was checked by low-strain pile-integrity testing (PIT). Fifty-five columns, spread across the site, were tested with a hand-held hammer and accelerometer (assuming a sound-concrete wave velocity of 4000 m/s); the reflected traces were reviewed against the sound-pile response to confirm continuity and the absence of significant reductions in cross-section. A representative reflectogram is shown in Figure 5-16.

| Details | Pile |
|---|---|
| 12/21/2024<br>C:4000m/s<br>Amp:10<br>F:0<br>Planned length:10.0m<br>Avg:15 | **572** |

Reflectogram

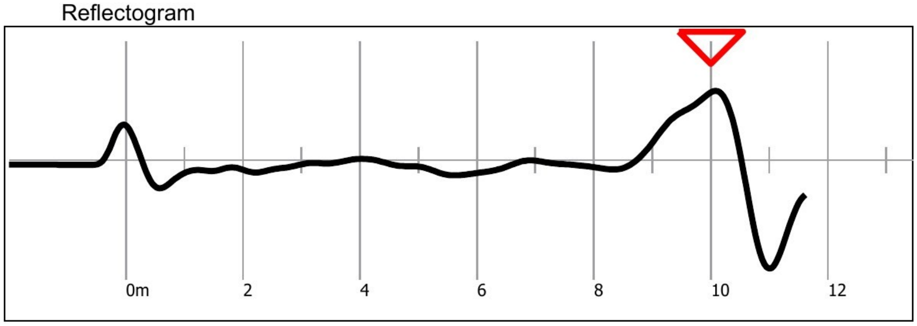



**Figure 5-16.** Sample low-strain pile-integrity (PIT) reflectogram for column 572 (planned length 10.0 m): a clear reflection from the toe and no significant intermediate reflection, indicating a sound, continuous column. All 55 columns tested fell within permissible limits. *(Applus / Menard FMSA-24047 report)*

Column concrete was sampled continuously through the works and tested for compressive strength. Across 41 casting batches covering the production columns, the 28-day strength averaged 46.6 MPa and ranged from 40.7 to 54.5 MPa — every batch comfortably above the 25 MPa minimum specified for the column material — with 7-day strengths averaging about 35 MPa. Figure 5-17 plots the batch results, and Table 5-13 summarizes the whole verification programme.

The structural continuity of the columns was checked by the low-strain pile-integrity test (PIT) method. Fifty-five columns were tested, using an instrumented hand-held hammer and an accelerometer with the PIT collector and a sound-concrete wave velocity of 4,000 m/s. Every one of the fifty-five records returned a clear, continuous reflectogram over the 10 to 18.5 m column lengths, with no indication of necking, cracking or discontinuity, confirming the integrity of the installed columns.

The placed concrete was verified by cylinder compressive-strength tests, in 41 batches representing the column population, tested at 7 and 28 days against the 25 MPa design requirement. The results, shown in Figure 5-17, are comfortably above the requirement in every batch: the 28-day strength ranges from 40.7 to 54.5 MPa with a mean of 46.6 MPa, about 1.9 times the specified value, while the 7-day strength already averages 35.2 MPa. All 41 batches passed. Taken together, the plate, zone, integrity and material tests provide the four pillars of a rigid-inclusion verification record, the as-built survey, the integrity testing, the material testing and the load testing, each of which the standardized post-installation data set of Chapter 4 is designed to carry. Table 5-13 summarizes the programme and its results.

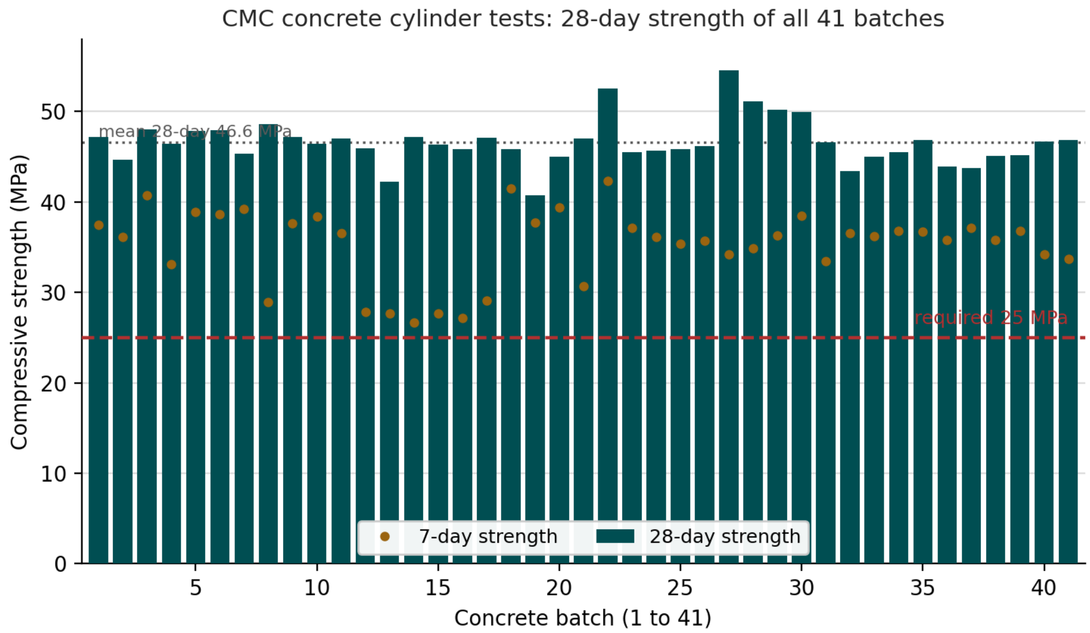



**Figure 5-17.** CMC concrete cylinder tests: the 28-day compressive strength of all 41 batches (bars) and the 7-day strength (points), against the 25 MPa design requirement. Every batch exceeds the requirement, with a 28-day mean of 46.6 MPa. *measured data plotted from the project concrete test records (anonymized).*

**Table 5-13.** Summary of the post-installation verification programme and results.

| Test | Number | Standard | Result |
|---|---|---|---|
| **Plate load test (single column)** | 4 | ASTM D-1194 / D-1196 | 0.25 to 0.61 mm at 150% design load; all pass |
| **Zone load test (4-column group)** | 1 | BS 1377 Part 9 | 3.36 mm at design; 6.29 mm at 150% design |
| **Pile integrity test (low-strain)** | 55 | Low-strain method | All 55 columns continuous; no defects (4,000 m/s) |
| **Concrete cylinder compressive test** | 41 batches | ASTM | 28-day 40.7 to 54.5 MPa (mean 46.6); all 41 pass |

## 5.8 Summary

This case study is a complete rigid-inclusion project on difficult ground: a settlement-sensitive electrical substation founded on a very loose, aggressive sabkha with shallow groundwater, improved with 1,075 controlled modulus columns beneath a load-transfer platform and verified by load and integrity testing. It exercises all four phases of the framework developed in this report. The pre-installation phase is represented by the borehole, SPT, CPT, geophysical and laboratory data and the aggressive-ground chemistry; the design phase by the ASIRI-based CMC design, the column geometry by zone and the platform thickness by structure; the installation phase by the drilled-displacement method and the depth-indexed per-column monitoring record; and the post-installation phase by the plate load, zone load, integrity and concrete tests, whose small measured settlements confirm the design. Read against Chapters 2 and 4, the project shows that the standardized parameter set and its mapping onto DIGGS can capture a real rigid-inclusion project end to end, from the ground model through the design and the as-built columns to their verification, which is precisely the demonstration the case study is intended to provide.

# References

ASIRI (2012). Recommendations for the design, construction and control of rigid inclusion ground improvements. IREX / Presses des Ponts, Paris.

ASTM D-1194. Standard test method for bearing capacity of soil for static load and spread footings (plate load test). ASTM International, West Conshohocken, PA.

ASTM D-1586. Standard test method for standard penetration test (SPT) and split-barrel sampling of soils. ASTM International, West Conshohocken, PA.

BSI (2010). BS 8006-1:2010, Code of practice for strengthened/reinforced soils and other fills. British Standards Institution, London.

BSI. BS 1377, Methods of test for soils for civil engineering purposes. British Standards Institution, London.

NF P94-262. Justification of geotechnical works, deep foundations. AFNOR, France.

SBC (2018). Saudi Building Code for Soils and Foundations, SBC 303-CR. Saudi Building Code National Committee, Riyadh.

**CHAPTER 6**

# DIGGS Schema Extensions for Rigid Inclusions

***This chapter will be developed by Dan after handling the XML.***

*The report has moved, chapter by chapter, from practice to data: the state of the art and the two contractors' methods in Chapter 3, the four-phase methodology and its 235 standardized parameters in Chapter 2, the essential elements in Chapter 4, and a fully documented project in Chapter 5. This chapter draws those threads together into the schema itself. It sets out the rigid-inclusion extension to DIGGS, which is being developed as the DIGGS 3.0 Extension, and explains, element by element, how a rigid-inclusion system is represented. It then delivers one DIGGS instance that encodes the Chapter 5 case study in that model and validates against the schema. The central idea is simple and is the key to everything that follows: a rigid inclusion is a constructed ground feature that is surveyed, logged and tested, so DIGGS models it as a sampling feature, the same family as a borehole, and not as a driven pile.*

## 6.1 Purpose and Scope

A data standard earns its place only when it can carry a real project without loss and be read back without ambiguity. Chapters 2 and 4 established what a rigid-inclusion project must record, 235 essential elements across the pre-installation, design, installation and post-installation phases, and found that most already have a home in DIGGS while a concentrated minority do not. This chapter turns that finding into a concrete schema, describing the rigid-inclusion module being developed in the DIGGS 3.0 Extension and how a project is expressed in it.

The chapter has two deliverables. The first is the description of the rigid-inclusion extension, element by element, presented in prose and diagrams rather than as raw schema text, so that the intent is clear to engineers and to the schema-development committee alike. The second is a single XML instance, delivered with this report as CaseStudy-RI-DIGGS.xml, that encodes the whole Chapter 5 substation project, all seven boreholes with their tests, every platform zone and every one of the roughly 1,075 columns with their construction, and the full verification programme, using the extension elements so that it validates against the 3.0 Extension schema. Reading the two together shows both the model and a complete worked example of it.

One principle governs the design and should be stated at the outset. The extension describes the rigid-inclusion system as it is designed, built and verified; it does not re-implement any design method. DIGGS carries the grid, the diameter, the target load, the placed grout volume, the measured settlement and the acceptance criterion; it does not carry the ASIRI or BS 8006 calculation that produced them. This keeps the schema stable across standards, as Chapter 4 required, and keeps the extension focused.

## References

APPENDIX A

# Essential Rigid-Inclusion Parameters

The standardized rigid-inclusion parameter set (RI-DIGGS_Essential_Parameters V2.xlsx), summarized by phase. For every element the workbook gives the symbol, the unit, the design standard(s) that require it, the element's location in the DIGGS schema, and whether that element already exists in DIGGS or is new.

# Essential Rigid-Inclusion Parameters by Phase

*Standardizing Rigid Inclusion Soil Improvement Data with DIGGS · Technical Report*

This workbook summarizes the 235 essential data elements of a rigid-inclusion project, one sheet per project phase. For every element it gives the symbol, the unit, the design standard(s) that require it, the element's location in the DIGGS schema, and whether that element already exists in DIGGS or is new.

| Phase | Parameters | Existing | Bridged | New | Notes |
|---|---|---|---|---|---|
| Pre-installation | 49 | 48 | 0 | 1 | |
| Design | 97 | 4 | 11 | 82 | |
| Installation | 52 | 17 | 1 | 34 | |
| Post-installation | 37 | 29 | 3 | 5 | |
| **Total** | **235** | **98** | **15** | **122** | |

## Legend

| | |
|---|---|
| **New** | New rigid-inclusion element that should be added to the schema ( DIGGS 3.0). |
| **Existing** | Carried by a DIGGS feature that already exists; reused unchanged. |
| **Bridge** | Hosted by an existing DIGGS feature with a minor, backward-compatible addition. |

*Standards: ASIRI (2012, France), BS 8006-1 (2010, UK), EBGEO (2011, Germany), FHWA GEC-13 (2017, USA); EC7 = Eurocode 7; ASTM / BS / EN = referenced test methods.*

| Group | Element (parameter) | Symbol | Unit | Required by (standard) | DIGGS schema location | New element? | Definition |
|---|---|---|---|---|---|---|---|
| A · Project & investigation context | **Project / job identification** | – | – | ASIRI, BS 8006-1, EBGEO, FHWA; EC7 | Project | **Existing** | Project name, number and owner under which the RI works are executed. |
| A · Project & investigation context | **Site location & coordinate system** | – | – | ASIRI, BS 8006-1, EBGEO, FHWA; EC7 | Project / PointLocation | **Existing** | Geographic position and the horizontal/vertical CRS (datum) of the works. |
| A · Project & investigation context | **Investigation program & standard** | – | – | ASIRI; FHWA; EC7 | DocumentInformation / SamplingActivity | **Existing** | Scope, methods and governing standard of the ground investigation (e.g. EC7-2, ASTM, ISO). |
| A · Project & investigation context | **Investigation point (borehole / sounding / pit) ID, type, coordinates, date** | – | – | EC7; ASTM D420 | Borehole / Sounding / TrialPit | **Existing** | Identity, kind, position and date of each exploration location. |
| A · Project & investigation context | **Investigation density / depth** | – | m | ASIRI | Borehole / Sounding | **Existing** | Number of points per area and termination depth (below deepest soft layer). |
| A · Project & investigation context | **Existing fill / made-ground / obstruction survey** | – | m | ASIRI | TrialPit / LithologyObservation | **Existing** | Presence, depth and nature of fill, boulders, cobbles or old foundations. |
| B · Stratigraphy, profile & groundwater | **Stratigraphic / lithological profile** | – | m | ASIRI, BS 8006-1, EBGEO, FHWA; EC7 | LithologyObservation / StratigraphyObservation / | **Existing** | Sequence, top/bottom levels and thickness of each soil & rock unit. |
| B · Stratigraphy, profile & groundwater | **Soft-layer thickness heterogeneity / depth-to-bearing variability** | – | m | ASIRI | GeoUnitObservation / StratigraphyObservation | **Existing** | Lateral variation of soft-layer thickness and bearing depth across the RI footprint. |
| B · Stratigraphy, profile & groundwater | **Soil description & classification** | – | – | EC7; ASTM D2487/D2488 | LithologyObservation / ConstituentObservation / ColorObservation | **Existing** | Field/lab description and classification group (e.g. USCS) of each layer. |
| B · Stratigraphy, profile & groundwater | **Groundwater / piezometric level** | – | m | ASIRI, BS 8006-1, EBGEO, FHWA | Well / WaterLevelMonitoring / Monitor | **Existing** | Depth (or head) of the water table and any perched/artesian conditions. |
| B · Stratigraphy, profile & groundwater | **Bearing (founding) stratum, level & description** | – | m | ASIRI, BS 8006-1, EBGEO, FHWA | GeoUnitObservation / StratigraphyObservation | **Existing** | Depth, description and competence of the layer that provides toe resistance. |
| C · Index & classification properties (laboratory) | **Natural water content** | $w$ | % | EC7; ASTM D2216 | WaterContentTest | **Existing** | Mass of pore water as a percentage of dry solids. |
| C · Index & classification properties (laboratory) | **Atterberg limits (LL, PL, PI)** | $w_L$, $w_P$, $I_P$ | % | EC7; ASTM D4318 | AtterbergLimitsTest | **Existing** | Liquid limit, plastic limit and plasticity index of fine soils. |
| C · Index & classification properties (laboratory) | **Particle-size distribution / fines content** | – | % | ASTM D6913/D7928 | ParticleSizeTest | **Existing** | Grain-size grading, including % passing 0.075 mm. |
| C · Index & classification properties (laboratory) | **Bulk & dry unit weight** | $\gamma$, $\gamma_d$ | $kN/m^3$ | ASIRI, BS 8006-1, EBGEO, FHWA; ASTM D7263 | LabDensityTest | **Existing** | Total and dry weight per unit volume of the soil. |
| C · Index & classification properties (laboratory) | **Specific gravity of solids** | $G_s$ | – | ASTM D854 | SpecificGravityTest | **Existing** | Ratio of solid density to water density. |
| C · Index & classification properties (laboratory) | **Initial void ratio** | $e_0$ | – | ASIRI | LabDensityTest / SpecificGravityTest | **Existing** | Ratio of void volume to solid volume before loading. |
| C · Index & classification properties (laboratory) | **Organic / peat content** | – | % | ASIRI; ASTM D2974 | LossOnIgnitionTest | **Existing** | Fraction of organic matter (loss on ignition). |
| D · Strength properties | **Undrained shear strength** | $c_u$ ($s_u$) | kPa | ASIRI, BS 8006-1, EBGEO, FHWA | TriaxialTest / InsituVaneTest / UnconfinedCompressiveStrengthTest | **Existing** | Undrained shear resistance of cohesive soft soil. |
| D · Strength properties | **Effective strength parameters** | $c'$, $\varphi'$ | kPa, ° | ASIRI, BS 8006-1, EBGEO, FHWA | TriaxialTest / DirectShearTest | **Existing** | Effective cohesion and friction angle from drained/CU tests. |
| D · Strength properties | **Critical-state / constant-volume friction angle** | $\varphi'_{cv}$ | ° | BS 8006-1; FHWA | TriaxialTest / DirectShearTest | **Existing** | Friction angle at large strain / constant volume. |
| D · Strength properties | **Sensitivity** | $S_t$ | – | EC7 | InsituVaneTest / LabVaneTest | **Existing** | Ratio of undisturbed to remoulded undrained strength. |
| D · Strength properties | **In-situ friction angle & relative density of granular / bearing strata** | $\varphi'$, $D_r$ | °, % | ASIRI; FHWA | StaticConePenetrationTest / DrivenPenetrationTest / | **Existing** | Interpreted effective friction angle and density of sands / the bearing stratum. |
| D · Strength properties | **At-rest earth-pressure coefficient** | $K_0$ | – | ASIRI; EC7 | FlatPlateDilatometerTest | **Existing** | Ratio of horizontal to vertical effective stress at rest. |
| E · Compressibility & consolidation (oedometer) | **Compression index** | $C_c$ | – | ASIRI; FHWA; ASTM D2435 | ConsolidationTest | **Existing** | Slope of e–log σ' on the virgin compression line. |
| E · Compressibility & consolidation (oedometer) | **Recompression / swelling index** | $C_s$ ($C_r$) | – | FHWA; ASTM D2435 | ConsolidationTest | **Existing** | Slope of e–log σ' on unload/reload. |
| E · Compressibility & consolidation (oedometer) | **Secondary-compression (creep) index** | $C_\alpha$ | – | ASIRI; EC7 | ConsolidationTest | **Existing** | Slope of e–log t after primary consolidation. |
| E · Compressibility & consolidation (oedometer) | **Preconsolidation pressure** | $\sigma'_p$ | kPa | ASIRI; FHWA | ConsolidationTest | **Existing** | Maximum past effective vertical stress. |
| E · Compressibility & consolidation (oedometer) | **Overconsolidation ratio** | OCR | – | ASIRI | ConsolidationTest | **Existing** | Ratio of preconsolidation to current effective stress. |
| E · Compressibility & consolidation (oedometer) | **Coefficient of volume compressibility** | $m_v$ | $MPa^{-1}$ ($m^2/MN$) | FHWA; EC7 | ConsolidationTest | **Existing** | Volumetric strain per unit effective-stress increment. |
| E · Compressibility & consolidation (oedometer) | **Constrained / oedometric modulus** | $E_{oed}$ | MPa | ASIRI | ConsolidationTest | **Existing** | One-dimensional (confined) stiffness modulus. |
| E · Compressibility & consolidation (oedometer) | **Coefficient of consolidation** | $c_v$ | $m^2/yr$ | ASIRI; EC7 | ConsolidationTest | **Existing** | Rate of primary (vertical) consolidation. |
| E · Compressibility & consolidation (oedometer) | **Coefficient of horizontal consolidation** | $c_h$ ($k_h$) | $m^2/yr$ (m/s) | EC7 | PorePressureDissipationTest / ConsolidationTest | **Existing** | Rate of radial consolidation / horizontal permeability. |
| E · Compressibility & consolidation (oedometer) | **Permeability** | $k$ ($k_v$) | m/s | ASIRI | InsituPermeabilityTest / LabPermeabilityTest / PumpingTest / | **Existing** | Hydraulic conductivity of the soft soil. |
| F · In-situ test data | **CPT/CPTu cone resistance** | $q_c$ ($q_t$) | MPa | ASIRI; ASTM D5778; Project spec/ITP | Sounding / StaticConePenetrationTest | **Existing** | (Corrected) cone tip resistance vs depth. |
| F · In-situ test data | **CPT sleeve friction & friction ratio** | $f_s$, $R_f$ | kPa, % | ASTM D5778 | StaticConePenetrationTest | **Existing** | Sleeve friction and its ratio to cone resistance. |
| F · In-situ test data | **CPTu pore pressure** | $u_2$ | kPa | ASTM D5778 | StaticConePenetrationTest / PorePressureDissipationTest | **Existing** | Penetration pore pressure behind the cone. |

| Group | Element (parameter) | Symbol | Unit | Required by (standard) | DIGGS schema location | New element? | Definition |
|---|---|---|---|---|---|---|---|
| F · In-situ test data | **SPT blow count** | N ($N_{60}$) | blows/300 mm | EC7; ASTM D1586 | Borehole / DrivenPenetrationTest | **Existing** | Standard penetration resistance. |
| F · In-situ test data | **Pressuremeter (Ménard) modulus** | $E_M$ | MPa | ASIRI; ISO | PressuremeterTest | **Existing** | Ménard pressuremeter deformation modulus. |
| F · In-situ test data | **Pressuremeter limit pressure** | $p_l$ ($p_l$*) | kPa | ASIRI | PressuremeterTest | **Existing** | (Net) limit pressure from the pressuremeter. |
| F · In-situ test data | **Field vane shear strength** | $c_{u,vane}$ | kPa | ASIRI; ASTM D2573 | InsituVaneTest | **Existing** | In-situ undrained strength by vane rotation. |
| F · In-situ test data | **Shear-wave velocity / small-strain modulus [I]** | $V_s$, $G_{max}$ | m/s, MPa | ASIRI | GeophysicalFieldSurvey / LabVelocityTest | **Existing** | Shear-wave velocity and derived small-strain shear modulus. |
| G · Environmental / durability inputs | **Sulfate content (soil & groundwater)** | $SO_4^{2-}$ | mg/kg, mg/L | ASIRI; Project spec/ITP | LabChemicalTest / EnvironmentalScreeningTest | **Existing** | Sulfate concentration governing concrete attack class. |
| G · Environmental / durability inputs | **Chloride content** | $Cl^-$ | mg/kg, mg/L | ASIRI | LabChemicalTest | **Existing** | Chloride concentration (corrosion of any reinforcement). |
| G · Environmental / durability inputs | **pH of soil / groundwater** | pH | – | ASIRI | LabChemicalTest / RedoxTest | **Existing** | Acidity/alkalinity of the ground environment. |
| G · Environmental / durability inputs | **Aggressivity / exposure classification** | – | – | ASIRI | LabChemicalTest | **Existing** | Derived aggressivity (exposure) class of soil & water. |
| H · Seismic & special inputs (conditional) | **Seismic site class / ground type** | – | – | ASIRI | GeophysicalFieldSurvey | **Existing** | Site classification for seismic action (Vs30 / soil type). |
| H · Seismic & special inputs (conditional) | **Design ground acceleration / hazard** | $a_g$ (PGA) | g | ASIRI | AbstractProgramDesign | **New** | Peak/desing ground acceleration for the site. |
| H · Seismic & special inputs (conditional) | **Liquefaction-susceptibility inputs** | – | – | ASIRI | StaticConePenetrationTest / DrivenPenetrationTest / ParticleSizeTest | **Existing** | Data (grading, N/q_c, GW) to assess liquefiable layers. |

| Group | Element (parameter) | Symbol | Unit | Required by (standard) | DIGGS schema location | New element? | Definition |
|---|---|---|---|---|---|---|---|
| A · Design basis & method (make the design self-describing) | **Governing design standard(s) & edition** | – | – | ASIRI, BS 8006-1, EBGEO, FHWA | AbstractProgramDesign | **New** | The code(s)/recommendations the design follows and their edition. |
| A · Design basis & method (make the design self-describing) | **Design method (coded)** | – | – | ASIRI, BS 8006-1, EBGEO, FHWA | AbstractProgramDesign | **New** | Analysis approach used (unit-cell / arching / FEM / LDC). |
| A · Design basis & method (make the design self-describing) | **Supported structure type (load source)** | – | – | ASIRI; FHWA; Han (2025) | AbstractProgramDesign | **New** | The structure imposing load on the RI system, isolated / strip footing, mat (raft), slab-on-grade, embankment (road / rail), retaining wall, tank / silo, box culvert, or bridge foundation / approach. |
| A · Design basis & method (make the design self-describing) | **RI type / installation-technique family (coded)** | – | – | FHWA; Han (2025) | AbstractProgramDesign | **New** | Product / technique family of the inclusion, drilled displacement column (DDC), auger-cast-in-place (ACIP), controlled modulus column (CMC), continuous flight auger (CFA), vibro concrete column (VCC), or driven / precast. |
| A · Design basis & method (make the design self-describing) | **Project delivery / contracting method** | – | – | Han (2025) | Project / BusinessAssociate | **Existing** | Contracting route, design-build or design-bid-build, and whose provisions govern (owner/agency vs contractor). |
| A · Design basis & method (make the design self-describing) | **Arching model (coded)** | – | – | ASIRI, BS 8006-1, EBGEO, FHWA | AbstractProgramDesign | **New** | Load-transfer model (Prandtl / Marston / Hewlett-Randolph / Zaeske / Terzaghi). |
| A · Design basis & method (make the design self-describing) | **Safety format / design approach** | – | – | ASIRI, BS 8006-1, EBGEO, FHWA | AbstractProgramDesign | **New** | EC7 DA2/DA3 · BS limit-state · DIN load cases · allowable+FS. |
| A · Design basis & method (make the design self-describing) | **Design working life** | $t_L$ | yr | BS 8006-1; EBGEO; FHWA | AbstractProgramDesign | **New** | Intended service life of the works. |
| A · Design basis & method (make the design self-describing) | **Consequence / reliability class** | – | – | ASIRI; BS 8006-1; EBGEO | AbstractProgramDesign | **New** | Domain (ASIRI) · f_n category (BS) · load case (EBGEO). |
| A · Design basis & method (make the design self-describing) | **Partial / global factor set** | γ, f | – | ASIRI, BS 8006-1, EBGEO, FHWA | AbstractProgramDesign | **New** | The named factors and values applied to actions, materials, resistances. |
| A · Design basis & method (make the design self-describing) | **Parameter provenance (PMT / CPT / SPT / lab)** | – | – | ASIRI; EBGEO; FHWA | Sounding / Borehole | **Existing** | Which test each soil parameter came from. |
| A · Design basis & method (make the design self-describing) | **Method-applicability gate & geometry envelope** | $k_{s,T}/k_s$ | – | ONLY EBGEO | AbstractProgramDesign | **New** | Stiffness-ratio gate (≥75) and geometric limits (h/(s−d), d/s, s_x/s_y, z) validating the method. |
| B · Geometry & layout | **Support (treated) area** | $A_{RI}$ | m² | ASIRI; EBGEO; FHWA | AbstractPlanarSamplingFeature | **New** | Plan extent actually treated with inclusions; commonly wider than the loaded footprint of the structure or embankment. |
| B · Geometry & layout | **Grid pattern (square / triangular)** | – | – | ASIRI; EBGEO; FHWA | AbstractProgramDesign | **New** | Plan arrangement of the inclusions. |
| B · Geometry & layout | **Spacing (centre-to-centre)** | s ($s_x$, $s_y$) | m | BS 8006-1; FHWA | AbstractProgramDesign | **New** | Distance between adjacent inclusions. |
| B · Geometry & layout | **Inclusion diameter** | D (d) | m | ASIRI; EBGEO; FHWA | AbstractProgramDesign | **New** | Nominal shaft diameter of the inclusion. |
| B · Geometry & layout | **Inclusion length / toe depth** | L | m | ASIRI | AbstractProgramDesign | **New** | Design length from cut-off to toe. |
| B · Geometry & layout | **Equivalent / tributary diameter** | $d_{Ers}$, $D_e$ | m | EBGEO; FHWA | AbstractProgramDesign | **New** | Diameter of the unit-cell area assigned to one inclusion. |
| B · Geometry & layout | **Head / cap present & size** | a | m | BS 8006-1; EBGEO; FHWA | AbstractProgramDesign | **New** | Presence and plan size of any head/cap. |
| B · Geometry & layout | **Area replacement ratio** | α ($a_s$) | % | ASIRI; EBGEO; FHWA | AbstractProgramDesign | **New** | Ratio of inclusion (cap) area to tributary area. |
| B · Geometry & layout | **Critical (full-arching) height** | $H_{crit}$ | m | BS 8006-1; EBGEO; FHWA | AbstractProgramDesign | **New** | Fill height above which surface differential vanishes. |
| B · Geometry & layout | **Embankment side-slope / batter** | n (β) | – (or °) | BS 8006-1; FHWA | AbstractProgramDesign | **New** | Side-slope ratio (n horizontal : 1 vertical) of the embankment. |
| B · Geometry & layout | **Position tolerance** | – | mm | ASIRI | AbstractProgramSpecificaton | **New** | Permitted plan deviation of the head. |
| B · Geometry & layout | **Verticality / inclination tolerance** | – | % | ASIRI | AbstractProgramSpecificaton | **New** | Permitted deviation from vertical. |
| C · Inclusion, material & structural | **Material (concrete / grout / mortar)** | – | – | ASIRI; EBGEO; FHWA | DesignGroutMix | **Existing** | Type of column material. |
| C · Inclusion, material & structural | **Characteristic compressive strength** | $f_{ck}$ ($f_{ck}$*) | MPa | ASIRI; FHWA | DesignGroutMix | **Existing** | Characteristic (reduced) compressive strength. |
| C · Inclusion, material & structural | **Design compressive strength** | $f_{cd}$ | MPa | ASIRI | AbstractProgramDesign | **New** | Factored design compressive strength. |
| C · Inclusion, material & structural | **Mean-stress limit (fixed cap)** | $\bar{\sigma}_{lim}$ | MPa | ASIRI | AbstractProgramDesign | **New** | Fixed limit on mean compressive stress (7 MPa). |
| C · Inclusion, material & structural | **Elastic modulus** | E | GPa | ASIRI; FHWA | DesignGroutMix / AbstractProgramDesign | **Bridge** | Young's modulus of the column material. |

| Group | Element (parameter) | Symbol | Unit | Required by (standard) | DIGGS schema location | New element? | Definition |
|---|---|---|---|---|---|---|---|
| C · Inclusion, material & structural | **Column axial capacity (structural)** | $R_c$ | kN | ASIRI; FHWA | AbstractProgramDesign | **New** | Structural axial resistance of the column. |
| C · Inclusion, material & structural | **Column axial utilization** | $N/R_c$ | – | ASIRI | AbstractProgramDesign | **New** | Ratio of axial demand to capacity. |
| C · Inclusion, material & structural | **Buckling check (soft-soil)** | – | – | ASIRI | AbstractProgramDesign | **New** | Verification against buckling in very soft soil. |
| C · Inclusion, material & structural | **Shear check** | $\tau_{cp}$ | MPa | ASIRI | AbstractProgramDesign | **New** | Shear stress verification of the section. |
| C · Inclusion, material & structural | **Column reinforcement (Domain 1 / tension zones)** | $\rho_s$, $A_s$, $L_{reinf}$ | %, mm², m | ASIRI; Han (2025) | AbstractProgramDesign | **New** | Steel ratio, area and reinforced length where the section is not fully compressed or where bending governs. |
| C · Inclusion, material & structural | **Lateral / bending demand on RI (eccentric · lateral · seismic)** | M, $H_{lat}$ | kN·m, kN | ASIRI; Han (2025) | AbstractProgramDesign | **New** | Bending moment and lateral force an RI develops under eccentric, lateral or seismic loading (edge columns, slopes, walls, seismic). |
| D · Shaft, soil interface & neutral plane | **Pressuremeter (Ménard) modulus (design input)** | $E_M$ | MPa | ASIRI | PressuremeterTest | **Bridge** | Soil deformation modulus adopted as a design input. |
| D · Shaft, soil interface & neutral plane | **Shaft load-transfer modulus** | $k_\tau$ | kN/m³ | ASIRI | AbstractProgramDesign | **New** | Mobilisation stiffness of shaft friction. |
| D · Shaft, soil interface & neutral plane | **Tip load-transfer modulus** | $k_q$ | kN/m³ | ASIRI | AbstractProgramDesign | **New** | Mobilisation stiffness of tip resistance. |
| D · Shaft, soil interface & neutral plane | **Negative skin-friction limit** | $q_s$ | kPa | ASIRI | AbstractProgramDesign | **New** | Limiting downdrag shear stress on the shaft. |
| D · Shaft, soil interface & neutral plane | **Neutral-plane depth** | $h_c$ ($z_n$) | m | ASIRI | AbstractProgramDesign | **New** | Depth where relative soil/column movement reverses. |
| D · Shaft, soil interface & neutral plane | **Max axial force / downdrag at neutral plane** | $Q_{max}$ | kN | ASIRI | AbstractProgramDesign | **New** | Peak axial force including downdrag. |
| E · Toe / base (end bearing) | **Bearing-stratum level** | – | m | ASIRI; FHWA | GeoUnitObservation | **Bridge** | Elevation of the competent founding layer. |
| E · Toe / base (end bearing) | **Toe / base resistance** | $R_b$ | kN | ASIRI, BS 8006-1, EBGEO, FHWA | AbstractProgramDesign | **New** | Ultimate end-bearing resistance at the toe. |
| E · Toe / base (end bearing) | **Shaft (positive) resistance** | $R_s$ | kN | ASIRI | AbstractProgramDesign | **New** | Ultimate skin-friction resistance along the shaft below the neutral plane. |
| E · Toe / base (end bearing) | **Minimum bearing-stratum embedment (design criterion)** | – | × D / m | ASIRI; Han (2025) | AbstractProgramDesign | **New** | Required penetration of the toe into the dense bearing stratum, e.g. ≥ 2× the RI diameter in US DOT provisions. |
| E · Toe / base (end bearing) | **Limit unit tip resistance / limit pressure** | $q_p$, $p_l$ | kPa | ASIRI | PressuremeterTest | **Bridge** | Unit tip resistance / pressuremeter limit pressure. |
| E · Toe / base (end bearing) | **Design axial load per column (demand)** | $Q_r$ (N, $Q_p(0)$) | kN | ASIRI; EBGEO; FHWA | AbstractProgramDesign | **New** | Required / nominal axial load delivered to one inclusion head. |
| F · Load-transfer platform (LTP) | **LTP present** | – | – | ASIRI; EBGEO; FHWA | AbstractPlanarSamplingFeature | **New** | Whether a load-transfer platform is used. |
| F · Load-transfer platform (LTP) | **LTP type / material** | – | – | ASIRI; EBGEO; FHWA | AbstractPlanarSamplingFeature | **New** | Platform composition: unreinforced granular fill, geogrid-reinforced fill, soil-mixed / cemented raft, or (extreme) structural concrete slab. |
| F · Load-transfer platform (LTP) | **LTP thickness** | $H_M$ (h, h*) | m | ASIRI; EBGEO; FHWA | AbstractPlanarSamplingFeature | **New** | Vertical thickness of the platform. |
| F · Load-transfer platform (LTP) | **LTP fill friction angle** | φ' | ° | ASIRI, BS 8006-1, EBGEO, FHWA | DirectShearTest / TriaxialTest | **Bridge** | Shear-strength angle of the platform / embankment fill. |
| F · Load-transfer platform (LTP) | **LTP fill unit weight** | γ | kN/m³ | BS 8006-1; FHWA | LabDensityTest | **Bridge** | Unit weight of platform fill. |
| F · Load-transfer platform (LTP) | **Arching coefficient (BS)** | $C_{c,arch}$ | – | BS 8006-1 | AbstractProgramDesign | **New** | Marston arching coefficient (BS). |
| F · Load-transfer platform (LTP) | **Cap-stress ratio (BS)** | $p'_c/\sigma'_v$ | – | BS 8006-1 | AbstractProgramDesign | **New** | Ratio of stress on cap to average vertical stress. |
| F · Load-transfer platform (LTP) | **Hewlett–Randolph arching efficiency & passive coefficient (BS)** | $_{rown}$, $E_{cap}$, $E_{min}$, | – | BS 8006-1; FHWA | AbstractProgramDesign | **New** | Dome-arching efficiencies and K_p=(1+sinφ'_p)/(1−sinφ'_p) in the BS Hewlett–Randolph model. |
| F · Load-transfer platform (LTP) | **Critical stress ratio (EBGEO)** | $K_{crit}$ | – | EBGEO | AbstractProgramDesign | **New** | tan²(45°+φ'/2) in Zaeske arching. |
| F · Load-transfer platform (LTP) | **Stress on soft soil between elements** | $\sigma_{zo,k}$ | kPa | EBGEO | AbstractProgramDesign | **New** | Residual vertical stress on soil between heads. |
| F · Load-transfer platform (LTP) | **Stress on element (head)** | $\sigma_{zs,k}$ | kPa | EBGEO | AbstractProgramDesign | **New** | Vertical stress delivered to the head. |
| F · Load-transfer platform (LTP) | **Adapted-Terzaghi earth-pressure coefficient** | K | – | FHWA | AbstractProgramDesign | **New** | Lateral earth-pressure coefficient in LDC (0.75). |
| F · Load-transfer platform (LTP) | **Head-punching bearing factor (ASIRI)** | $N_q$ | – | ASIRI | AbstractProgramDesign | **New** | Prandtl bearing factor at critical-state φ'. |
| F · Load-transfer platform (LTP) | **Head stress on inclusion** | $q_p^+$ | kPa | ASIRI | AbstractProgramDesign | **New** | Vertical stress transferred onto the head. |
| G · Geosynthetic reinforcement (conditional) | **Geosynthetic present** | – | – | ASIRI, BS 8006-1, EBGEO, FHWA | AbstractProgramDesign | **New** | Whether basal reinforcement is used. |
| G · Geosynthetic reinforcement (conditional) | **Number of reinforcement layers** | $n_{geo}$ | – | EBGEO; FHWA | AbstractProgramDesign | **New** | Count of geosynthetic layers. |
| G · Geosynthetic reinforcement (conditional) | **Tensile stiffness** | J ($J_k$) | kN/m | BS 8006-1; EBGEO; FHWA | AbstractProgramDesign | **New** | Secant (isochronous) tensile stiffness at design strain/life. |

| Group | Element (parameter) | Symbol | Unit | Required by (standard) | DIGGS schema location | New element? | Definition |
|---|---|---|---|---|---|---|---|
| G · Geosynthetic reinforcement (conditional) | **Long-term design strength** | $T_D$ ($R_{B,d}$) | kN/m | BS 8006-1; EBGEO; FHWA | AbstractProgramDesign | New | Factored long-term tensile strength. |
| G · Geosynthetic reinforcement (conditional) | **Reduction factors (creep, damage, durability)** | A1–A5 / $f_m$ | – | BS 8006-1; EBGEO | AbstractProgramDesign | New | Partial/reduction factors on reinforcement strength. |
| G · Geosynthetic reinforcement (conditional) | **Reinforcement design tension** | $T_{rp}$ · $E_{M,geo}$ · T | kN/m | BS 8006-1; EBGEO; FHWA | AbstractProgramDesign | New | Tensile force the reinforcement must carry. |
| G · Geosynthetic reinforcement (conditional) | **Reinforcement anchorage / bond length** | $L_e$, $L_b$ ($L_A$) | m | BS 8006-1; EBGEO; FHWA | AbstractProgramDesign | New | Embedment / bond length of reinforcement beyond the outer row. |
| G · Geosynthetic reinforcement (conditional) | **Allowable / design strain** | $\varepsilon$ | % | BS 8006-1; EBGEO; FHWA | AbstractProgramDesign | New | Permitted tensile strain in reinforcement. |
| G · Geosynthetic reinforcement (conditional) | **Creep-strain limit over design life** | $\Delta\varepsilon_{kr}$ | % | BS 8006-1; EBGEO | AbstractProgramDesign | New | Additional creep strain allowed over service life. |
| G · Geosynthetic reinforcement (conditional) | **Minimum reinforcement load fraction** | – | % | BS 8006-1 | AbstractProgramDesign | New | Floor on load carried by reinforcement (≥15%). |
| H · Soft soil, ground model & loading | **Undrained shear strength (design)** | $c_u$ | kPa | ASIRI; FHWA | TriaxialTest / InsituVaneTest | Bridge | Soft-soil undrained strength adopted in design. |
| H · Soft soil, ground model & loading | **Effective strength (design)** | c', φ' | kPa, ° | ASIRI; BS 8006-1 | TriaxialTest / DirectShearTest | Bridge | Design effective-strength parameters of soft soil. |
| H · Soft soil, ground model & loading | **Compressibility parameters (design)** | $_d$, $C_c$, $C_s$, $\sigma'_p$, O | – | ASIRI; FHWA | ConsolidationTest | Bridge | Soft-soil compressibility set adopted in design. |
| H · Soft soil, ground model & loading | **Subgrade reaction modulus (Winkler)** | $k_s$ | kN/m³ | EBGEO | AbstractProgramDesign | New | Elastic support stiffness of soft soil. |
| H · Soft soil, ground model & loading | **Groundwater level (design)** | – | m | ASIRI | Well / WaterLevelMonitoring | Bridge | Design water table. |
| H · Soft soil, ground model & loading | **Embankment height** | H | m | BS 8006-1; FHWA | AbstractProgramDesign | New | Height of embankment / fill. |
| H · Soft soil, ground model & loading | **Fill / embankment unit weight** | $\gamma$ | kN/m³ | BS 8006-1 | LabDensityTest | Bridge | Unit weight of embankment fill. |
| H · Soft soil, ground model & loading | **Surcharge / live load** | $w_s$ (q, $p_k$) | kPa | BS 8006-1; FHWA | AbstractProgramDesign | New | Applied surcharge / traffic / live load. |
| H · Soft soil, ground model & loading | **Structure / slab load** | q | kPa | ASIRI; FHWA | AbstractProgramDesign | New | Distributed structural load (slab, tank, footing). |
| I · Load transfer & verification (ULS / stability) | **Load (stress) efficacy** | E | – | ASIRI | AbstractProgramDesign | New | Fraction of total load carried by the inclusions. |
| I · Load transfer & verification (ULS / stability) | **Settlement efficacy / settlement-reduction ratio** | $E_{tass}$ | – | ASIRI | AbstractProgramDesign | New | Reduction of settlement relative to untreated ground. |
| I · Load transfer & verification (ULS / stability) | **Active earth-pressure coefficient (fill)** | $K_a$ | – | BS 8006-1; FHWA | AbstractProgramDesign | New | tan²(45°−φ'/2) of the embankment fill. |
| I · Load transfer & verification (ULS / stability) | **Global-stability analysis method (coded)** | – | – | ASIRI, BS 8006-1, EBGEO, FHWA; Han (2025) | AbstractProgramDesign | New | Method for RIS-embankment slope stability, Column-Wall Method (CWM, reference), Equivalent Strength (ESM), Stress Reduction (SRM), Pile Support |
| I · Load transfer & verification (ULS / stability) | **Global / slope stability factor** | FS | – | BS 8006-1; EBGEO; FHWA; Han (2025) | AbstractProgramDesign | New | Factor of safety on overall / rotational stability; US DOT provisions require FS ≥ 1.5. |
| I · Load transfer & verification (ULS / stability) | **Lateral spreading demand** | $T_{ds}$ / $P_{Lat}$ | kN/m | BS 8006-1; EBGEO; FHWA | AbstractProgramDesign | New | Outward thrust at the embankment base. |
| I · Load transfer & verification (ULS / stability) | **Lateral-spreading factor of safety** | $FS_{ls}$ | – | FHWA | AbstractProgramDesign | New | Safety on spreading resistance vs demand. |
| I · Load transfer & verification (ULS / stability) | **Pile-group lateral extent** | $L_p$ | m | BS 8006-1; FHWA | AbstractProgramDesign | New | Outward extent columns must cover beyond crest. |
| I · Load transfer & verification (ULS / stability) | **Seismic / liquefaction outcome** | – | – | ASIRI; Han (2025) | AbstractProgramDesign | New | Result of seismic / liquefaction verification; RIs may mitigate liquefaction by soil densification but their stiffness effect is limited. |
| J · Serviceability & design-set acceptance | **Total settlement** | $s_{tot}$ | mm | ASIRI; FHWA | AbstractProgramDesign | New | Predicted total settlement of the system. |
| J · Serviceability & design-set acceptance | **Differential settlement** | $\Delta s$ | mm | ASIRI; FHWA | AbstractProgramDesign | New | Predicted differential/surface settlement. |
| J · Serviceability & design-set acceptance | **Serviceability criterion** | – | – | ASIRI; BS 8006-1; Han (2025) | AbstractProgramSpecificaton | New | Owner limit on total / differential settlement or deflection; US DOT provisions cite max differential ≈ 0.25–0.75 in (unit cell) or 0.5 in/100 ft to 1.5 in/1000 ft. |
| J · Serviceability & design-set acceptance | **Reinforcement mid-span deflection** | y | mm | BS 8006-1 | AbstractProgramDesign | New | Sag of reinforcement between heads. |
| J · Serviceability & design-set acceptance | **Durability provisions** | – | – | ASIRI | AbstractProgramDesign / DesignGroutMix | New | Cover, cement type, RF chain, corrosion measures. |
| J · Serviceability & design-set acceptance | **Load-test regime & acceptance (set at design)** | – | – | ASIRI; FHWA | AbstractProgramSpecificaton | New | Type, number and pass criteria of load tests. |
| J · Serviceability & design-set acceptance | **Integrity-testing regime** | – | – | ASIRI | AbstractProgramSpecificaton | New | Extent and method of integrity testing. |
| J · Serviceability & design-set acceptance | **Monitoring / instrumentation plan** | – | – | ASIRI; EBGEO; FHWA; Han (2025) | AbstractProgramSpecificaton / Monitor | New | Instruments, locations and frequencies, settlement plates, horizontal shape-arrays (SAAs), strain gauges in test RIs, inclinometer tubes and vibration monitoring. |

| Group | Element (parameter) | Symbol | Unit | Required by (standard) | DIGGS schema location | New element? | Definition |
|---|---|---|---|---|---|---|---|
| A · Column identity & installation context | **Column / element ID** | – | – | EN 12699/14679; Project spec/ITP | AbstractLinearSamplingFeature | **New** | Unique identifier of the installed inclusion (e.g. CMC-EB-127). |
| A · Column identity & installation context | **Project / contract** | – | – | Project spec/ITP | Project | **Existing** | Project and contract the column belongs to. |
| A · Column identity & installation context | **Rig / machine ID** | – | – | Project spec/ITP | Equipment / DrillRig | **Existing** | Identifier of the rig / data logger (e.g. DRL-BAU-OmniBox-202). |
| A · Column identity & installation context | **Operator / crew** | – | – | Project spec/ITP | BusinessAssociate | **Existing** | Rig operator and supervisor. |
| A · Column identity & installation context | **Installation date & start/end time** | – | date-time | EN 12699/14679; Project spec/ITP | AbstractConstructionActivity | **New** | Calendar date and clock times of installation. |
| A · Column identity & installation context | **Installation method (coded)** | – | – | ASIRI; EN 12699/14679; Project spec/ITP | AbstractConstructionActivity | **New** | Technique used (full-displacement auger, partial, drilled-displacement, pre-drill). |
| A · Column identity & installation context | **Installation sequence / order** | – | – | EN 12699/14679; Han (2025) | AbstractConstructionActivity | **New** | The order in which inclusions are installed relative to neighbours and to any adjacent structures. |
| A · Column identity & installation context | **Tool / auger diameter** | – | mm | Project spec/ITP | Equipment | **Existing** | Diameter of the displacement tool / auger. |
| B · As-built geometry | **As-built diameter** | D | mm | ASIRI; Project spec/ITP | AbstractLinearSamplingFeature / ConcretePile | **New** | Achieved column diameter (tool + volume check). |
| B · As-built geometry | **As-built length** | L | m | EN 12699/14679; Project spec/ITP | AbstractLinearSamplingFeature / ConcretePile | **New** | Achieved length from cut-off to toe. |
| B · As-built geometry | **Maximum depth reached** | $z_{max}$ | m | Project spec/ITP | AbstractLinearSamplingFeature / MeasurementWhileDrilling | **New** | Deepest tip depth attained. |
| B · As-built geometry | **Working-platform / rig-reference level (depth datum)** | – | m | EBGEO; Project spec/ITP | AbstractConstructionActivity | **New** | Level of the platform the rig stands on, the datum for all recorded depths. |
| B · As-built geometry | **Working-platform material** | – | – | EBGEO; Project spec/ITP | AbstractConstructionActivity | **New** | Material of the working platform that carries the installation plant, typically compacted granular or cemented soil; usually retained as the first LTP layer. |
| B · As-built geometry | **Working-platform thickness** | $t_{wp}$ | m | EBGEO; Project spec/ITP | AbstractConstructionActivity | **New** | Thickness of the working-platform layer built to support the rig; retained as part of the LTP once construction is complete. |
| B · As-built geometry | **Top (cut-off) elevation** | – | m | ASIRI; Project spec/ITP | AbstractLinearSamplingFeature | **New** | Level of the finished column top / trim. |
| B · As-built geometry | **Toe elevation** | – | m | ASIRI; FHWA | AbstractLinearSamplingFeature | **New** | Level of the column base. |
| B · As-built geometry | **As-built head position (x, y)** | – | m | ASIRI | PointLocation | **Existing** | Plan coordinates of the installed head. |
| B · As-built geometry | **Verticality / inclination (as-built)** | – | ° or % | ASIRI | MeasurementWhileDrilling | **New** | Achieved deviation from vertical. |
| C · Depth-indexed process record (the coverage) | **Penetration rate vs depth** | $v_{pen}$ | m/h (cm/min) | Project spec/ITP | MeasurementWhileDrilling | **New** | Downward advance speed of the tool. |
| C · Depth-indexed process record (the coverage) | **Extraction rate vs depth** | $v_{ext}$ | m/h | Project spec/ITP | MeasurementWhileDrilling | **New** | Withdrawal speed during concreting. |
| C · Depth-indexed process record (the coverage) | **Rotation speed** | $n_{rot}$ | rev/min | Menard panel (GIW p.41) | MeasurementWhileDrilling | **New** | Auger rotation rate. |
| C · Depth-indexed process record (the coverage) | **Rotation torque vs depth** | C ($M_t$) | kN·m (t·m) | Menard anchorage V<250 m/h & C>5 t·m/0.5 | MeasurementWhileDrilling | **New** | Torque resisting rotation. |
| C · Depth-indexed process record (the coverage) | **Crowd / down-thrust force** | $F_{crowd}$ | kN (bar) | HBI DAQ (crowd/torque); Menard down-pull | MeasurementWhileDrilling | **New** | Static down-force pushing the tool. |
| C · Depth-indexed process record (the coverage) | **Grout / concrete pressure vs depth** | $P_B$ | bar (kPa) | Menard CMC brochure (controlled pressure | MeasurementWhileDrilling | **New** | Pumping pressure of the fluid concrete/grout. |
| C · Depth-indexed process record (the coverage) | **Grout / concrete volume vs depth** | V | m³ | HBI neat & actual grout volume | MeasurementWhileDrilling | **New** | Cumulative placed volume against depth. |
| C · Depth-indexed process record (the coverage) | **Pump strokes** | – | count | HBI pump stroke count; Menard pump blows | MeasurementWhileDrilling | **New** | Concrete-pump stroke count. |
| C · Depth-indexed process record (the coverage) | **Volume overconsumption** | – | % | Menard CMC record (GIW p.42) | MeasurementWhileDrilling | **New** | Actual vs theoretical volume (Menard). |
| C · Depth-indexed process record (the coverage) | **Adjacent-element / ground-movement check during install** | – | mm / obs | Keller HBI QC (monitor adjacent location | Monitor / MonitorResult | **Existing** | Communication, heave, grout rise or top-deflection of neighbouring fresh columns. |
| D · Concrete / grout as-placed | **Total concrete / grout volume placed** | $V_{tot}$ | m³ | EN 12699/14679; Project spec/ITP | GroutBatch / AbstractConstructionActivity | **Bridge** | Total material placed in the column. |
| D · Concrete / grout as-placed | **Grout-level monitoring / grout drop (uncovered column)** | – | – | Project spec/ITP; DFI (2023) | AbstractConstructionActivity | **New** | Monitoring of the grout level in freshly placed, uncovered columns (especially below groundwater or in karst) and the corrective action (topping-off, re-drilling / |
| D · Concrete / grout as-placed | **Wet-squeeze / grout-return volume** | $V_{ws}$ | m³ | Project spec/ITP; DFI (2023) | AbstractConstructionActivity | **New** | Volume of fluid grout or concrete returning to the surface after placement or during adjacent installation; a small return is normal, an excessive or sudden |
| D · Concrete / grout as-placed | **Concrete mix / strength class** | – | – | Project spec/ITP | DesignGroutMix | **Existing** | Specified mix and class (e.g. C20). |
| D · Concrete / grout as-placed | **Cement type** | – | – | Project spec/ITP | DesignGroutMix | **Existing** | Cement type used (e.g. Type V sulfate-resisting). |
| D · Concrete / grout as-placed | **Fresh-concrete slump** | – | mm | ASTM C143; Project spec/ITP | SlumpTest | **Existing** | Workability of the delivered concrete. |
| D · Concrete / grout as-placed | **Delivery-to-placement elapsed time (transport window)** | – | min | Project spec/ITP | GroutBatch | **Existing** | Time from batching / loading to placement in the column. |

| Group | Element (parameter) | Symbol | Unit | Required by (standard) | DIGGS schema location | New element? | Definition |
|---|---|---|---|---|---|---|---|
| D · Concrete / grout as-placed | **Fresh grout density / flow (grout-based RI)** | – | kg/m³ (s) | ASTM C939/D4380 | MudBalanceTest / FlowConeTest / MarshFunnelTest | **Existing** | Density (mud balance) or flow of a cement grout where grout, not concrete, is used. |
| D · Concrete / grout as-placed | **Concrete temperature** | – | °C | ASTM C1064; Project spec/ITP | MaterialTest / GroutBatch | **Existing** | Temperature of delivered concrete. |
| D · Concrete / grout as-placed | **Air content** | – | % | ASTM C231 | MaterialTest | **Existing** | Entrained/entrapped air of fresh concrete. |
| D · Concrete / grout as-placed | **Batch-ticket data** | – | – | IR A-1 | GroutBatch / AssociatedFile | **Existing** | Plant, ticket #, truck, load time, m³, purchaser. |
| D · Concrete / grout as-placed | **Fresh-concrete sampling (cylinders cast)** | – | – | ASTM C172/C31; Project spec/ITP | Sample / GroutSpecimen | **Existing** | Cylinders taken for strength testing. |
| D · Concrete / grout as-placed | **Grout-pump / plant calibration** | – | – | EN 12699/14679 | GroutPump / GroutPlant | **Existing** | Calibration record of the concrete/grout pump and batching plant, pressure gauges, stroke volume, flow meter and batcher scales, with date and certificate reference. |
| E · Termination & anchorage | **Indicator / pre-production test programme** | – | – | Project spec/ITP; DFI (2023) | AbstractConstructionActivity | **New** | Pre-production trial that correlates the drilling response to the stratigraphy and establishes and validates the termination criteria before production. |
| E · Termination & anchorage | **Refusal / anchorage criterion (coded)** | – | – | Project spec/ITP | AbstractConstructionActivity | **New** | Rule used to stop driving (rate / rate+torque / design depth). |
| E · Termination & anchorage | **Achieved termination outcome** | – | – | Project spec/ITP | AbstractConstructionActivity | **New** | Whether design depth or refusal governed. |
| E · Termination & anchorage | **As-built toe embedment into competent stratum** | – | m (× D) | ASIRI; Han (2025) | AbstractLinearSamplingFeature | **New** | Length of column penetrating the bearing / competent layer; checked against the design minimum (e.g. ≥ 2× diameter). |
| E · Termination & anchorage | **Drilling time, compressible vs competent layer** | – | min | Project spec/ITP | AbstractConstructionActivity / MeasurementWhileDrilling | **New** | Time spent drilling each stratum (CSS split). |
| F · Reinforcement (conditional) | **Reinforcement present & type** | – | – | ASIRI | AbstractLinearSamplingFeature | **New** | Any bar/cage/profile installed (usually none). |
| F · Reinforcement (conditional) | **Reinforcement length / section** | – | m | ASIRI | AbstractLinearSamplingFeature | **New** | Embedded length and section of reinforcement. |
| G · Durations & production | **Drilling duration** | – | min | Project spec/ITP | AbstractConstructionActivity | **New** | Time to advance to depth. |
| G · Durations & production | **Concreting duration** | – | min | Project spec/ITP | AbstractConstructionActivity | **New** | Time to place concrete during extraction. |
| G · Durations & production | **Remarks / downtime (breakdown, standby, move)** | – | – | Project spec/ITP | AbstractConstructionActivity | **New** | Non-productive events noted by the crew. |
| G · Durations & production | **Attached rig-log file (PDF)** | – | – | Project spec/ITP | AssociatedFile | **Existing** | The original per-column log document. |

| Group | Element (parameter) | Symbol | Unit | Required by (standard) | DIGGS schema location | New element? | Definition |
|---|---|---|---|---|---|---|---|
| A · As-built verification survey | **Post-construction head coordinates** | – | m | ASIRI; Project spec/ITP | PointLocation | **Existing** | Surveyed plan position of each finished head. |
| A · As-built verification survey | **Position deviation vs tolerance** | – | mm | ASIRI; Project spec/ITP | PointLocation / AbstractProgramSpecificaton | **Bridge** | Difference between as-built and design position. |
| A · As-built verification survey | **Verticality check** | – | ° or % | ASIRI | PointLocation | **Existing** | Verified inclination of the column. |
| A · As-built verification survey | **Post-construction diameter verification** | – | m | Project spec/ITP | AbstractLinearSamplingFeature / MaterialTest | **New** | Exposed/measured head diameter. |
| A · As-built verification survey | **Cut-off / trim level (verified)** | – | m | ASIRI; Project spec/ITP | AbstractLinearSamplingFeature | **New** | Confirmed finished top level after trimming. |
| B · Integrity & material verification | **Low-strain integrity test** | – | – | ASIRI; ASTM D5882; Han (2025) | Test / AbstractInsituTestProcedure | **Existing** | Sonic-echo / impedance test of column continuity; US DOT: on ≈ 5% of production RIs. |
| B · Integrity & material verification | **Thermal integrity profiling (TIP)** | – | – | DFI (2023); ASTM D7949 | Test / AbstractInsituTestProcedure | **Existing** | Thermal (curing-heat) integrity test of the column cross-section and continuity; a non-destructive method complementary to the low-strain test. |
| B · Integrity & material verification | **Concrete cylinder / core compressive strength** | $f_c$ (7 d / 28 d) | MPa | ASTM C39/C42; Project spec/ITP | UnconfinedCompressiveStrengthTest / MaterialTest | **Existing** | Measured compressive strength of samples/cores. |
| C · Load testing (performance verification) | **Test-element class & test-load level** | – | / % design loa | ASIRI; EC7 | Test / AbstractInsituTestProcedure | **Existing** | Whether the tested column is a trial / sacrificial or a working column, and the load level (quality vs capacity; to 150% or to failure). |
| C · Load testing (performance verification) | **Static load test on single column** | Q–s | kN, mm | ASIRI; FHWA; ASTM D1143; Han (2025) | Test / AbstractInsituTestProcedure | **Existing** | Maintained-load test giving load–settlement; US DOT: verification tests to 200% and production tests to 150% of design load. |
| C · Load testing (performance verification) | **Load-test setup / reaction & procedure** | – | – | BS 8006-1; ASTM D1143 | Test / AbstractInsituTestProcedure | **Existing** | Reaction system (kentledge / anchored / beam), load schedule, hold increments, datum, toe tell-tales. |
| C · Load testing (performance verification) | **Critical creep load** | $R_{c,cr}$ | kN | ASIRI | Test / AbstractInsituTestProcedure | **Existing** | Load at onset of creep from the test curve. |
| C · Load testing (performance verification) | **Plate Load Test on column head (PLT)** | – | kN, mm | ASTM D1194/D1196; Project spec/ITP | Test / AbstractInsituTestProcedure | **Existing** | Plate test on the head to a multiple of design load. |
| C · Load testing (performance verification) | **Zone Load Test on LTP (ZLT)** | – | kN, mm | BS 8006-1; Project spec/ITP | Test / AbstractInsituTestProcedure | **Existing** | Large-plate test on the platform over a group. |
| C · Load testing (performance verification) | **Dynamic load test (if used)** | – | kN, mm | FHWA; ASTM D4945 | Test / PDARecord | **Existing** | High-strain dynamic test of capacity. |
| C · Load testing (performance verification) | **Measured column stiffness / capacity** | – | kN, kN/mm | ASIRI; FHWA | Test / AbstractInsituTestProcedure | **Existing** | Capacity & stiffness back-figured from tests. |
| D · Load-transfer platform verification | **LTP thickness (as-built)** | – | m | ASIRI; EBGEO; FHWA | AbstractPlanarSamplingFeature | **New** | Constructed platform thickness. |
| D · Load-transfer platform verification | **LTP compaction / degree of compaction** | $D_{Pr}$ | % | EBGEO; FHWA | InsituDensityTest / LabCompactionTest | **Existing** | Achieved relative compaction of fill. |
| D · Load-transfer platform verification | **LTP fill friction angle (verified)** | φ' | ° | EBGEO; FHWA | DirectShearTest / TriaxialTest | **Existing** | Confirmed shear strength of platform fill. |
| D · Load-transfer platform verification | **Geosynthetic installation record** | – | – | EBGEO; FHWA | AssociatedFile / AbstractPlanarSamplingFeature | **Bridge** | Layers, orientation, tension, overlap, cover. |
| D · Load-transfer platform verification | **LTP special-inspection frequency / regime** | – | – | DFI (2023); Project spec/ITP | AbstractProgramSpecificaton | **New** | The special-inspection regime for the LTP: continuous visual inspection, and compaction / density testing at a frequency set by foundation type and area (per |
| E · Instrumentation & monitoring | **Settlement monitoring** | $s_{meas}$ | mm | ASIRI; EBGEO; FHWA | Monitor / MonitorResult | **Existing** | Measured settlement (plates, extensometers). |
| E · Instrumentation & monitoring | **Differential settlement (measured)** | $\Delta s_{meas}$ | mm | ASIRI; FHWA | Monitor / MonitorResult | **Existing** | Measured surface differential settlement. |
| E · Instrumentation & monitoring | **Measured settlement reduction (treated vs untreated)** | – | – / mm | ASIRI | Monitor / MonitorResult | **Existing** | Improvement factor of measured settlement vs an untreated baseline. |
| E · Instrumentation & monitoring | **Pore-water pressure** | u | kPa | ASIRI | Monitor / MonitorResult | **Existing** | Measured excess/steady pore pressure. |
| E · Instrumentation & monitoring | **Load on head / soil (earth-pressure cells)** | – | kPa | ASIRI; EBGEO | Monitor / MonitorResult | **Existing** | Measured stress on head and on soft soil. |
| E · Instrumentation & monitoring | **Reinforcement / geosynthetic strain** | $\varepsilon_{meas}$ | % | BS 8006-1; EBGEO | Monitor / MonitorResult | **Existing** | Measured tensile strain in reinforcement. |
| E · Instrumentation & monitoring | **Lateral movement (inclinometer)** | – | mm | EBGEO; FHWA | Monitor / MonitorResult | **Existing** | Horizontal ground/toe movement. |
| E · Instrumentation & monitoring | **Adjacent-structure / adjacent-RI deformation (installation effect)** | – | mm | Han (2025) | Monitor / MonitorResult | **Existing** | Measured lateral displacement, heave or deformation of neighbouring structures and previously installed RIs caused by displacement installation. |
| E · Instrumentation & monitoring | **Composite shear-wave velocity / seismic Site Class (post-improvement)** | $V_{s,comp}$ | m/s | DFI (2023); IBC (Site Class) | GeophysicalFieldSurvey / LabVelocityTest | **Existing** | Shear-wave velocity of the improved composite ground measured after installation, used to confirm or revise the seismic Site Class. |
| E · Instrumentation & monitoring | **LTP shear-deformation monitoring (shape array)** | – | mm | DFI (2023) | Monitor / MonitorResult | **Existing** | Monitoring of shear deformation within the load-transfer platform using a shape array or inclinometer, typically during a group / area load test. |
| F · Acceptance & records | **Acceptance criteria (settlement / differential / load test)** | – | – | ASIRI; FHWA; Project spec/ITP | AbstractProgramSpecificaton | **New** | Pass/fail thresholds from the design. |
| F · Acceptance & records | **Inspection sign-off & non-conformance (NCR) records** | – | – | ASIRI; Project spec/ITP | DocumentInformation / AssociatedFile | **Existing** | Per-element hold / witness sign-off and any non-conformance / corrective-action trail. |
| F · Acceptance & records | **Measured vs predicted settlement** | – | mm | ASIRI; FHWA | Monitor | **Bridge** | Comparison of monitored to design settlement. |
| F · Acceptance & records | **Test frequency / coverage achieved** | – | – | ASIRI; Project spec/ITP | DocumentInformation | **Existing** | Numbers of PLT/ZLT/integrity/strength tests done. |
| F · Acceptance & records | **As-built / verification report** | – | – | ASIRI; EC7 | DocumentInformation / AssociatedFile | **Existing** | Final documentation of the built system & tests. |
| F · Acceptance & records | **Monitoring duration & frequency** | – | – | ASIRI; EBGEO; FHWA | Monitor | **Existing** | How long and how often the system is monitored. |